Dynamics 4

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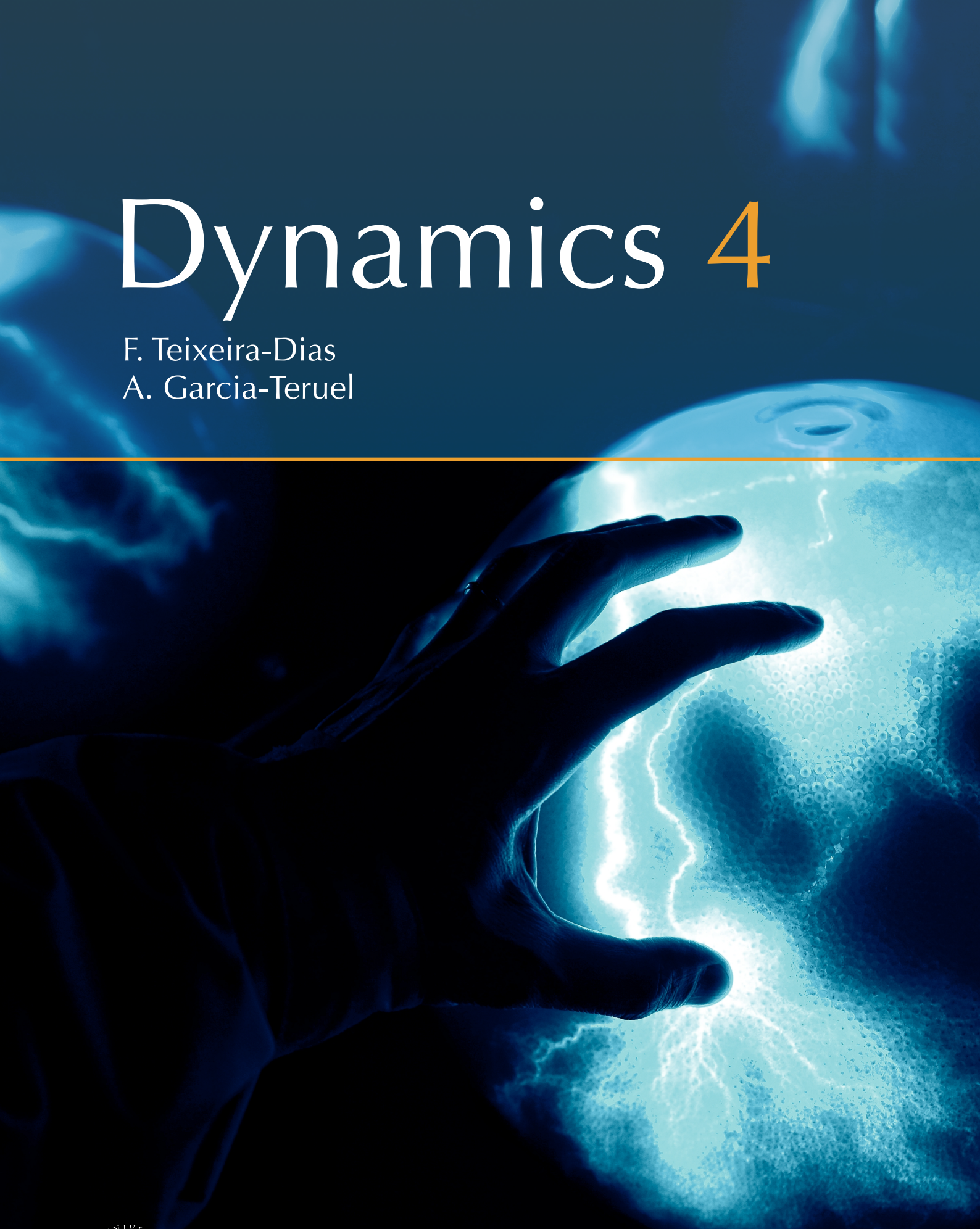

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The authors acknowledge and thank the invaluable contribution to these notes from the Dynamics 4 tutors and students. Your continued help in finding typos, correcting mistakes and suggesting and contributing with new content has been crucial in making these notes a useful and a major component of the learning experience. Thank you classes of 2013 to 2025. Thank you Nicole, Kayleigh, Thomas, Maisie, Katie, James, Sam, Aldo and Thibaut.

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1 Course information and structure

1.1 Introduction

Welcome to Dynamics 4 — this course is part of the Mechanical Engineering degrees at the School of Engineering of The University of Edinburgh. It provides an understanding of core aspects of advanced dynamic analysis, including system modelling, dynamic response and vibration analysis, linear and non-linear structural dynamics, wave propagation, some aspects of impulsive dynamics and the dynamics of continuous and multi-degree of freedom systems. The main aims and objectives of this course are to obtain an understanding and appreciation of the potential and of the limits of analytical solutions, and the value of these, in underpinning modern computer methods for simulating the mechanical response of complex dynamic structural systems.

The course format presented here is for the academic year 2026-27. The course delivery methods have changed significantly in recent years. The course content and learning outcomes, however, have remained mostly unchanged.

1.2 General information and prerequisites

Dynamics 4 is a compulsory core course for the MEng (hons) Mechanical Engineering undergraduate degree and optional for other degrees, such as Civil Engineering, Electrical and Mechanical Engineering and BEng (hons) Mechanical Engineering. Although not a prerequisite for Dynamics 5 or any other optional Level 10/11 courses, which covers structural impact, ballistics and crashworthiness, and high strain rate material behaviour, in Dynamics 4 you will also learn basic concepts of impact on structures and how impact may affect the mechanical response of structures. Dynamics 4 is a 10 credit course and students are expected to put, on average, a total of approximately 100 hours of work into the course. Details about the opportunities for feedback are discussed in Section 1.7.

The main topics covered in Dynamics 4 focus on free and forced vibration of continuous systems. There are no compulsory prerequisites for previous courses in structural dynamics, which often mainly cover vibration of single- and multiple-degree-of-freedom systems. The scope of the present course is thus wider as we will be covering methods that can be applied to all kinds of dynamic behaviour in both simple and complex engineering systems and structures. A good and solid understanding of engineering mathematics is, however, beneficial.

The academic level of the Dynamics 4 course is generally higher than that of previous courses. Added complexities may arise in three different areas: (i) the maths, which may require correct execution of somewhat lengthy and complex calculations and/or derivations. In particular, students need skill in taking derivatives of complex functions and in doing matrix calculations; (ii) the ability to analyse and correctly describe complex motions in engineering systems; (iii) a certain level of abstract thinking is also required to understand concepts such as mode shape functions in continuous systems, or the concept of virtual displacements in Lagrangian dynamics. This, however, may have been partly covered in previous courses.

Important note Dynamics 4 is strongly based on intermediate and advanced mathematics and physics concepts. Therefore, if this course is optional for you, you are advised to reconsider taking Dynamics 4 if you have not taken and passed engineering mathematics courses (e.g. Engineering Mathematics 1A/1B, and Engineering Mathematics 2A/2B), or other courses equivalent to these.

These Course Notes give a practical view of the covered topics, showing and discussing real-life applications. However, because of the somewhat abstract nature of some of the course material, it is doubly important for students to get a practical grasp on how to use the methods introduced in the lectures. The course makes extensive use of worked examples of different levels of complexity. Examples and detailed step-by-step solutions are presented in grey boxes. When going through these examples and solutions, it is very important that you not only look at the methods of solution employed, but also try to understand any relevant engineering assumptions and how these would apply to different systems. It is indispensable that students additionally try to solve all the tutorial examples included in the separate exercise sheets.

1.3 Teaching team

The Dynamics 4 teaching team has extensive experience in the application of theoretical and design methods to understand dynamic engineering systems. Don't hesitate to contact any of them if you want to know more about their engineering, academic or research backgrounds. They can be contacted by e-mail or by sending a request for an online or in-person meeting through the University of Edinburgh EASE Office 365 system.

Teaching team The Dynamics 4 instructors for the current academic year are:

- Filipe Teixeira-Dias (Course Organiser) · f.teixeira-dias@ed.ac.uk
  3.09 Alexander Graham Bell (AGB) building
- Nicole Kretz (Tutor) · n.c.kretz@sms.ed.ac.uk
  James Clerk Maxwell building (JCMB)

For issues related to your enrolment in the course, marks and grading systems, teaching timetables and room locations, you should contact the Course Secretary at the Engineering Teaching Organisation (ETO) by email or meet them at the ETO in the Engineering Forum building.

Course secretary The Dynamics 4 course secretary for the current academic year is:

- Catherine Davidson · c.davidson@ed.ac.uk, eto@eng.ed.ac.uk
  Engineering Teaching Organisation (ETO)
  1.002 Engineering Forum

1.4 Course content

These Dynamics 4 Course Notes cover three main subject areas. Each one of these is covered in a separate chapter. Please note that not all subject areas might be covered during the lectures and assessed. Below is a short and somewhat generic description of each one of these subject areas.

- The Lagrange method of analytical dynamics is the first topic and will be covered in Chapter 2, titled *Analytical dynamics*. This is a formal approach for setting up equations of motion for complex dynamic systems with dynamic constraints (i.e. constrained motions). Free Body Diagrams (FBD), discussed in Structural Mechanics and Dynamics 3 and other

courses, can also be used but prove quite difficult when dealing with complex systems which operate under linear or non-linear dynamic constraints. The Lagrange method, however, allows the derivation of correct sets of Equations of Motion (EoM) through formal calculations from the energy functions of the system. Hopefully, the advantages of using this method will become obvious early in this course.

- Wave propagation in continuous systems is the second topic, discussed in detail in Chapter 3 — *Dynamics of continuous systems*. This module covers the understanding of wave propagation in continuous solids, such as long bars and beams, through a systematic approach for deriving the parameters of lumped-parameter descriptions. The properties of wave propagation, including sound propagation, and the standing waves which characterise the fundamental vibration modes of continuous systems with boundaries, will also be discussed.

- The third and final topic expands on the previous module to explore the effects of impacts on structures, namely to understand how impulsive loads may change the dynamic response of slender structures such as beams. This is covered in Chapter 4 — *Impulsive dynamics*.

1.5 Course structure

The chapters described above are taught in seven Learning Modules, labelled LM1 to LM7. Each Dynamics 4 Learning Module covers a specific topic or sub-topic, it will address one or more of the course Learning Outcomes and will consist of a number of different learning elements. These might include short videos describing the theoretical background in detail, walked-through step-by-step example solutions, tutorial sheets and solutions, focused discussions, external URL links, recommendations for additional reading, etc. You will be able to find all these on Learn in the *Course Materials and Tools* folder. The Learning Modules should be followed in sequence, starting with LM1. Make sure you download and read the Dynamics 4 Quick Guide, which you can also find on the course Learn pages. This guide covers most of what you need to know about how the course will be delivered in 2026-27.

A learning guide is provided in Figure 1.1, complementing the Dynamics 4 Quick Guide, which you can find on Learn. Note that the time needed to complete each learning module — and the recommended *before date* — is estimated and may differ from person to person. You should not see these as targets but instead as a way to help you plan your semester and the preparation for exams. The estimated times include going through all the materials (i.e. videos, exercise sheets, solutions, etc.), interaction in the discussion boards and self-study time.

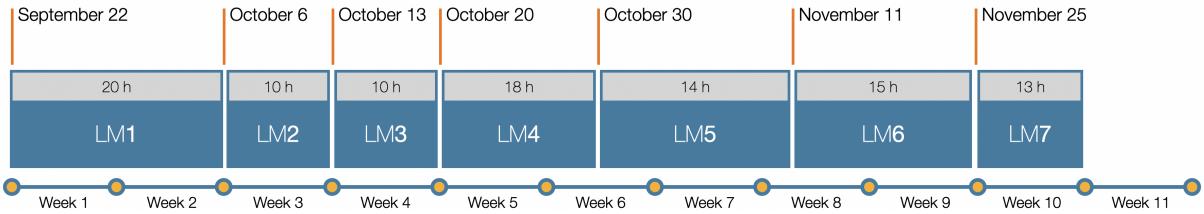


Figure 1.1: Teaching structure of Dynamics 4 for the academic year 2026-27: sequence of Learning Modules, estimated time and suggested dates for completion of modules. Note that the time needed to complete each Learning Module and the recommended *before date*, is estimated and may differ from person to person. These should not be seen as targets but instead as a way to help you plan your semester and the preparation for exams. The estimated times include going through all the materials, interaction in the discussion boards and self-study time.

You will find a short description of each Dynamics 4 Learning Module in the diagram on page 11 of these notes. This includes the topics covered and where you can find them in these course notes. The course lectures covering the theoretical background use slides, which are also

indicated in the course structure on page 11, along with the exercise sheets and examples that will help you consolidate these topics. Each module has a dedicated thread in the Dynamics 4 Discussion Board, which you can access from Learn.

1.6 Learning outcomes

Learning Outcomes do not necessarily have to be comprehensive. They should instead constitute a select group of examples of things — skills, understanding and knowledge — students are expected to learn and acquire during the course. The Dynamics 4 Learning Outcomes, listed below, cover a range of different skills, methods and engineering topics.

Learning outcomes On completion of this course, students will be able to:

- ✓ Apply virtual work-based methods to dynamical systems, relating between Lagrangian and Newtonian Mechanics;
- ✓ Derive system differential equations of motion for dynamical systems from energy-based approaches (e.g. Lagrange's method);
- ✓ Recognise forms of advanced dynamical behaviour, such as system instability and non-linearity, and appreciate their effects on the dynamical response and methods used to analyse them;
- ✓ Identify structural dynamic instability causes and propose solutions; and
- ✓ Know the common wave equations for basic structural elements and be able to use these to find natural frequencies and mode shapes.

1.7 Assessment and feedback

Dynamics 4 is assessed by exam only, which is closed bookwith three compulsory questions. When revising, note that all core material from relevant courses in previous years is assumed known.

Assessment Assessment is by exam only (100% written exam). This will be a closed book exam with three compulsory questions. Core material from relevant courses in previous years is assumed to be known and, as such, may not be explicitly covered in Dynamics 4.

Students have a number of formal opportunities to get feedback from the instructors on how they are matching their expectations and the learning outcomes. These include the Mid-Course Feedback and a number of formative feedback moments in the second half of the semester. You are also encouraged to give feedback to the instructors about all aspects related to the course (i.e. teaching, delivery, materials, etc.) Constructive criticism/suggestions are welcome and the Mid-Course Feedback and the final course feedback questionnaire are ideal moments to do this. You can also use the *course_logistics* thread in the Discussion Board for this.

All course materials are made available to students on Learn from the start of the semester. This includes these course notes, the lecture slides and lecture notes, lecture recordings, the tutorial exercise sheets, and worked out examples. More information about Dynamics 4 can be found on the course selection and programme building tool Path (https://edin.ac/4xfB5eA) and on the University's course catalogue, which is part of the Degree Regulations and Programmes of Study (DRPS 2026-27, https://edin.ac/3NxmxGh).

LM1 — Learning Module 1: Lagrange's method for conservative systems

- The contents of this module are covered in **Sections 2.1 and 2.2 (pages 15 to 30)**
- Use **Slides LM1.3 to LM1.20** (and watch the lecture recordings if you want).
- Solve **examples 1.1 to 1.4 in Exercise Sheet 1**
- The discussion board thread for this module is labelled **analytical_dynamics**
- The estimated time needed to complete the module is **20 hours**, ideally **by 6 October**

LM2 — Learning Module 2: Generalised forces and non-conservative systems

- The contents of this module are covered in **Section 2.3 (pages 30 to 36)**
- Use **slides LM2.3 to LM2.10** (and watch the lecture recordings if you want).
- Solve **examples 2.1 to 2.5 in Exercise Sheet 2**
- The discussion board thread for this module is labelled **analytical_dynamics**
- The estimated time needed to complete the module is **10 hours**, ideally **by 13 October**

LM3 — Learning Module 3: Dynamic stability

- The contents of this module are covered in **Section 2.4 (pages 36 to 42)**
- Use **slides LM3.3 to LM3.12** (and watch the lecture recordings if you want).
- Solve **examples 3.1 to 3.5 in Exercise Sheet 3**
- The discussion board thread for this module is labelled **dynamic_stability**
- The estimated time needed to complete the module is **10 hours**, ideally **by 20 October**

LM4 — Learning Module 4: Lagrange's method for continuous systems

- The contents of this module are covered in **Section 2.5 (pages 42 to 56)**
- Use **slides LM4.3 to LM4.28** (and watch the lecture recordings if you want).
- Solve **examples 4.1 to 4.4 in Exercise Sheet 4**
- The discussion board thread for this module is labelled **continuous_systems**
- The estimated time needed to complete the module is **18 hours**, ideally **by 30 October**

LM5 — Learning Module 5: Longitudinal waves

- The contents of this module are covered in **Sections 3.1 and 3.2 (pages 61 to 70)**
- Use **slides LM5.3 to LM5.14** (and watch the lecture recordings if you want).
- Solve **examples 5.1 to 5.6 in Exercise Sheet 5**
- The discussion board thread for this module is labelled **longitudinal_waves**
- The estimated time needed to complete the module is **14 hours**, ideally **by 11 November**

LM6 — Learning Module 6: Transverse vibration and flexural waves on beams

- The contents of this module are covered in **Section 3.3 (pages 70 to 80)**
- Use **slides LM6.3 to LM6.18** (and watch the lecture recordings if you want).
- Solve **examples 6.1 to 6.8 in Exercise Sheet 6**
- The discussion board thread for this module is labelled **transverse_waves**
- The estimated time needed to complete the module is **15 hours**, ideally **by 25 November**

LM7 — Learning Module 7: Impact on flexible structures

- The contents of this module are covered in **Sections 4.1 and 4.2 (pages 91 to 100)**
- Use **slides LM7.3 to LM7.32** (and watch the lecture recordings if you want).
- Solve **examples 7.1 to 7.8 in Exercise Sheet 7**
- The discussion board thread for this module is labelled **impact_on_structures**
- The estimated time needed to complete the module is **13 hours**, ideally **by 4 December**

1.8 Using these course notes

The main purpose of these course notes is to free students, if they wish, from copying material off the board or the projection and slides. These notes cover all the essential course content and should as such also serve as reference. All notes that the instructors may write on the *blackboard* will also be made available to you on LEARN. You should use the time and mental capacity that are thus freed during lectures in order to think about the material, and try to follow and understand the procedures as they are explained. Please be aware, however, that reading the notes does not replace attendance, since the present document cannot cover in detail the practicalities of problem analysis and solution, the problems encountered in practice by students, the useful tricks that may help in overcoming these problems and all the discussions that naturally arise during lectures, tutorial sessions and examples classes.

Course notes Having these course notes with you during lectures and tutorials may be helpful, since the worked examples can, to some extent, serve as *templates* for finding solutions to the tutorial problems. Reading through worked examples, however, can never substitute for the mental engagement needed to do the work by yourself. The nature of the problems posed is such that it is unlikely the notes will help you unless you have fully engaged with the lectures, tutorials and similar problems during the course.

This course might feel more advanced than those from previous academic years, and most of the material is addressed towards students with a genuine interest in dynamics problems. Thus, certain examples of considerable complexity are worked out in detail, using methods that are at times not trivial. Whenever possible, and to help visualise the application of the theoretical concepts, a section with real-life engineering applications is included at the end of each chapter.

To assist the learning process, these notes include a set of different types of coloured boxes designed to help identify and highlight relevant definitions, derivations, worked out and discussed examples, and numerical methods and code to assist in the solution methods. These boxes are colour-coded and examples are shown below.

Highlight box Highlight boxes are used in these course notes to highlight definitions, notes or any relevant approaches or methods of analysis. These boxes have a blue vertical line and light blue background. You will have seen these in the paragraphs and sections above.

Exercise 1.1 — Title of the exercise. This is the layout of an exercise box (dark orange vertical line and light grey background), which will include the question, any relevant data or sketches and the corresponding detailed example solutions and discussion. ■

Code box This is an example of a code snippet (*e.g.* PYTHON) box. These will be used in the course notes whenever it becomes useful to highlight auxiliary code to help solve or analyse a problem or derivation. Code boxes are light grey and have a black vertical line. Below is an example of a code snippet:

```
def greet(name):  return f"Hello, name!"
name = "Alice"
print(greet(name))
```

The final sections of these course notes include an appendix with a list of useful detailed mathematical derivations (Appendix A, page 101). A list of acronyms, variables and symbols used during the course (Notation, page 111) is also available, which may be useful during the

lectures, tutorials or when revising and preparing for the exam. Also at the end of this document is a glossary of definitions (Glossary, page 107), a word index that may help you locate a specific topic in the notes (Index, page 113) and a list of references for additional reading, including books, scientific papers and web-page links (Bibliography, page 117). This is not meant to be an extensive list of supporting references; each listed reference addresses in more or less detail one or more of the topics covered in these course notes. Although not strictly necessary, these final sections of the course notes will hopefully be useful and assist you in the learning process.

If at all possible, students are advised not to print these notes but instead to use and read them in a computer or portable device such as a tablet. One reason for this is because this PDF document has both external hyperlinks (i.e. to pages on the internet and e-mail addresses) and internal links such as figures and tables, equations, page numbers, chapter and section numbers, etc. Have you noticed the wee triangle next to all page numbers? Try clicking on it!

Thank you! This document is — and will surely continue to be — a *work in progress* and as such may (will certainly!) contain typos and other things worth correcting, changing and improving. Please do make the instructors aware of any such issues you may find and give suggestions for improvements. This will help us in making sure these course notes are in constant evolution and will undoubtedly help your colleagues in future editions of the course.

Thank you... and we hope you enjoy Dynamics 4!

2

Analytical dynamics

2.1 Introduction

In this course you will learn more about Engineering Structural Dynamics. The main aim of this engineering area is to analyse the dynamic behaviour of engineering systems such as machines, structures, etc. A three-step process, such as the one shown in Figure 2.1, can be used to achieve this. The first step is to develop a conceptual model of the system.

A good conceptual model must cover all relevant aspects of the real system, while discarding non-essential and irrelevant details. As such, the model development process starts with a series of decisions, namely deciding which features of the real system should be retained and which should be discarded. The ideal representation of the features that are retained is also very important. Unfortunately, the skill of developing good conceptual models is not taught at university, even though model development is one of the most important, fascinating, challenging and rewarding aspects of engineering.

Once the conceptual engineering model is done, the following step is the derivation of the corresponding mathematical model. This means to use the conceptual model to derive a set of equations — i.e. ordinary or partial differential equations in the scope of this course — that mathematically describe the dynamic behaviour of the system. The final steps are the mathematical or computational/numerical analysis of these equations and the interpretation of the results. The question to answer now is: what do these equations and solutions/results mean for the behaviour of the real-world system that we are trying to understand and model?

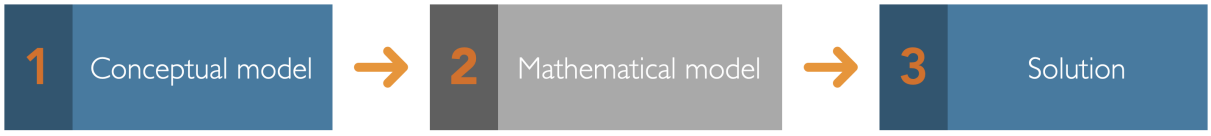


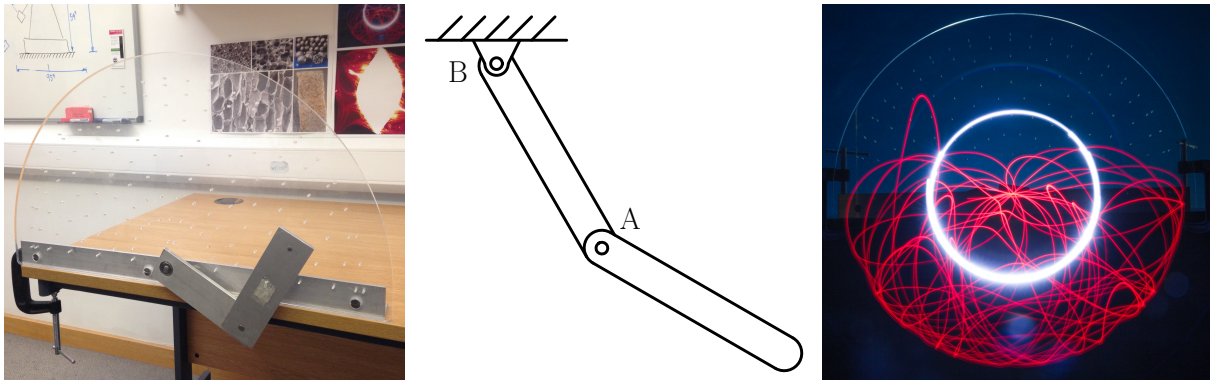
Figure 2.1: Schematic representation of the three-step process used to analyse the dynamical behaviour of complex engineering systems.

In this chapter we will discuss Lagrange's method, which deals exclusively with the second step, i.e. with the derivation of system equations for a given conceptual model of a real system. Lagrange's method can be envisaged as a *machine* for formulating correct equations of motion. It is based on energy and virtual work considerations, meaning that no free-body diagrams (FBD) and no force or moment analyses are required. As we will see later, this method is particularly suitable for systems with reaction forces caused by dynamic constraints.

The concept of dynamic constraints is illustrated in Figure 2.2. A double pendulum consists of two bars connected at point A and suspended from a pinned support at point B. This system exhibits two dynamic constraints since both bars must move in such a manner that their endpoints remain connected at A, and that the other endpoint of the upper bar remains attached to the support at B. Because of these constraints, reaction forces that depend in a complicated manner

on the motion of both bars will be acting at A and B.

A simple system such as the one shown in Figure 2.2 can have chaotic motion depending on the initial release conditions. This chaotic motion is shown by the red curved line in Figure 2.2(c), which tracks the mid-point of the lower arm of the double pendulum. Lagrange's method automatically accounts for all relevant reaction forces and offers a systematic approach for obtaining correct equations of motion for systems with complex constraints.



(a) (b) (c)

Figure 2.2: A double pendulum: (a) system with fixed positions for initial release, (b) schematic representation, and (c) motion tracking of upper (white light) and lower (red light) arms illustrating the chaotic motion of the system. A detailed analysis of this system will be presented later in Section 2.6.1.

Lagrange's method: Advantages The main practical advantages of Lagrange's method are:

- ✓ It deals with scalar quantities and as such requires less engineering intuition than needed in FBD approaches for getting the correct direction and balancing of forces, determining reaction forces caused by constraints, etc. There is no need to get the orientation of the forces right.
- ✓ It is the only straightforward method to derive equations of motion for systems with complex constraints.
- ✓ For complex problems the method is much more efficient than FBD: it is easy to get lost in a tangle of forces and moments. Stepping back and setting up the energy functions is often much more efficient and simple.

Lagrange's method: Disadvantages The main disadvantages of Lagrange's method are:

- ✓ Lagrange's method is not as intuitive as the free body diagram approach. There is no easy way to check whether the equations are correct. Relying on precise algebra is essential for getting things right.
- ✓ Very good differentiation skills are also essential.
- ✓ System velocities need to be correct in order to correctly set up the kinetic energy functions. You don't need a *feel* for forces and moments, but a good feel for the way things move is essential.
- ✓ In some simple problems and systems (such as those treated in previous courses) the method can be more complicated and take longer than FBD.

In the following section we will start by explaining how Lagrange's method can be applied to simple conservative mechanical systems.

2.2 Lagrange's method for conservative systems

In general terms, a system is said to be conservative if the total energy of the system does not change (or can be assumed not to change). Although in practice all systems are non-conservative, it is almost always possible to define a conservative system or sub-system provided the correct boundaries are chosen. On a mechanical system, this often means that all friction and damping forces acting on the system are non-existent or can be neglected. Additionally, there must be no external forces that do any work. It should be noted however, that reaction forces perpendicular to the motion are still possible as they don't do work. In these cases Lagrange's theorem can be used to derive an equation of motion or a system of equations of motion. This is described in more detail in Section 2.2.1 below.

2.2.1 Lagrange's theorem

Consider a system with n degrees-of-freedom (d.o.f.) that can be described by a set of n independent generalised coordinates q_i, with $i = 1, \ldots, n$. These are called generalised coordinates, or generalised displacements, because they do not necessarily have dimensions of length. An angle of rotation could, for example, also be a generalised coordinate. As an example, the most obvious coordinate describing the movement of a pendulum is the angle between the pendulum arm and the vertical. This is in fact the degree-of-freedom of the pendulum. The total kinetic energy of the n-d.o.f. system is the generic function

$$T = T(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_n) \tag{2.1}$$

where $\dot{q}_i$, with $i = 1, \ldots, n$ are the generalised velocities. Assuming as a simplification that the potential energy only depends on the generalised coordinates and not their velocities, the total potential energy can be described by the generic function

$$V = V(q_1, \ldots, q_n) \tag{2.2}$$

Consider also the following combination of the kinetic and potential energies:

$$\mathcal{L} = T - V = \mathcal{L}(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_n) \tag{2.3}$$

Function $\mathcal{L}$ is called the Lagrangian of the system. This is not the total energy of the system but the difference between the kinetic energy and potential energy. The Euler-Lagrange equation for the dynamic system is[1]

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) = \frac{\partial \mathcal{L}}{\partial q_i} \tag{2.4}$$

Note the use of partial derivatives because the kinetic energy T is a function of more than one generalised coordinate and velocity, and the potential energy V is a function of more than one generalised coordinate. Substituting the Lagrangian (equation 2.3) into the Euler-Lagrange equation 2.4 leads to

$$\begin{aligned} &\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial (T-V)}{\partial \dot{q}_i}\right] = \frac{\partial (T-V)}{\partial q_i} \\ \Rightarrow \quad &\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial V}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} + \frac{\partial V}{\partial q_i} = 0 \end{aligned} \tag{2.5}$$

[1] Although not required for this course, further information on how to derive the Euler-Lagrange equation can be found here: https://goo.gl/dmsDYf

Recalling the initial assumption that the potential energy does not depend on the velocities $\dot{q}_i$, the set of n differential equations for a free vibration conservative system — i.e. with no external forces or dissipative actions, such as friction or damping forces — is given by

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} + \frac{\partial V}{\partial q_i} = 0$$

More information on total and partial derivatives can be found in Appendix A.1.

Lagrange's theorem for conservative systems If a free vibration conservative system can be described by a set of n independent generalised coordinates q_i, with $i = 1, \ldots, n$, which can be displacements or angular rotations, and

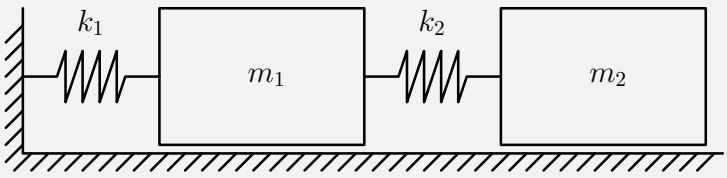
$$T(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_n) \qquad \text{and} \qquad V(q_1, \ldots, q_n)$$

are the total kinetic and total potential energy functions of the system, respectively, the corresponding equations of motion can be determined with the following set of differential equations:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} + \frac{\partial V}{\partial q_i} = 0 \qquad \text{with} \qquad i = 1, \ldots, n \tag{2.6}$$

As will be demonstrated in the following sections of these notes, this method can be applied to almost any dynamical system. An additional advantage of this method is that the complexity of its application does not scale up with the complexity of the system. The only two things that may get more complicated are the energy functions T and V, and the process of differentiating these functions. We now illustrate the application of this method with a very simple 2 degree-of-freedom dynamic system.

Exercise 2.1 Consider the 2 degree-of-freedom system shown in the figure below and formulate the corresponding equations of motion using Lagrange's theorem. Neglect friction between the masses and the horizontal surface.



Solution The first step is always to choose the appropriate generalised coordinates (degrees-of-freedom) that describe the movement of the system. In this case this is straightforward: x_1 and x_2 are the horizontal displacements of the two masses from their equilibrium positions. Hence, $q_1 = x_1$ and $q_2 = x_2$. The equations of motion as obtained from a free body diagram (FBD) are

$$m_1\ddot{x}_1 + k_1 x_1 + k_2(x_1 - x_2) = 0$$
$$m_2\ddot{x}_2 + k_2(x_2 - x_1) = 0$$

Using Lagrange's method described above, it is first necessary to determine the two energy functions of the system. The kinetic energy T is the sum of the kinetic energies of the two

masses:

$$T = \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2$$

The potential energy V is the sum of the potential (elastic) energies of the two springs, noting that the elongation of the second spring is given by the relative displacement of the two masses,

$$V = \frac{1}{2}k_1x_1^2 + \frac{1}{2}k_2(x_1 - x_2)^2$$

Applying Lagrange's method described above (see equation 2.6) it should first be noted that

$$\frac{\partial T}{\partial x_1} = \frac{\partial T}{\partial x_2} = 0$$

The remaining terms in the Lagrange equation are

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{x}_1}\right) &= m_1\ddot{x}_1 \\
\frac{\partial V}{\partial x_1} &= k_1x_1 + k_2(x_1 - x_2) \\
\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{x}_2}\right) &= m_2\ddot{x}_2 \\
\frac{\partial V}{\partial x_2} &= -k_2(x_1 - x_2)
\end{aligned}$$

Combining the individual terms according to Lagrange's equations leads to

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{x}_1}\right) + \frac{\partial V}{\partial x_1} &= 0 \\
\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{x}_2}\right) + \frac{\partial V}{\partial x_2} &= 0
\end{aligned}$$

Introducing the derivatives produces the correct equations of motion, which are

$$\begin{aligned}
m_1\ddot{x}_1 + k_1x_1 + k_2(x_1 - x_2) &= 0 \\
m_2\ddot{x}_2 - k_2(x_1 - x_2) &= 0
\end{aligned}$$

As an exercise, you should try to determine these equations using the free body diagram method. As expected, these equations of motion are the same that can be obtained using Lagrange's approach. ■

2.2.2 Setting up the Lagrange formalism

The problem of setting up Lagrange's equations described in the previous section, can be reduced to writing down correct expressions for the kinetic and potential energies as functions of the system coordinates. The first step to achieve this is to choose the best set of coordinates. Here an intelligent and adequate choice often helps. In particular, the number of variables used to describe the motion of the system must never exceed the number of degrees-of-freedom of the system. The following paragraphs describe how to obtain these energy terms in detail.

Kinetic energy

The kinetic energy is a non-negative function of the generalised velocities — the time-derivatives of the system coordinates. The kinetic energy of a system is the sum of the kinetic energies of all

its individual components. The most simple example is the motion of a point particle of mass m (a lumped mass) moving with velocity v, with kinetic energy

$$T = \frac{1}{2}mv^2 \tag{2.7}$$

It should be noted that in general the velocity is a vector. However, v in the above equation is the magnitude of the velocity vector. As such and when in doubt, write the position of the particle as a Cartesian vector $\mathbf{x} = (x_1, x_2, x_3)$ and differentiate the coordinates, yielding

$$v^2 = \dot{x}_1^2 + \dot{x}_2^2 + \dot{x}_3^2 \qquad \text{or} \qquad v = \sqrt{\dot{x}_1^2 + \dot{x}_2^2 + \dot{x}_3^2} \tag{2.8}$$

Generalising for the motion of a rigid body of mass M, the kinetic energy for pure linear motion (translation) is

$$T = \frac{1}{2}Mv^2 \tag{2.9}$$

and for fixed axis rotation about an axis through a point A is

$$T = \frac{1}{2}I_{\mathrm{A}}\dot{\varphi}^2 \tag{2.10}$$

where φ is the angle of rotation. The kinetic energy of general plane motion — simultaneous linear and angular motion (translation and rotation) — can be derived simply by combining the two previous expressions, that is,

$$T = \frac{1}{2}Mv_{\mathrm{G}}^2 + \frac{1}{2}I_{\mathrm{G}}\dot{\varphi}^2 \tag{2.11}$$

where v_{G} is the magnitude of the velocity of the centre of gravity (CoG) of the rigid body and I_{G} is the moment of inertia about an axis going through the centre of gravity.

Potential energy

By definition, potential energy is the energy of a body by virtue of its position relative to others, stresses within itself, electric charge, field effects or other factors. Potential energy is associated with forces that act on the body in a way that depends only on the body's position in space. These forces can be represented by a vector at every point in space forming a vector field of forces, or a force field. The derivation of potential energy functions for general continuous systems will be discussed later in these notes. For simpler lumped mass systems, useful cases include the elastic energy of a linear spring

$$V_{\mathrm{e}} = \frac{1}{2}k\Delta x^2 \tag{2.12}$$

where k is the elastic stiffness of the spring and Δx is the extension of the spring, which should be expressed in terms of the generalised coordinates of the system. Also useful is the elastic energy of a torsion spring

$$V_{\mathrm{e}} = \frac{1}{2}k_{\mathrm{T}}\Delta\alpha^2 \tag{2.13}$$

where k_{T} is the torsional stiffness and $\Delta\alpha$ is the twist angle (in radians). For a cylindrical bar with uniform cross section, for example, the torsional stiffness is

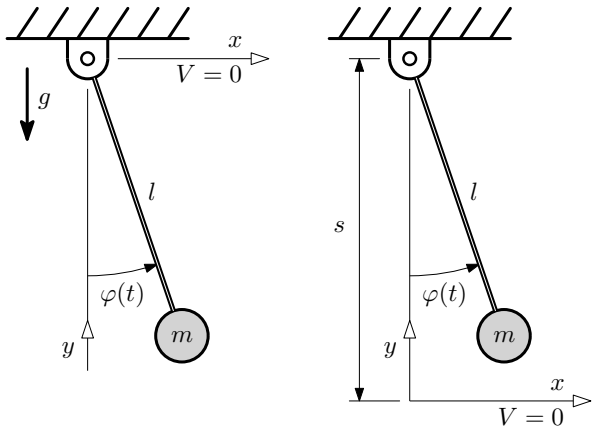
$$k_{\mathrm{T}} = \frac{G\pi d^4}{32L} \tag{2.14}$$

where L and d are the length and the diameter of the bar, respectively, and G is the shear modulus of the material of the bar.

The gravitational potential energy of a point mass or a rigid body is

$$V_{\mathrm{g}} = mgh \tag{2.15}$$

where m is the mass of the body or point mass and h is the height (altitude) above an arbitrary fixed reference. Care should be taken when finding height h. This height is absolute, meaning that it should be negative if the reference altitude is above the point mass or above the centre of gravity of the body, and positive otherwise. Figure 2.3 shows the gravitational potential energy of a lumped mass pendulum for the two cases when the mass is below and above the chosen reference altitude, assuming Oy always positive upwards. Note that changing the reference altitude affects the equation of motion because it changes the point of view from which the system is analysed. The equations of motion describe the behaviour of an object in terms of forces, accelerations and motion relative to a reference point. When this reference point is changed, we are essentially changing the frame of reference from which observations and measurements are made.



(a) $V_{\mathrm{g}} = -mgl\cos\varphi$ (b) $V_{\mathrm{g}} = mg(s - l\cos\varphi)$

Figure 2.3: Gravitational potential energy for a lumped mass pendulum with: (a) mass below the chosen reference altitude and (b) mass above the chosen reference altitude.

Gravitational potential energy Finding the gravitational potential energy terms of a mechanical system is not always trivial and frequently causes confusion, mostly because V_{g} depends on three distinct things, namely:

- ✓ The chosen orientation for the vertical (Oy) axis;
- ✓ The location of the arbitrary reference altitude ($V = 0$) and
- ✓ Whether the mass is moving upwards or downwards within the gravitational field.

The following steps can help ensure the gravitational potential energy of a body or point mass is correctly set up:

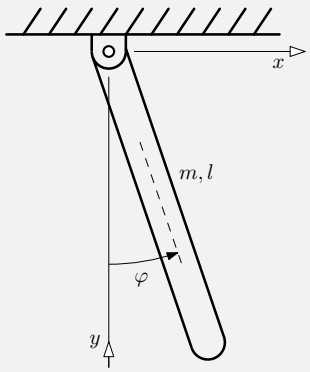
1. Whenever possible, assume Oy positive upwards.
2. If at all possible, put the horizontal reference altitude as low as possible. Ideally, all masses and their trajectories should be above this reference.

Exercise 2.2 below gives an example where Oy is positive upwards and the reference altitude for the potential energy was chosen to be above the centre of gravity of the body. The potential energy of the mass is, consequently, negative.

Once all contributions to the kinetic and potential energy are determined, it is essential,

before using the formalism of Lagrange, to derive the equations of motion, to express all variables entering these energies in terms of the generalised coordinates of the system.

Exercise 2.2 Consider the single degree-of-freedom system shown in the figure below and formulate the corresponding equation of motion using Lagrange's method. The pendulum has mass m and length l. Neglect friction at the pivot point.



Solution This is another simple example that illustrates what to do when we have rotational motion. The key problem here is not to *forget* any terms in the kinetic energy function T. Again, it is easy to see that the system is described by a single coordinate: the rotation angle φ. It is then necessary to express the kinetic and potential energies in terms of this generalised coordinate. To do this start by defining the x and y coordinates of the CoG of the bar, which are given by

$$x = \frac{1}{2} l \sin\varphi \quad \text{and} \quad y = -\frac{1}{2} l \cos\varphi$$

The vertical coordinate y is negative because the vertical Oy axis was chosen to be pointing upwards. The moment of inertia of the bar about its end point — the pivot point — is

$$I_{\mathrm{E}} = \frac{1}{3} m l^2$$

The kinetic and potential energies are then, respectively,

$$T = \frac{1}{2} I_{\mathrm{E}} \dot{\varphi}^2 = \frac{1}{6} m l^2 \dot{\varphi}^2$$

and

$$V = mgy = -\frac{1}{2} mgl \cos\varphi$$

Alternatively we could have separated the kinetic energy into its linear and angular motion components, that is, into the sum of the energy of translation of the bar's centre of gravity and the energy of rotation of the bar around its centre of gravity, with the same result,

$$\begin{aligned} T &= \frac{1}{2} m \left(\dot{x}^2 + \dot{y}^2\right) + \frac{1}{2} I_{\mathrm{G}} \dot{\varphi}^2 \\ &= \frac{1}{8} m l^2 \left(\sin^2\varphi + \cos^2\varphi\right) \dot{\varphi}^2 + \frac{1}{2 \times 12} m l^2 \dot{\varphi}^2 \\ &= \frac{1}{6} m l^2 \dot{\varphi}^2 \end{aligned}$$

This is here a more complicated approach. However, in the next example we will see that this is the only method which works when the bar endpoint is not fixed. Finally, we apply Lagrange's equations and derive the equation of motion of the system. The differential terms are

$$\frac{\partial T}{\partial \dot{\varphi}} = \frac{1}{3}ml^2\dot{\varphi}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{\varphi}}\right) = \frac{1}{3}ml^2\ddot{\varphi}, \qquad \frac{\partial T}{\partial \varphi} = 0, \qquad \frac{\partial V}{\partial \varphi} = \frac{1}{2}mgl\sin\varphi$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{\varphi}}\right) - \frac{\partial T}{\partial \varphi} + \frac{\partial V}{\partial \varphi} = 0$$

and, consequently,

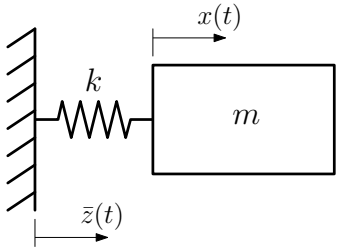
$$\frac{1}{3}ml^2\ddot{\varphi} + \frac{1}{2}mgl\sin\varphi = 0$$

is the equation of motion of the single degree-of-freedom pendulum system. ■

2.2.3 Systems with prescribed boundary motions

We are now going to look at systems where the boundary (or boundaries) are not fixed. Moving boundary excitation may have been discussed in previous courses and happens when the support or boundary of a system moves in a prescribed manner. Examples of such systems include the vibration transmission problem, ground shaking due to an earthquake, etc. When applying the Lagrange method to such problems it is important to understand that the coordinate describing the moving boundary, that we will here designate by $\bar{z}(t)$, is neither a force nor a generalised coordinate of the system. It is actually a dynamic constraint. The bar over the variable is meant to indicate this.

To account for this constraint, $\bar{z}(t)$ is simply included into the energy functions where it mixes with the system's coordinates. Its time dependence must nonetheless be taken into account when doing derivatives to obtain the Lagrange equations. An example of such a system is shown in Figure 2.4(a) where the potential energy is defined by the relative deformation of the spring of stiffness k and the kinetic energy by the relative velocity of mass m. A system such as this one could be used as a *very* simplified conceptual model to describe the movement of a building on an earthquake, as shown in Figure 2.4(b).



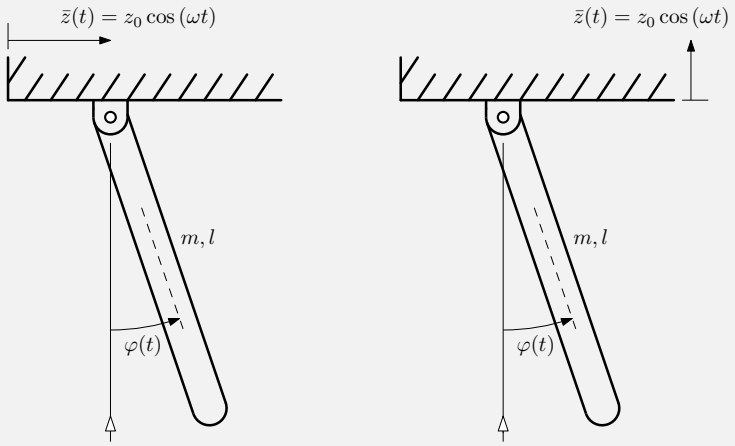
(a)

(b)

Figure 2.4: Systems with prescribed motion: (a) a generic single degree-of-freedom dynamic system with a prescribed boundary motion $\bar{z}(t)$ and (b) shake tables used to analyse the response of buildings during an earthquake (image reproduced under Creative Commons license CC BY-SA 3.0, author: Shustov).

In the following example we apply Lagrange's method to the single-arm pendulum-system of Exercise 2.2, now with added vertical and horizontal boundary displacements $\bar{z}(t) = z_0\cos(\omega t)$.

Exercise 2.3 Use Lagrange's formalism to obtain the equations of motion for the two distributed mass pendulum systems shown in the figure below: (a) horizontal motion of the pendulum support, (b) vertical motion of the pendulum support.



Solution These two cases are very similar to the previous example, Exercise 2.2 in page 22, the only difference being the fact that now the boundary is not fixed. Again, we chose the angle of the pendulum relative to the vertical axis to be the generalised coordinate $\varphi(t)$. We first consider case (a), where the boundary has a prescribed horizontal motion $\bar{z}(t) = z_0 \cos(\omega t)$. The coordinates of the CoG of the bar are (note the additional $\bar{z}(t)$ term)

$$x = \bar{z}(t) + \frac{1}{2} l \sin\varphi$$
$$y = -\frac{1}{2} l \cos\varphi$$

The kinetic energy becomes

$$T = \frac{1}{2} m \left(\dot{x}^2 + \dot{y}^2\right) + \frac{1}{2} I_G \dot{\varphi}^2 = \frac{1}{2} m \left[\dot{\bar{z}}^2 + l\dot{\bar{z}}\dot{\varphi}\cos\varphi + \frac{1}{4} l^2 \left(\cos^2\varphi + \sin^2\varphi\right)\dot{\varphi}^2\right] + \frac{1}{2}\frac{ml^2}{12}\dot{\varphi}^2$$
$$= \frac{1}{2} m \left(\dot{\bar{z}}^2 + l\dot{\bar{z}}\dot{\varphi}\cos\varphi\right) + \frac{1}{6} m l^2 \dot{\varphi}^2$$

Note that if only the rotation around the endpoint was considered, the first two terms wouldn't be part of this equation. For this case, as adding a horizontal boundary motion does not change the gravitational potential energy of the system, the potential energy is the same as before

$$V = mgy = -\frac{1}{2} mgl \cos\varphi$$

The Lagrange calculation is now more interesting and challenging. The differential terms become

$$\frac{\partial T}{\partial \dot{\varphi}} = \frac{1}{2} m l \dot{\bar{z}} \cos\varphi + \frac{1}{3} m l^2 \dot{\varphi}$$
$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{\varphi}}\right) = \frac{1}{2} m l \left(\ddot{\bar{z}}\cos\varphi - \dot{\bar{z}}\dot{\varphi}\sin\varphi\right) + \frac{1}{3} m l^2 \ddot{\varphi}$$

$$\frac{\partial T}{\partial \varphi} = -\frac{1}{2}ml\dot{\bar{z}}\dot{\varphi}\sin\varphi, \qquad \frac{\partial V}{\partial \varphi} = \frac{1}{2}mgl\sin\varphi$$

and the equation of motion is then

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{\varphi}}\right) - \frac{\partial T}{\partial \varphi} + \frac{\partial V}{\partial \varphi} = 0 \qquad \Rightarrow \qquad \frac{1}{3}ml^2\ddot{\varphi} + \frac{1}{2}ml\ddot{\bar{z}}\cos\varphi + \frac{1}{2}mgl\sin\varphi = 0$$

The simple fact that the system has a boundary motion creates a *strange* non-linear term in the equation of motion. This simple system can thus lead to extremely complex behaviour.

Case (b) works in a similar manner. The coordinates of the CoG of the bar are in this case

$$x = \frac{1}{2}l\sin\varphi$$
$$y = \bar{z}(t) - \frac{1}{2}l\cos\varphi$$

The kinetic energy function becomes

$$T = \frac{1}{2}m\left(\dot{\bar{z}}^2 + l\dot{\bar{z}}\dot{\varphi}\sin\varphi\right) + \frac{1}{6}ml^2\dot{\varphi}^2$$

As the boundary is moving vertically, the potential energy function now also has a $\bar{z}$-dependent term,

$$V = mgy = -\frac{1}{2}mgl\cos\varphi + mg\bar{z}(t)$$

The Lagrange differential terms become

$$\frac{\partial T}{\partial \dot{\varphi}} = \frac{1}{2}ml\dot{\bar{z}}\sin\varphi + \frac{1}{3}ml^2\dot{\varphi}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{\varphi}}\right) = \frac{1}{2}ml\left(\ddot{\bar{z}}\sin\varphi + \dot{\bar{z}}\dot{\varphi}\cos\varphi\right) + \frac{1}{3}ml^2\ddot{\varphi}$$
$$\frac{\partial T}{\partial \varphi} = \frac{1}{2}ml\dot{\bar{z}}\dot{\varphi}\cos\varphi, \qquad \frac{\partial V}{\partial \varphi} = \frac{1}{2}mgl\sin\varphi$$

Combining the terms above, the equation of motion is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{\varphi}}\right) - \frac{\partial T}{\partial \varphi} + \frac{\partial V}{\partial \varphi} = 0 \qquad \Rightarrow \qquad \frac{1}{3}ml^2\ddot{\varphi} + \frac{1}{2}ml\ddot{\bar{z}}\sin\varphi + \frac{1}{2}mgl\sin\varphi = 0$$

■

When we calculate the equation of motion from the pivot point in the simple pendulum with a moving boundary in Exercise 2.3, we measure angles and distances relative to that fixed point. In the reference frame, we accounted for the relative motion of the pendulum with respect to the pivot point.

If the reference point changes to the centre of mass or any other location, we must measure angles and distances relative to that new point/reference. The centre of mass or any other reference point will move differently than the pivot point, and this relative motion needs to be considered in the equations of motion. In essence, it is like changing our perspective, noting that different perspectives can yield different equations. The physical behaviour of the system remains the same, but how it is mathematically described changes based on the chosen reference point. The equations will be different because the coordinates, angles and distances are measured differently in each reference frame.

2.2.4 Non-linear behaviour of dynamic systems

System equations derived through the Lagrange method (or otherwise) often contain non-linear terms, as we have seen in previous sections.

Non-linear terms are not proportional to a system's generalised coordinate and not proportional to any of its derivatives. Such terms contain, however, generalised coordinates and/or their derivatives to higher powers or products, or are inside non-linear mathematical functions such as trigonometric functions (e.g. sin or cos). If any such terms are present in the equation of motion, the equations are said to be non-linear.

Before we move on to the analysis of the behaviour of non-linear dynamic systems, we should look at the advantages and disadvantages of having to deal with non-linear equations of motion and decide when and in what circumstances these can — or, more accurately, should — be simplified.

Linear systems The following are characteristics of linear dynamic systems:

- ✓ Mechanical systems with linear behaviour are often easy to analyse.
- ✓ The equations of motion have explicit solutions.
- ✓ The principle of superposition is always valid.
- ✓ Solutions are proportional to *stimuli*.
- ✓ The physical and mechanical responses of the system can be completely known.

Non-linear systems The following are characteristics of non-linear dynamic systems:

- ✓ The solutions to the equations of motion of mechanical systems with non-linear behaviour are often impossible to find. In most cases a numerical solution is necessary.
- ✓ The principle of superposition may not be valid.
- ✓ Solutions are not proportional to *stimuli*.
- ✓ The above limitations lead to the physical behaviour and mechanical response of the non-linear system to often be complex and impossible to fully determine.

Since linear systems are significantly easier to analyse and often give an acceptable description of the response of many engineering systems, it is often possible to use a procedure called linearisation to go from non-linear to linear equations of motion. As we will see later, by studying the stability of solutions of the resulting linearised equations it is sometimes possible to decide whether or not linearisation is acceptable.

Linearisation

When linearising a set of equations of motion, we are in fact approximating a non-linear system to a linear one. This is possible and reasonable only if the deviation of the system coordinates from some reference point is small, and remains small. The first task in this process of linearisation is to choose an appropriate reference coordinate (point). This will usually be a static equilibrium configuration of the system. We then take this point as the origin of the system's generalised coordinates. As the deviation from the reference point is assumed to be small, all non-linear terms involving powers or products of the system coordinates and/or their derivatives can (and should) be neglected. Additionally, all non-linear functions of the generalised coordinates should be replaced by their expansions up to linear terms. All constant terms are also retained. More information on linearisation can be found in Appendix A.3.

Linearisation Non-linear functions are replaced by their expansions up to linear terms and all constant terms are retained. For example, the following generic non-linear mathematical functions of x are linearised, around $x = 0$, as indicated:

$$\cos(x) \approx \cosh(x) \approx 1$$
$$\cos(x + \pi) \approx -1$$
$$\cos\left(x + \frac{\pi}{2}\right) \approx -x$$
$$\cos\left(x - \frac{\pi}{2}\right) \approx x$$
$$\sin(x) \approx \sinh(x) \approx x$$
$$\sin(x + \pi) \approx \sin(x - \pi) \approx -x$$

As a remark it should be noted that the process of linearisation of the equations of motion of a dynamic system should always be followed with care. Not all dynamic systems can be linearised. If a system is prone to large displacements — the definition of *large* being somewhat subjective — its equations of motion should not be linearised, otherwise these equations of motion will describe a system significantly different from the real one.

If the equations of motion include combinations of non-linear functions, the above process should be applied step-by-step. For instance, to linearise the mathematical function $(x + y + 1)^2 \cos(x)$ assuming that x and y are two system coordinates and the linearisation is around $(x, y) = (0, 0)$, the first step should be to linearise the $\cos(x)$ function and simultaneously expand the polynomial function, that is,

$$(x + y + 1)^2 \cos(x) \approx (x + y + 1)^2 \times 1 = \left[(x + y)^2 + 2(x + y) + 1\right] \times 1$$

The final step is to linearise the polynomial terms, leading to

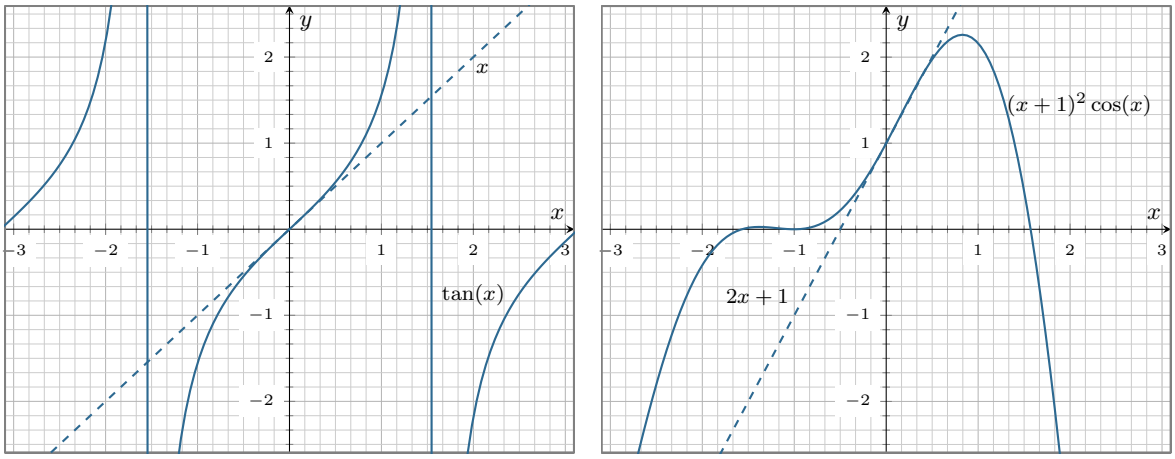
$$(x + y + 1)^2 \cos(x) \approx 2x + 2y + 1$$

Figure 2.5 gives a graphical representation of how the linearisation process works with two more examples. $\tan(x)$ and its linearised version x are plotted in Figure 2.5(a), and function $(x + 1)^2 \cos(x)$ and its linearised version $2x + 1$ are plotted in Figure 2.5(b). The limits within which these linearisations are reasonable can be roughly estimated from these images to be $[-0.8, 0.8]$ for the first function and $[-0.4, 0.6]$ for the second one. The PYTHON code in the following code box illustrates the linearisation of $(x + 1)^2 \cos(x) \approx 2x + 1$ in a plot. You can use similar code to visualise the linearisation of other mathematical functions.

Linearisation PYTHON code snippet to linearise $(x + 1)^2 \cos(x) \approx 2x + 1$ in a plot:

```
x = np.arange(-3, 3.1, 0.1)
f = (x + 1)**2 * np.cos(x)
g = 2 * x + 1
plt.plot(x, f, label='f(x) = (x+1)^2 * cos(x)')
plt.plot(x, g, label='g(x) = 2x + 1')
plt.show()
```

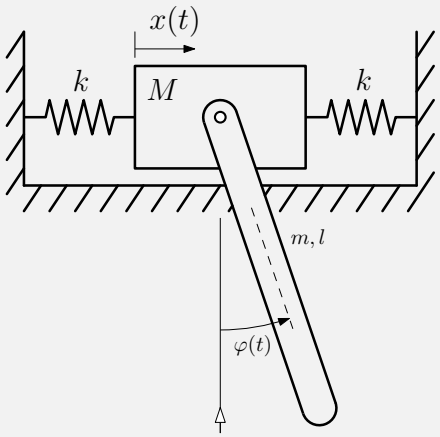
The following example illustrates the use of Lagrange's method to determine the equations of motion of a non-linear dynamic system. This example uses the linearisation methods described above to obtain the approximated linear response of the system.



(a) $\tan(x)$ (solid line) $\approx x$ (dashed line) (b) $(x+1)^2\cos(x)$ (solid line) $\approx 2x+1$ (dashed line)

Figure 2.5: Graphical examples of the linearisation of mathematical functions around $x = 0$.

Exercise 2.4 Use Lagrange's method to set up the equations of motion of the system shown below, identify any non-linear terms and linearise the equations of motion.



Solution Here it is crucial to first figure out *what is going on*, meaning that we should first start by understanding the movement of all the components of the system. The system has two degrees of freedom since mass M can move horizontally and the pendulum bar can rotate around its point of suspension.

We use the two generalised coordinates, $x(t)$ for the position of the mass and $\varphi(t)$ for the angle of rotation of the bar. The two degrees-of-freedom are, however, coupled since the motion of the mass also moves the pinned support of the bar. Hence, we should also expect to find x-related terms in the kinetic energy function of the bar.

The kinetic energy of mass M is straightforward but that of the bar is more complicated. We first write down the coordinates of the CoG of the bar. Note that these should not be called x and y because x has already been used to denote the degree-of-freedom of mass M. As such, the coordinates of the CoG of the bar are

$$x_\mathrm{b} = x + \frac{1}{2} l \sin\varphi \qquad \text{and} \qquad y_\mathrm{b} = -\frac{1}{2} l \cos\varphi$$

Taking into account both the rotation of the bar around its CoG and the linear motion of the CoG, the total kinetic energy of the system becomes

$$\begin{aligned} T &= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\left(\dot{x}_\mathrm{b}^2 + \dot{y}_\mathrm{b}^2\right) + \frac{1}{2}I_\mathrm{G}\dot{\varphi}^2 \\ &= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\left[\dot{x}^2 + l\dot{x}\dot{\varphi}\cos\varphi + \frac{1}{4}l^2\left(\cos^2\varphi + \sin^2\varphi\right)\dot{\varphi}^2\right] + \frac{1}{2}\times\frac{ml^2}{12}\dot{\varphi}^2 \\ &= \frac{1}{2}(M+m)\dot{x}^2 + \frac{1}{2}ml\dot{x}\dot{\varphi}\cos\varphi + \frac{1}{6}ml^2\dot{\varphi}^2 \end{aligned}$$

The potential energy has elastic contributions from the two springs and from the gravitational energy of the bar. These terms added together give

$$V = 2\left(\frac{1}{2}kx^2\right) - \frac{1}{2}mgl\cos\varphi$$

We now apply Lagrange's method to get the first equation of motion, for the x degree-of-freedom,

$$\begin{aligned} &\frac{\partial T}{\partial \dot{x}} = (M+m)\dot{x} + \frac{1}{2}ml\dot{\varphi}\cos\varphi, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{x}}\right) = (M+m)\ddot{x} + \frac{1}{2}ml\left(\ddot{\varphi}\cos\varphi - \dot{\varphi}^2\sin\varphi\right) \\ &\frac{\partial T}{\partial x} = 0, \qquad \frac{\partial V}{\partial x} = 2kx \end{aligned}$$

The resulting equation of motion for $x(t)$ is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{x}}\right) - \frac{\partial T}{\partial x} + \frac{\partial V}{\partial x} = 0 \quad \Rightarrow \quad (M+m)\ddot{x} + \frac{1}{2}ml\left(\ddot{\varphi}\cos\varphi - \dot{\varphi}^2\sin\varphi\right) + 2kx = 0$$

Doing the same for φ yields

$$\begin{aligned} &\frac{\partial T}{\partial \dot{\varphi}} = \frac{1}{2}ml\dot{x}\cos\varphi + \frac{1}{3}ml^2\dot{\varphi}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{\varphi}}\right) = \frac{1}{2}ml\left(\ddot{x}\cos\varphi - \dot{x}\dot{\varphi}\sin\varphi\right) + \frac{1}{3}ml^2\ddot{\varphi} \\ &\frac{\partial T}{\partial \varphi} = -\frac{1}{2}ml\dot{x}\dot{\varphi}\sin\varphi, \qquad \frac{\partial V}{\partial \varphi} = \frac{1}{2}mgl\sin\varphi \end{aligned}$$

The equation of motion for $\varphi(t)$ is then

$$\begin{aligned} &\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{\varphi}}\right) - \frac{\partial T}{\partial \varphi} + \frac{\partial V}{\partial \varphi} = 0 \\ &\Rightarrow \quad \frac{1}{3}ml^2\ddot{\varphi} + \frac{1}{2}ml\ddot{x}\cos\varphi + \frac{1}{2}mgl\sin\varphi = 0 \end{aligned}$$

The non-linear terms in the above equations are all the terms that involve either sin or cos functions. There are no other non-linear terms in this example. To linearise the equations of motion we first have to specify what is the reference state.

Reference states can be any static equilibrium positions of the system. In this example this is $x = 0$ and either $\varphi = 0$ or $\varphi = \pi$. We first look at the reference state with angle zero, that is, with the bar hanging downwards, since this is a stable state. We insert $\cos\varphi \approx 1$, $\sin\varphi \approx \varphi$ to get

$$\begin{aligned} &(M+m)\ddot{x} + \frac{1}{2}ml\left(\ddot{\varphi} - \dot{\varphi}^2\varphi\right) + 2kx = 0 \\ &\frac{1}{3}ml^2\ddot{\varphi} + \frac{1}{2}ml\ddot{x} + \frac{1}{2}mgl\varphi = 0 \end{aligned}$$

The equation of motion for x still contains one non-linear term. However, for small values of x and φ these can be neglected to obtain

$$(M+m)\ddot{x} + \frac{1}{2}ml\ddot{\varphi} + 2kx = 0$$
$$\frac{1}{3}ml^2\ddot{\varphi} + \frac{1}{2}ml\ddot{x} + \frac{1}{2}mgl\varphi = 0$$

This is the linearised system of equations of motion. It is a 2-degree-of-freedom system which can be solved using matrix methods, as discussed in previous courses. It should however be noted that here the mass matrix looks rather unusual since it has non-diagonal terms.

This exercise can be extended by repeating the linearisation process by using the unstable equilibrium position $\varphi = \pi$ as reference state. This can be done by making $\varphi = \pi + \phi$ and linearising for the new variable ϕ. ■

2.3 Generalised forces and non-conservative systems

Friction or damping forces, and external forces that do work, change the energy of a system. Such forces lead to generalised forces in the Lagrange equations of motion. Similarly to what was described above relative to generalised coordinates, here we use the term generalised forces because the force terms may not always be forces, they can also be moments, which are the generalised forces associated to angular coordinates.

2.3.1 Generalised form of Lagrange's equations

The equations of motion of a dynamic system described by n independent generalised coordinates q_i, with $i = 1, \ldots, n$, can be determined using the following form of the Lagrange equations:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} + \frac{\partial V}{\partial q_i} = Q_i$$

where Q_i is the generalised force associated to the system coordinate q_i.

Lagrange's theorem for non-conservative systems If a free vibration non-conservative system can be described by a set of n independent generalised coordinates q_i, with $i = 1, \ldots, n$, which can be linear displacements or angular rotations, and

$$T(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_n) \qquad \text{and} \qquad V(q_1, \ldots, q_n)$$

are the total kinetic and total potential energy functions of the system, respectively, the corresponding equations of motion can be determined with the following set of differential equations:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} + \frac{\partial V}{\partial q_i} = Q_i \qquad \text{with} \qquad i = 1, \ldots, n \tag{2.16}$$

where Q_i are the generalised forces acting on the system.

The definition and determination of generalised forces is conceptually a complex and often not trivial task. It is thus worthwhile spending some time trying to grasp this concept of generalised force. We start with a more formal definition.

Generalised forces Generalised forces find use in Lagrangian mechanics, where they play a role conjugate to the generalised coordinates. They are obtained from the applied forces or moments acting on a system that has its configuration defined in terms of generalised coordinates.

The generalised force Q_i acting on a system's generalised coordinate q_i can be evaluated from the virtual work δW_i defined as

$$\delta W_i = Q_i \delta q_i \tag{2.17}$$

done by the system's non-conservative forces during a virtual displacement δq_i of the i-th coordinate. The generalised force can then be determined as the virtual work divided by the virtual displacement, that is,

$$Q_i = \frac{\delta W_i}{\delta q_i} \tag{2.18}$$

Thus, the evaluation of generalised forces reduces to determining the relation between virtual work and virtual displacement. These *virtual* quantities will be described in more detailed in the following sections.

2.3.2 Virtual displacements and virtual work

By definition, a virtual displacement is an imagined, possible, instantaneous, small displacement or set of displacements, imposed on a system in a given state of motion or rest. Below are a few characteristics of virtual displacements:

- A virtual displacement is a mental exercise — we assume the system coordinates and velocities as given and virtually move one coordinate in one direction, assuming that this does not change the velocities or the remaining coordinates.
- These virtual displacements must be physically possible, that is, they must not violate any constraints imposed on the system.
- We usually take one coordinate and virtually displace it while keeping all others fixed. A virtual displacement of the i-th system coordinate is denoted as δq_i.

If such a virtual displacement is actually forced upon the system, the non-conservative forces will do work on the system. Thus, the virtual work is the work that would be done by the actual system's forces if a virtual displacement would actually be imposed on the system. The conceptual difficulty here is that this displacement does not actually occur — it does not matter whether or not the system actually moves in such a manner. The practical difficulty is that it is sometimes surprisingly difficult to work out the work that would occur if one coordinate is changed by a small amount while all others are not. The practicalities of evaluating virtual work and virtual forces will be explained later with examples 2.5 through 2.7.

2.3.3 Damping forces

For conceptual reasons and simplification we often consider discrete viscous dampers that create velocity-proportional forces as the main dissipation terms in a dynamic system. These can represent almost any energy being lost from the system, such as, actual viscous dampers, heat lost by friction, etc. Citing the *Encyclopædia Britannica*[2], damping, by definition is

> "[...] *a restraint of vibratory motion, such as mechanical oscillations and noise by dissipation of energy. There are many types of mechanical damping. Friction, which*

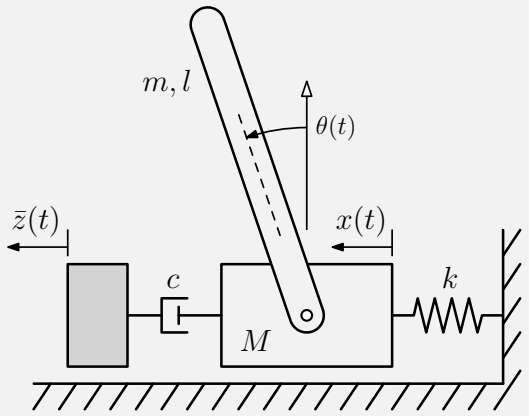
[2]Although not required, you can revisit the Dynamics 3 course notes for more details on this.

may also be called dry or Coulomb damping, arises from the electrostatic forces of attraction between the sliding surfaces and converts mechanical energy of motion, or kinetic energy, into heat. [...] *Viscous damping is caused by such energy losses as occur in liquid lubrication between moving parts or in a fluid forced through a small opening by a piston, as in automobile shock absorbers. The viscous damping force is directly proportional to the relative velocity between the two ends of the damping device.*

There are other types of damping. Resonant electric circuits, in which an alternating current is surging back and forth, as in a radio or television receiver, are damped by electric resistance. The signal to which the receiver is tuned supplies energy synchronously to maintain resonance. [...] *In magnetic damping, energy of motion is converted to heat by way of electric eddy currents induced in either a coil or an aluminium plate (attached to the oscillating object) that passes between the poles of a magnet.*"

In a virtual work calculation, dampers are simply replaced by the forces they create. These forces depend on the generalised velocities which, as discussed above, are assumed fixed during a virtual displacement. In the following example of a dynamic system, the only generalised force arises from a viscous damper.

Exercise 2.5 The dynamic system in the figure below has one viscous damper with damping constant c and a horizontal prescribed boundary motion $\bar{z}(t)$. This boundary motion $\bar{z}(t)$ is imposed on mass M by the left wagon through the viscous damper. Under these conditions, identify an appropriate set of system coordinates (i.e. the generalised coordinates) and formulate the equations of motion for these coordinates using Lagrange's method.



Solution The key here is that the motion of the wagon is forced upon the system. As such, the function $\bar{z}(t)$ is given and, consequently, not a system coordinate. To completely describe the state of the system, it is first necessary to specify the position of mass M — which we denote by $x(t)$ — and the angle of rotation $\theta(t)$ of the bar. In this solution we assume x to point to the left, and the angle of rotation as shown in the figure above (ACW).

The system's kinetic and potential energies are defined as functions of these two coordinates. The kinetic energy of mass M and the elastic potential energy of the spring are straightforward. To find the kinetic and potential energy of the bar, however, we first write the coordinates of the centre of gravity of the bar, which shouldn't be called x and y to

avoid confusion with the generalised coordinates of the system,

$$x_\mathrm{b} = x + \frac{1}{2} l \sin\theta$$
$$y_\mathrm{b} = \frac{1}{2} l \cos\theta$$

The kinetic energy function is then

$$\begin{aligned} T &= \frac{1}{2} M \dot{x}^2 + \frac{1}{2} m \left(\dot{x}_\mathrm{b}^2 + \dot{y}_\mathrm{b}^2 \right) + \frac{1}{2} \frac{m l^2}{12} \dot{\theta}^2 \\ &= \frac{1}{2} M \dot{x}^2 + \frac{1}{2} m \left[\dot{x}^2 + \dot{x} \dot{\theta} l \cos\theta + \frac{1}{4} l^2 \left(\sin^2\theta + \cos^2\theta \right) \dot{\theta}^2 \right] + \frac{1}{2} \times \frac{m l^2}{12} \dot{\theta}^2 \\ &= \frac{1}{2} (M+m) \dot{x}^2 + \frac{1}{2} m l \dot{x} \dot{\theta} \cos\theta + \frac{1}{6} m l^2 \dot{\theta}^2 \end{aligned}$$

and the potential energy function is

$$V = \frac{1}{2} k x^2 + \frac{1}{2} m g l \cos\theta$$

This system has damping and as such will have a generalised force associated to the damper. The damping force depends on the relative velocity of both ends of the damper. This force is $-c(\dot{x} - \dot{\tilde{z}})$ and acts directly on mass M. If this mass is displaced by δx, it does the virtual work $-c(\dot{x} - \dot{\tilde{z}})\delta x = Q_x \delta x$. Hence the generalised force on the coordinate $x(t)$ is

$$Q_x = -c(\dot{x} - \dot{\tilde{z}})$$

There is no generalised force acting on the angular coordinate $\theta(t)$. Collecting all the terms and applying Lagrange's method to determine the equations of motion, in much the same way as in previous examples, leads to the equation of motion for $x(t)$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{x}}\right) - \frac{\partial T}{\partial x} + \frac{\partial V}{\partial x} = Q_x \quad \Rightarrow \quad (M+m)\ddot{x} + \frac{1}{2} m l \left(\ddot{\theta} \cos\theta - \dot{\theta}^2 \sin\theta \right) + kx = -c\left(\dot{x} - \dot{\tilde{z}}\right)$$

and the equation of motion for $\theta(t)$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{\theta}}\right) - \frac{\partial T}{\partial \theta} + \frac{\partial V}{\partial \theta} = 0 \quad \Rightarrow \quad \frac{1}{3} m l^2 \ddot{\theta} + \frac{1}{2} m l \ddot{x} \cos\theta - \frac{1}{2} m g l \sin\theta = 0$$

■

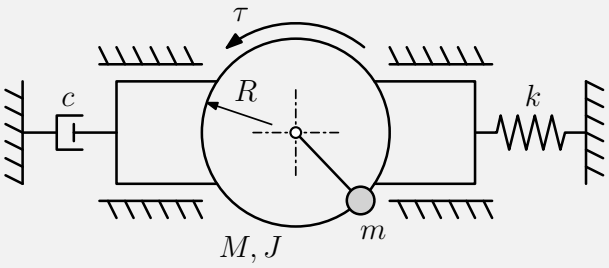
2.3.4 Other generalised forces

Any action upon a system that can change the system's energy will in general lead to generalised forces on one or multiple system coordinates. As was discussed before, examples of actions that lead to generalised forces include externally applied forces — provided they do work —, damping and friction. Torques in rotating systems will also lead to generalised forces. This is illustrated in Example 2.6 below.

Exercise 2.6 Consider the dynamic behaviour of a rotary engine with a vertical axis unbalanced rotor. The figure below represents a schematic top view of the engine and its mounting. The unbalance is represented by a small eccentric point mass m sitting on the radius R of the rotor. The total mass of the engine (including the eccentric mass) is M and the moment of inertia of the rotor (also including the eccentric mass) is J. The rotor is

driven by a constant torque τ.

Identify a set of system coordinates, determine the corresponding generalised forces and formulate the equations of motion for this system using Lagrange's method.



Solution The first step is to determine the generalised coordinates of the system. To specify the position of the rotor and its unbalanced mass we need an angle of rotation $\varphi(t)$ which can be taken anticlockwise (ACW) from the downward vertical in the figure. The horizontal position of the engine casing is denoted by $x(t)$, which is also the elongation of the elastic spring.

The potential energy function is straightforward. As indicated in the description above, the figure is a top view of the engine — vertical axis rotor —, hence there is no gravitational energy associated with the positions of the system masses. The only potential energy term is the elastic energy of the spring,

$$V = \frac{1}{2}kx^2$$

For the total kinetic energy function we consider separately the following three terms: (i) the horizontal motion of the engine without the eccentric mass m, (ii) the rotation of the rotor without the eccentric mass m, and (iii) the motion of the eccentric mass itself. The kinetic energy of linear motion of the engine is

$$T_1 = \frac{1}{2}(M-m)\dot{x}^2$$

Note that $(M-m)$ is the inertia (i.e. mass) of the motor excluding the eccentric mass m. The rotation kinetic energy of the rotor is

$$T_2 = \frac{1}{2}\left(J - mR^2\right)\dot{\varphi}^2$$

and here $(J-mR^2)$ is the inertia of the rotor excluding the eccentric mass m (the inertia of rotation of the mass is mR^2). As the position of the unbalanced mass is given by $x_m = x + R\sin\varphi$ and $y_m = -R\cos\varphi$, its kinetic energy is

$$T_3 = \frac{1}{2}m\left(\dot{x}_m^2 + \dot{y}_m^2\right) = \frac{1}{2}m\left(\dot{x}^2 + 2R\dot{x}\dot{\varphi}\cos\varphi + R^2\dot{\varphi}^2\right)$$

The total kinetic energy of the system is then obtained by adding all three contributions, yielding

$$T = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}J\dot{\varphi}^2 + mR\dot{x}\dot{\varphi}\cos\varphi$$

The next step is to determine the generalised forces. These arise from the torque acting on the rotor and from the viscous damper. As the angle φ increases by a virtual amount $\delta\varphi$,

the torque does the work $\tau\delta\varphi$. Consequently, the generalised force on φ is simply τ. The damper creates no force on this coordinate, only on $x(t)$. The damper force $-c\dot{x}$ does the work $-c\dot{x}\delta x$ when the engine coordinate x is increased by δx. Consequently, the generalised force on x is $-c\dot{x}$. We are now in a position to write down the equations of motion. Skipping the intermediate steps, the equation of motion for the coordinate $x(t)$ becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{x}}\right) - \frac{\partial T}{\partial x} + \frac{\partial V}{\partial x} = Q_x \quad \Rightarrow \quad M\ddot{x} + mR\left(\ddot{\varphi}\cos\varphi - \dot{\varphi}^2\sin\varphi\right) + kx = -c\dot{x}$$

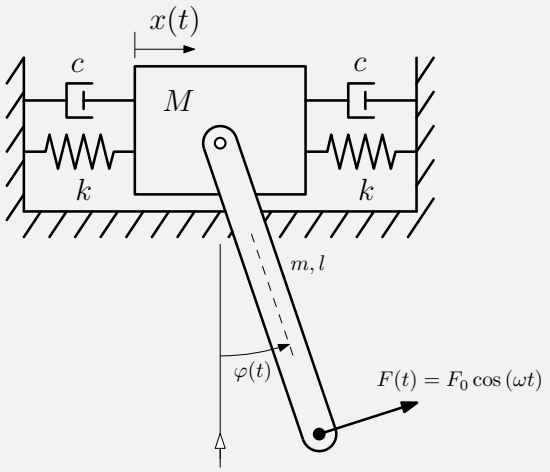
and the equation of motion for coordinate $\varphi(t)$ is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{\varphi}}\right) - \frac{\partial T}{\partial \varphi} + \frac{\partial V}{\partial \varphi} = \tau \quad \Rightarrow \quad J\ddot{\varphi} + mR\ddot{x}\cos\varphi = \tau$$

■

We now turn to a slightly more complex example where the generalised forces act on multiple system coordinates but are created by a single external force. In systems that are described by multiple generalised coordinates, care must be taken when working out the virtual work terms, since sometimes virtual displacements of one coordinate may cause forces acting on another part of the system to also do virtual work. Thus, even *remote* external forces can lead to generalised forces on a given coordinate. This is illustrated in Example 2.7 below.

Exercise 2.7 In the system shown in the figure below, a pendulum is connected to a mass M and acted by an external force $F(t)$, always perpendicular to the pendulum bar. The mass moves horizontally without friction and is connected to rigid boundaries by springs and viscous dampers. Determine the equations of motion of the system.



Solution This is a similar problem to exercise 2.4, with the only difference being the two viscous dampers acting on the mass and the external force acting on the bar. We can therefore use the same system variables $x(t)$ and $\varphi(t)$ as in Exercise 2.4 and get the same kinetic and potential energy functions

$$T = \frac{1}{2}(M+m)\dot{x}^2 + \frac{1}{2}ml\dot{x}\dot{\varphi}\cos\varphi + \frac{1}{6}ml^2\dot{\varphi}^2 \quad \text{and} \quad V = 2\left(\frac{1}{2}kx^2\right) - \frac{1}{2}mgl\cos\varphi$$

We now assume that the system is in a state characterised by the set of values $(x, \varphi, \dot{x}, \dot{\varphi})$ of the generalised coordinates and the corresponding velocities.

Virtual work done upon virtual displacement of the variable x

To create a virtual displacement in x we move mass M by δx while keeping the other coordinate (the angle φ) fixed. Noting that both dampers create forces that act in opposite direction to the velocity $\dot{x}$, the damper forces are $-2c\dot{x}$. If we impose a virtual displacement δx, these forces do the work $-2c\dot{x}\delta x$.

The virtual displacement does not change the velocity. This is, however, not the only virtual work related to a change of the x coordinate. If mass M is moved and the angle φ is kept fixed, then also the end point of the bar travels by δx along the x direction. The x component of the force on the bar is $F_0\cos(\omega t)\cos\varphi$. It does the work $F_0\cos(\omega t)\cos\varphi\delta x$. Hence, the total work is

$$[F_0\cos(\omega t)\cos\varphi - 2c\dot{x}]\,\delta x = Q_x\delta x \quad \text{with} \quad Q_x = F_0\cos(\omega t)\cos\varphi - 2c\dot{x}$$

Virtual work done upon virtual displacement of the variable φ

If we increase the angle φ by an amount $\delta\varphi$, then the endpoint of the bar moves by $l\delta\varphi$ in the tangential direction. The tangential force acting on the bar's end then does the work $F_0\cos(\omega t)l\delta\varphi = Q_\varphi\delta\varphi$. Consequently, the generalised force acting on coordinate φ is

$$Q_\varphi = lF_0\cos(\omega t)$$

which has the dimensions of a moment (e.g. [Nm] in SI units). We now write down the Lagrange equations of motion, first for the $x(t)$ coordinate

$$\begin{aligned}&\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial\dot{x}}\right) - \frac{\partial T}{\partial x} + \frac{\partial V}{\partial x} = Q_x\\ \Rightarrow\quad &(M+m)\ddot{x} + \frac{1}{2}ml\left(\ddot{\varphi}\cos\varphi - \dot{\varphi}^2\sin\varphi\right) + 2kx = F_0\cos(\omega t)\cos\varphi - 2c\dot{x}\end{aligned}$$

and finally, for the $\varphi(t)$ coordinate

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial\dot{\varphi}}\right) - \frac{\partial T}{\partial\varphi} + \frac{\partial V}{\partial\varphi} = Q_\varphi \quad \Rightarrow \quad \frac{1}{3}ml^2\ddot{\varphi} + \frac{1}{2}ml\ddot{x}\cos\varphi + \frac{1}{2}mgl\sin\varphi = lF_0\cos(\omega t)$$

■

2.4 Dynamic stability

Equations of motion are widely used to determine if a structure or system is stable or not. Dynamic instabilities in engineering systems and structures often lead to undesirable dynamic responses such as shaking, rattling, wobbling and undesirable noise. These instabilities may occur under *steady operating conditions* with no obvious external source of dynamic excitation.

Some examples of engineering systems that are prone to dynamic instability are shown in Figure 2.6 (top three images) and include (i) disk brake squeal; (ii) machine tool chattering and (iii) *wobbly* structures such as bridges. A whole branch of science is devoted to understanding and mitigating undesirable oscillations and instabilities that may occur due to the interaction between a steady state fluid flow and a flexible structure. Examples of such situations are shown in the bottom three images in Figure 2.6 and include: (iv) *singing* ship propellers; (v) *galloping* of overhead transmission lines and (vi) flutter instabilities of bridges and aircraft wings.

By definition, in mathematics, structural stability is a fundamental property of any dynamic system. This means that the qualitative behaviour of the trajectories, i.e. the mechanical response of the system, is unaffected by small perturbations. The following quote[3] explains this:

[3] V.V. Bolotin. *Nonconservative problems of the theory of elastic stability*, Pergamon Press, Oxford, 1963.

Figure 2.6: Examples of engineering systems prone to dynamic instability. Clockwise from top-left: a bicycle disk brake, a conventional lathe, the Millennium Bridge in London, the NASA X-29 research aircraft, overhead power lines and one of the main USS Churchill propellers. Reproduced under Creative Commons license CC BY-SA 3.0 (authors: Rics, Greudin, Hossinger, NASA, Kreuzschnabel, Hogan).

> "*The basic method of investigating non-conservative problems in the theory of elastic stability is the dynamic method, which is based on the study of the oscillation of the system close to its position of equilibrium. This draws the theory of elastic stability closer to the general theory of stability of motion and its applications in the theory of automatic control, in the hydrodynamics of a viscous liquid and in other fields of applied mechanics and engineering.*"

The conditions for the onset of instability can be found by considering a steady solution of the system's equations of motion and investigating the response of the system under small deviations from this solution, using the linearisation method. If the linearised equations show exponentially growing perturbations, the solution is said to be unstable and is likely to lead to one of the types of undesired behaviour discussed above or, in extreme cases, the catastrophic collapse or failure of a structure or system.

Stability problems may have been briefly discussed in previous courses (recall self-excited vibration). In this course we discuss the general method for the analysis of stability in the context of Lagrange's method, using a few simple examples. Below is a step-by-step approach that can be followed when analysing the stability of a mechanical system or structure.

Analysis of dynamic stability Analysing the dynamic stability conditions of complex system is not always a straightforward task. Doing this in a systematical way, however, can help identify the potentially unstable modes of a mechanical system or structure. The steps described below can be followed to achieve this.

1. Develop a model of the system, obtain the corresponding equations of motion and determine potential steady-state/stationary positions.

2. Define the solutions describing the investigated steady-state behaviour:

$$\mathbf{q} = \{q_1, \ldots, q_N\} = \{q_1^{\mathrm{s}}, \ldots, q_N^{\mathrm{s}}\}$$

3. Identify any non-linear terms and linearise them around the investigated solutions:

$$\mathbf{q} = \mathbf{q}^{\mathrm{s}} + \delta\mathbf{q} = \{q_1^{\mathrm{s}} + \delta q_1, \ldots, q_N^{\mathrm{s}} + \delta q_N\}$$

4. Insert these into the equations of motion and linearise them under the assumption that all δq_i are small.
5. Assume the solution for the deviations in the form

$$\delta\mathbf{q}(t) = \{\delta q_1^0, \ldots, \delta q_N^0\} \exp(\lambda t)$$

6. This will give you a characteristic equation for the eigenvalues λ and will determine if the system is stable or unstable.

The practicality and efficiency of this method depends crucially on the linearisation of the equations of motion around the stable/stationary position. By doing this we avoid having to trace the ensemble of time histories for every conceivable dynamic deviation from the equilibrium position, which for a system with many degrees-of-freedom would clearly be a forbidding task. Below is a mathematical definition of dynamic stability.

Dynamic stability A dynamic system is said to be *asymptotically stable* if its free-vibration response approaches zero as time approaches infinity. Consequently, a system is said to be *unstable* if its free-vibration response grows without bound as time approaches infinity. Finally, a system is *marginally stable* or *borderline stable* if its free-vibration response neither decays nor grows as time approaches infinity.

In mathematical terms stability can be interpreted in terms of the roots λ of the characteristic equations of the system. The possible solutions λ_i will have a real and imaginary part. Within these, any roots lying in the left half plane of the complex plane (i.e. where $\mathrm{Re} < 0$) correspond to either exponential decay or damped sinusoidal free-vibration responses, which will approach zero as time increases. Consequently, these will be asymptotically stable (see Figure 2.7).

Systems whose characteristic roots lie in the right half plane (i.e. where $\mathrm{Re} > 0$) will be unstable because they will yield exponentially increasing free-vibration responses, approaching infinity as time approaches infinity. Roots lying on the imaginary axis (i.e. $\mathrm{Re} = 0$) yield pure sinusoidal free-vibration responses with constant amplitude, corresponding to a marginally stable behaviour. In conclusion, a mechanical or structural system is said to be dynamically unstable if any one of the eigenvalues obtained from its characteristic equation has a positive real part. The diagram in Figure 2.7 illustrates this.

For an asymptotically stable system it is then not possible to find a set of initial conditions for which the response approaches infinity. It should however, be noted that for unstable systems it is possible to find a set of initial conditions that might make the response approach zero as time increases. As an example of this behaviour, let us assume the characteristic roots of a system are $\lambda_{1,2} = \pm 1$. When a characteristic equation has distinct real roots $r_1, \ldots, r_n$, the general solution is $x(t) = C_1 e^{r_1 t} + \ldots + C_n e^{r_n t}$, meaning that in this case the response of the system is

$$x(t) = C_1 e^{-t} + C_2 e^{+t}$$

where C_1 and C_2 are constants that can be determined from the initial conditions. For the particular case where $x(0) = 1$ and $\dot{x}(0) = -1$ these constants become $C_1 = 1$ and $C_2 = 0$ and

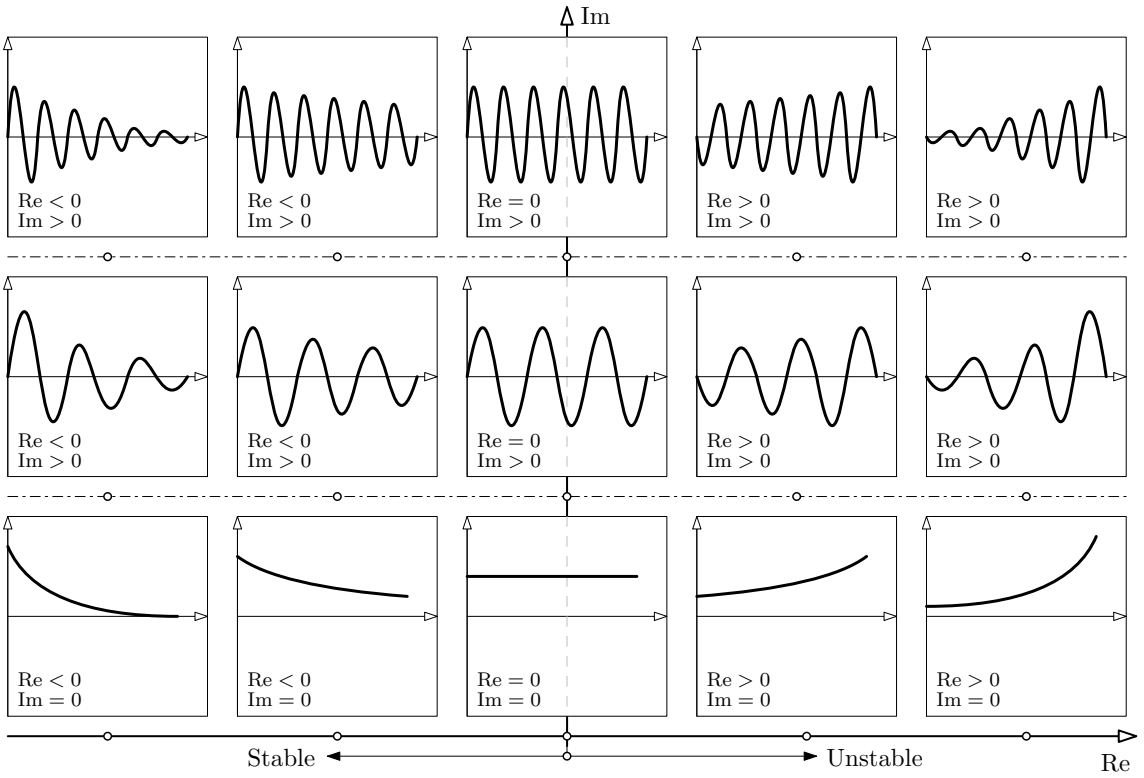


Figure 2.7: Typical response of a dynamic system in relation to its characteristic roots, with indication of ranges of stable, unstable and marginally stable responses.

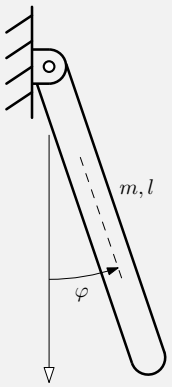
hence the response becomes

$$x(t) = e^{-t}$$

which approaches zero as time increases, meaning that the system will respond with a stable behaviour if it is perturbed from these particular initial conditions.

To illustrate this method, we use first an extremely simple example: a simple pendulum moving by the sole action of the gravitational field and without damping or friction.

Exercise 2.8 Define a system variable, obtain the equations of motion for this variable and analyse the stability of the bar (pendulum) in its two steady state configurations: (i) hanging vertically and (ii) standing vertically.



Solution This example is Exercise 2.2 revisited. The system is described by a single coordinate, the rotation angle φ. The kinetic and potential energy functions are

$$T = \frac{1}{2} I_{\mathrm{E}} \dot{\varphi}^2 = \frac{1}{6} m l^2 \dot{\varphi}^2 \quad \text{and} \quad V = mgy = -\frac{1}{2} mgl \cos\varphi$$

respectively, and the Lagrange equation of motion is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{\varphi}}\right) + \frac{\partial V}{\partial \varphi} = 0 \quad \Rightarrow \quad \frac{1}{3} m l^2 \ddot{\varphi} + \frac{1}{2} mgl \sin\varphi = 0$$

This equation of motion has stationary solutions for $\varphi^{\mathrm{s}} = n\pi$ where n is an arbitrary integer. Of these, only $\varphi^{\mathrm{s}} = 0$ and $\varphi^{\mathrm{s}} = \pi$ have independent meaning because consecutive ones are repetitions of these two. To study the stability of these solutions, we start by considering $\varphi^{\mathrm{s}} = 0$ and put $\varphi = 0 + \delta\varphi$. Thus $\ddot{\varphi} = \delta\ddot{\varphi}$ and $\sin\varphi \approx \delta\varphi$, and we get the linearised equation

$$\frac{1}{3} m l^2 \delta\ddot{\varphi} + \frac{1}{2} mgl \delta\varphi = 0$$

with the associated characteristic equation

$$\frac{1}{3} m l^2 \lambda^2 + \frac{1}{2} mgl = 0$$

The corresponding roots

$$\lambda = \pm i \sqrt{\frac{3g}{2l}}$$

of this equation are purely complex, hence have zero real part. Thus, the state $\varphi^{\mathrm{s}} = 0$ is (marginally) stable. We now repeat the analysis for the stationary state $\varphi^{\mathrm{s}} = \pi$. With $\varphi = \pi + \delta\varphi$ we get $\ddot{\varphi} = \delta\ddot{\varphi}$ and $\sin\varphi = \sin(\pi + \delta\varphi) \approx -\delta\varphi$, and the linearised equation of motion now becomes

$$\frac{1}{3} m l^2 \delta\ddot{\varphi} - \frac{1}{2} mgl \delta\varphi = 0$$

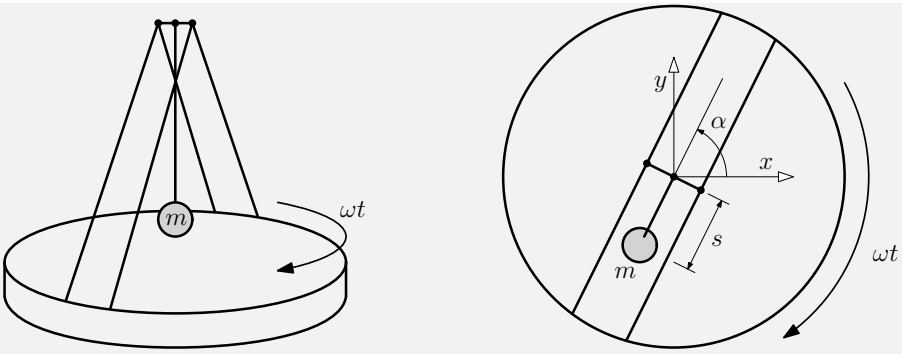
with the corresponding characteristic equation (note the negative sign on the gravitational term)

$$\frac{1}{3} m l^2 \lambda^2 - \frac{1}{2} mgl = 0$$

The roots $\lambda = \pm\sqrt{3g/(2l)}$ are now real and there is one positive root, hence the state $\varphi^{\mathrm{s}} = \pi$ is unstable, which is intuitively clear: the pendulum is standing vertically upwards in this state and will fall down at the slightest perturbation. ■

We now turn to a much more challenging example where setting up the equations of motion is not an obvious task. To this end, we simply put the pendulum on a table that rotates with constant angular velocity ω.

Exercise 2.9 A lumped mass pendulum is mounted at the centre of a horizontal plate with no mass, which rotates with constant angular velocity ω, as shown in the figure below (left). Define a system variable describing the dynamics of the pendulum, obtain an equation of motion for this variable and find the steady state solutions. Analyse the stability of these solutions for different rotation velocities of the disk.



Solution This time the pendulum is centrally mounted on a rotating disk. The oscillation of the pendulum is constrained to one direction. However, the axis of rotation of the pendulum now rotates with the angular velocity of the disk ω.

The first step is to analyse the kinematics of the mechanical system. The pendulum is again described by a single coordinate φ which is the angle from the vertical. In addition we introduce the angle of rotation of the disk, which we denote as α. We choose the direction of the vertically hanging pendulum to be the z-axis. The x and y-axes are contained in the plane of the disk as shown in the top view of the system in the figure above (right).

The distance of the mass from the vertical z-axis is denoted by s. This is related to the deflection angle φ of the pendulum by $s = l\sin\varphi$ where l is the pendulum length. The (x, y, z) coordinates of the mass m are then given by

$$\begin{aligned} x &= l\sin\varphi\cos\alpha \\ y &= l\sin\varphi\sin\alpha \\ z &= z_0 - l\cos\varphi \end{aligned}$$

Noting that $\dot{\alpha} = \omega$, the components of the velocity are

$$\begin{aligned} \dot{x} &= l(\dot{\varphi}\cos\varphi\cos\alpha - \omega\sin\varphi\sin\alpha) \\ \dot{y} &= l(\dot{\varphi}\cos\varphi\sin\alpha + \omega\sin\varphi\cos\alpha) \\ \dot{z} &= l\dot{\varphi}\sin\varphi \end{aligned}$$

The total kinetic energy function is

$$\begin{aligned} T &= \frac{1}{2}m\left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2\right) \\ &= \frac{1}{2}ml^2\left[(\dot{\varphi}\cos\varphi\cos\alpha - \omega\sin\varphi\sin\alpha)^2 + (\dot{\varphi}\cos\varphi\sin\alpha + \omega\sin\varphi\cos\alpha)^2 + \dot{\varphi}^2\sin^2\varphi\right] \\ &= \frac{1}{2}ml^2\left(\dot{\varphi}^2 + \omega^2\sin^2\varphi\right) \end{aligned}$$

and the potential energy function is

$$V = mgz = mg(z_0 - l\cos\varphi)$$

Applying Lagrange's method, the equation of motion becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial\dot{\varphi}}\right) - \frac{\partial T}{\partial\varphi} + \frac{\partial V}{\partial\varphi} = 0 \quad \Rightarrow \quad ml^2\ddot{\varphi} - ml^2\omega^2\sin\varphi\cos\varphi + mgl\sin\varphi = 0$$

To analyse the dynamic stability of the system we should first ask what are the stationary solutions of this equation. Stationarity means that all time derivatives must be zero, hence

$$-ml^2\omega^2 \sin\varphi \cos\varphi + mgl \sin\varphi = 0$$

which is fulfilled if either $\sin\varphi = 0$ or $\cos\varphi = g/(l\omega^2)$. It is interesting to note that for small angular velocities ω, the second condition cannot be fulfilled since the cosine cannot get bigger than 1. As such, we are left with the classical stationary solutions for the non-rotating pendulum, $\varphi^{\mathrm{s}} = 0$ (hanging vertically downwards) and $\varphi^{\mathrm{s}} = \pi$ (standing vertically upwards). However, for $\omega^2 > g/l$ new solutions emerge. For these solutions, the centrifugal force on the mass balances the gravitational restoring force in such a way that the mass remains stationary at a finite non-zero angle.

We now analyse the stability of the solution $\varphi^{\mathrm{s}} = 0$. Using the same linearisation as in Exercise 2.8 and approximating $\cos\varphi \approx 1$ for small φ, we obtain the linearised equation

$$ml^2\delta\ddot{\varphi} + \left(mgl - ml^2\omega^2\right)\delta\varphi = 0$$

The roots of the corresponding characteristic equation are

$$\lambda = \pm\sqrt{\omega^2 - \frac{g}{l}}$$

This gives two complex (purely imaginary) roots for small angular velocities ω, and a positive and negative real root for large angular velocities. The system is thus unstable for rotation velocities above $\omega^2 = g/l$ or, in other words, when the rotation frequency of the disk exceeds the natural frequency of the pendulum without rotation. ■

As an additional exercise it is interesting to try and show that the vertically upwards position of the pendulum is always unstable irrespective of the rotation frequency of the disk.

2.5 Lagrange's method for continuous systems

Until now we have considered systems which can be described by a small (and thus finite) number of generalised coordinates — essentially lumped parameter models of systems that consist of one or more rigid bodies. The concept of *rigid body* is, however, only theoretical as in real life bodies and structures are all deformable, albeit to different extents. If loads are sufficiently small, the behaviour of most structures and solid bodies under dynamic loads is governed by continuous elastic deformations and can thus be described by elastic behaviour/constitutive laws. An important part of this course, which will be discussed in the remainder of this chapter, deals with the dynamic behaviour and mechanical response of systems where mass and elasticity are continuously distributed. To achieve this we will focus on the simplest case of continuous and slender structural elements, i.e. bars and beams.

It is important to know the difference between a bar and a beam, and this will be relevant for the following sections of this chapter and also for Chapter 3. By definition, bars are structural elements that are only subjected to axial loads, in compression and/or in tension. What distinguishes bars from beams is the fact that the latter are, in addition to axial forces, subjected to transverse loads, i.e. perpendicular to the beam's longitudinal axis.

Bar and beam models can be used as approximations for modelling many engineering systems, such as (i) beams and frames in buildings; (ii) turbine and wind generator blades; (iii) aircraft wings and tailplanes; (iv) bridges; (v) ship hulls; (vi) pillars, masts, chimneys and pipework. Some such examples are shown in Figure 2.8.

In theory, to fully describe the dynamic response of a continuous body, it is necessary to specify the motion of every single point of that same body. Given the continuous nature of

Figure 2.8: Examples of engineering dynamic continuous systems that can be modelled using Lagrange's method. All images reproduced under Creative Commons license CC BY-SA 3.0 (authors, clockwise from top-left: L. Allorge, H. McDaniel, R. Roletschek, N. Micheva, P. Selim and the NOAA Photo Library).

solid bodies, there are an infinite number of points in a continuous body. Consequently, strictly speaking such systems have an infinite number of degrees-of-freedom. The equations of motion for such systems are partial differential equations, and methods for their analysis and solution will be discussed in Chapter 3 of these course notes.

In many engineering applications, however, the relevant dynamic behaviour can be approximated in terms of a few modes of motion only. By using engineering intuition, the infinite number of variables needed to describe an engineering system can often be reduced to a small number of variables of interest. The basic idea is to identify the likely/relevant motions of the system under the acting loads, and then formulate that into a mathematical equation, or set of equations. The Lagrange formalism then allows us to evaluate the likely dynamic behaviour associated with the motion that was *guesstimated*. Formally, this method is often called the Assumed Modes Method (AMM). In the context of estimating the lowest natural frequency of a system, it is also known as Rayleigh's method (RM). This method is not exact but an approximation, and the quality of the results depends strongly on the initial guesses and assumptions. The method can nonetheless be envisaged as a systematic approach to deriving lumped parameter descriptions. This method can actually be used when, in order to analyse the vibration behaviour of a mechanical system or structure, the continuous mass and distributed elasticity of a cantilever beam is replaced by a point mass attached to a massless flexural spring. The formalisms that are presented here have the advantage that they allow us to systematically derive the lumped masses and stiffnesses from an estimation of how the motion of the slender structure (e.g. beam) will look like.

2.5.1 Energy functions for continuous systems

As we know already from the previous sections, one of the essential steps when applying Lagrange's method is to obtain the appropriate kinetic and potential energy functions of the system. When dealing with continuous systems, these functions represent the kinetic and potential energies of all mass points. To better explain how this can be dealt with we will first look at a simple structural element. In the following we consider the specific case of a simply supported beam of uniform cross-section and homogeneous material properties, under the effect of transverse loads, as shown in Figure 2.9. The length of the beam is denoted as L, the area of the cross-section is A and the bending stiffness of the beam is EI. Note that in this particular case EI is constant because the

material is homogeneous and the shape and area of the cross-section are constant. The density of the beam material is ρ and the total mass is $M = \rho AL$.

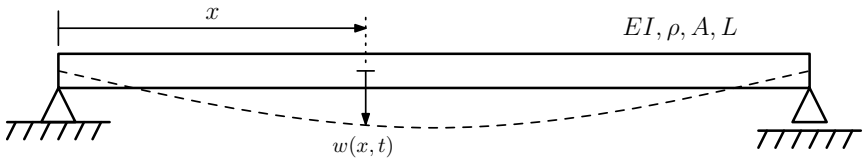


Figure 2.9: Simply supported beam of bending stiffness EI, length L, density ρ and cross-section area A, under a transverse displacement $w(x,t)$. The dashed line represents the deflection of the beam.

The transverse displacement of the beam is described by the displacement field $w(x,t)$, which can be envisaged as a function of x (the longitudinal coordinate along the beam) which describes the shape of the beam at time t. The kinetic energy of the beam can then be expressed as

$$\begin{aligned} T &= \frac{1}{2}\int_0^L \dot{w}^2 \mathrm{d}m = \frac{1}{2}\int_0^L \dot{w}^2 \rho A \mathrm{d}x \\ &= \int_0^L \frac{\rho A}{2}\left(\frac{\partial w}{\partial t}\right)^2 \mathrm{d}x \end{aligned} \tag{2.19}$$

For small transversal displacements ($w \ll L$), the potential energy of the beam is the work done in deforming the beam, which is

$$V = \frac{1}{2}\int_0^L M_\mathrm{b} \mathrm{d}\theta \tag{2.20}$$

where M_b is the bending moment on the beam

$$M_\mathrm{b} = M_\mathrm{b}(x,t) = EI\frac{\partial^2 w(x,t)}{\partial x^2} \tag{2.21}$$

and $\theta = \partial w/\partial x$ is the slope of the deformed beam. The elastic potential energy then becomes

$$\begin{aligned} V &= \frac{1}{2}\int_0^L \left(EI\frac{\partial^2 w}{\partial x^2}\right)\frac{\partial^2 w}{\partial x^2}\mathrm{d}x \\ &= \int_0^L \frac{EI}{2}\left(\frac{\partial^2 w}{\partial x^2}\right)^2 \mathrm{d}x \end{aligned} \tag{2.22}$$

To evaluate these kinetic and potential energy functions we need information about the deflection function $w(x,t)$ or, in other words, information about the shape of the deforming beam. In the following sections this problem will be solved using an educated guess. Later, in Chapter 3, more systematic methods to evaluate this function will be discussed.

2.5.2 Single degree-of-freedom model approximation

The simplest possible case is when we want to study only one type of deformation, i.e. only one mode shape of the beam. For the simply supported beam shown in Figure 2.9, this may be a symmetrical deflection of the beam as shown in Figure 2.10(a). The crucial first step is then to propose a mathematical function $\varphi_1(x)$ that describes the shape of the dynamic deflection of the beam — this function is the first mode shape of the beam.

The choice of the mode shape function is often down to engineering intuition. The only requirement is that it must comply with the support conditions — i.e. the boundary conditions.

Thus, for the beam in Figure 2.9, the mode shape function must be zero at both ends of the beam since these ends rest on the supports or, in mathematical notation

$$w(0,t) = w(L,t) = 0$$

Once the first mode shape of the beam is known the generic deflection function can be written as

$$w(x,t) = \varphi_1(x) q_1(t) \tag{2.23}$$

where the mode amplitude $q_1(t)$ is the generalised coordinate.

This last step requires some degree of abstract thinking. A *coordinate* in Lagrangian dynamics is understood as a time-dependent variable which characterises the motion of the system. It neither has to have the dimensions of length, nor does it have to be associated with a simple geometrical feature of the system. In equation 2.23 above, the generalised coordinate is then the amplitude of the mode shape function. The physical meaning of this for the motion of the beam is shown graphically in Figure 2.10(b). To characterise the shape of the beam as it undergoes an oscillatory up-and-down motion we have chosen a sine function which is zero at the beam ends (taken as $x = 0$ and $x = L$) and which also has unit amplitude ($\varphi_1 \in [-1, +1]$). The values of q_1 then give the actual deflection of the centre point of the beam.

An equation of motion for the generalised coordinate q_1 can be obtained by simply plugging equation 2.23 into equations 2.19 and 2.22, and using Lagrange's method.

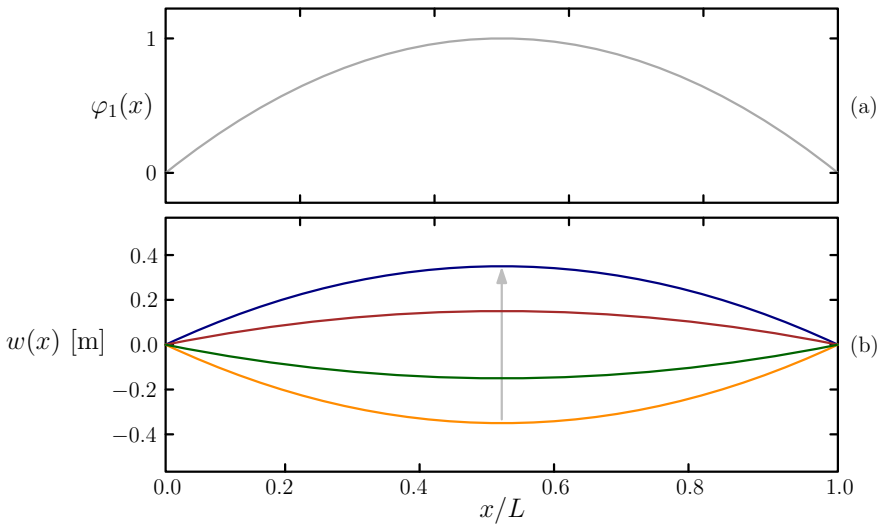


Figure 2.10: (a) An example of mode shape function for the beam shown in Figure 2.9 and (b) four different configurations of the beam, corresponding to the generalised coordinate values $q_1 = -0.3$ m (orange line), -0.1 m (green line), 0.1 m (red line), 0.3 m (blue line).

Lagrange's method for continuous beams The following steps should be followed to determine the equation of motion of a continuous beam:

1. Obtain the functions that give the kinetic and potential energy of the beam due to the time-dependent deflection.
2. Evaluate any other energy terms associated with the deflection of the beam.
3. Use Lagrange's method to derive the equation of motion of the beam containing the generalised coordinate $q_1(t)$.

For the simply supported beam in Figure 2.9 the kinetic energy function is

$$T = \int_0^L \frac{\rho A}{2}\left(\frac{\partial w}{\partial t}\right)^2 \mathrm{d}x = \frac{\rho A}{2}\dot{q}_1^2 \int_0^L [\varphi_1(x)]^2 \mathrm{d}x \qquad \text{and} \qquad T = \frac{1}{2}M_1\dot{q}_1^2 \tag{2.24}$$

and the elastic potential energy function is

$$V = \int_0^L \frac{EI}{2}\left(\frac{\partial^2 w}{\partial x^2}\right)^2 \mathrm{d}x = \frac{EI}{2}q_1^2 \int_0^L \left(\frac{\partial^2 \varphi_1}{\partial x^2}\right)^2 \mathrm{d}x \qquad \text{and} \qquad V = \frac{1}{2}K_1 q_1^2 \tag{2.25}$$

More information on integration and a review of integration rules can be found in Appendix A.4.

Once the mode shape function $\varphi_1(x)$ is specified, the integrals simply produce constants that represent the effective mass M_1 and the effective stiffness K_1 associated with the up-and-down (transverse) motion of the beam. Consequently, the corresponding equation of motion is the equation of a simple oscillator, that is,

$$M_1\ddot{q}_1 + K_1 q_1 = 0 \tag{2.26}$$

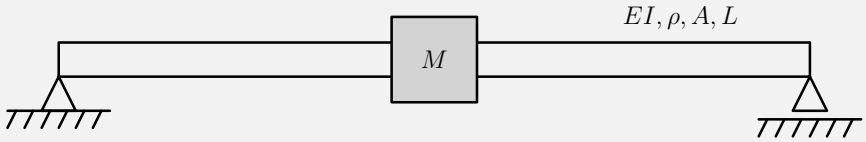
and the solution is a sinusoidal vibration with frequency

$$w = \sqrt{\frac{K_1}{M_1}} \tag{2.27}$$

Rayleigh's theorem When applying the assumed modes method to calculate natural frequencies of vibrating systems, as in the above description, the exact natural frequency will be a lower bound to the calculated frequencies. This bound is attained if the mode shape is the correct (exact) function.

The following example illustrates the application of the assumed modes method to a simply supported beam with an attached lumped mass, using a single degree-of-freedom model approximation.

Exercise 2.10 Define a mode shape function and derive the equation of motion for the free transverse vibration of a simply supported homogeneous beam of length L with an attached lumped mass M at mid-span ($x = L/2$).



Solution The boundary conditions are: (i) beam deflection equal to zero at both beam ends and (ii) bending moment equal to zero at both beam ends or, in a more formal way,

$$w(0,t) = w(L,t) = 0 \qquad \Rightarrow \qquad \varphi_1(x=0) = \varphi_1(x=L) = 0$$

$$w''(0,t) = w''(L,t) = 0 \qquad \Rightarrow \qquad \left.\frac{\mathrm{d}^2\varphi_1}{\mathrm{d}x^2}\right|_{x=0} = \left.\frac{\mathrm{d}^2\varphi_1}{\mathrm{d}x^2}\right|_{x=L} = 0$$

A possible mode shape function that fulfils all the above boundary conditions is

$$\varphi_1(x) = \sin\left(\frac{\pi x}{L}\right)$$

As the assumed displacement is given by the product of the mode shape function by an amplitude function that is the generalised coordinate in the Lagrange formalism,

$$w(x,t) = \varphi_1(x) q_1(t)$$

the deflection of the central mass M is

$$w\left(\frac{L}{2}, t\right) = \varphi_1\left(\frac{L}{2}\right) q_1(t)$$

The next step is to find the energy functions. The kinetic energy is the kinetic energy of the central mass added to the distributed kinetic energy of the beam:

$$\begin{aligned} T &= \frac{1}{2} M \left[\dot{w}\left(\frac{L}{2}, t\right)\right]^2 + \frac{\rho A}{2} \int_0^L [\dot{w}(x,t)]^2 \mathrm{d}x \\ &= \frac{1}{2} M \dot{q}_1^2 + \frac{\rho A}{2} \dot{q}_1^2 \int_0^L \varphi_1^2 \mathrm{d}x \\ &= \frac{1}{2}\left(M + \frac{\rho A L}{2}\right) \dot{q}_1^2 \end{aligned}$$

The potential energy has three distinct components: the gravitational energy of the central mass, the distributed gravitational energy of the elastic beam, and the flexural/elastic energy of the beam (due to bending). In mathematical terms, this yields

$$\begin{aligned} V &= Mgw\left(\frac{L}{2}, t\right) + \rho A g \int_0^L w(x,t) \mathrm{d}x + \frac{EI}{2} \int_0^L \left(\frac{\partial^2 w}{\partial x^2}\right)^2 \mathrm{d}x \\ &= Mgq_1 + \rho A g q_1 \int_0^L \varphi \mathrm{d}x + \frac{EI}{2} q_1^2 \int_0^L \left(\frac{\partial^2 \varphi}{\partial x^2}\right)^2 \mathrm{d}x \\ &= Mgq_1 + \rho A g q_1 \int_0^L \sin\left(\frac{\pi x}{L}\right) \mathrm{d}x + \frac{EI}{2} q_1^2 \left(\frac{\pi}{L}\right)^4 \int_0^L \sin^2\left(\frac{\pi x}{L}\right) \mathrm{d}x \\ &= \left(M + \frac{2\rho A L}{\pi}\right) g q_1 + \frac{EI\pi^4}{4L^3} q_1^2 \end{aligned}$$

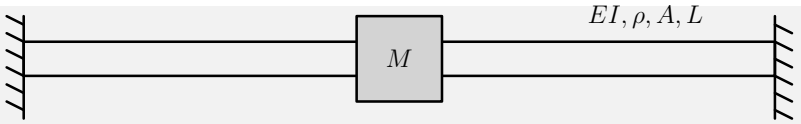
Deriving the equation of motion leads to

$$\begin{aligned} &\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_1}\right) + \frac{\partial V}{\partial q_1} = 0 \\ \Rightarrow \quad &\left(M + \frac{\rho A L}{2}\right) \ddot{q}_1 + \left(M + \frac{2\rho A L}{\pi}\right) g + \frac{EI\pi^4}{2L^3} q_1 = 0 \end{aligned}$$

Note that the effective inertial mass of the beam in the above equation of motion is not its actual mass ρAL, but only half of it. This is because the outer parts of the beam move less than the centre and thus create smaller inertial forces. ■

To better illustrate the importance of boundary conditions and how these can completely change the behaviour of a structure, in the next example we consider the same beam but with fully clamped ends.

Exercise 2.11 Define a mode shape function and derive the equation of motion for the free transverse vibration of a fully clamped homogeneous beam of length L with an attached lumped mass M at mid-span ($x = L/2$).



Solution This problem is similar to Example 2.10 but with fully clamped ends. The boundary conditions are now: (i) beam deflection equal to zero at both beam ends and (ii) deflection slope equal to zero at both beam ends:

$$w(0,t) = w(L,t) = 0 \qquad \Rightarrow \qquad \varphi_1(x=0) = \varphi_1(x=L) = 0$$

$$w'(0,t) = w'(L,t) = 0 \qquad \Rightarrow \qquad \left.\frac{\mathrm{d}\varphi_1}{\mathrm{d}x}\right|_{x=0} = \left.\frac{\mathrm{d}\varphi_1}{\mathrm{d}x}\right|_{x=L} = 0$$

The problem is very similar to the previous exercise. However, as the boundary conditions are different we need a different mode shape function. It can be easily verified that the following function verifies all the above boundary conditions:

$$\varphi_1(x) = \frac{1}{2}\left[1 - \cos\left(\frac{2\pi x}{L}\right)\right]$$

The deflection of the central mass is again given by

$$w\left(\frac{L}{2}, t\right) = q_1(t)\varphi_1\left(\frac{L}{2}\right) = q_1(t)$$

The kinetic energy and potential energy calculations are also similar to the previous case. The total kinetic and potential energy functions are

$$\begin{aligned}
T &= \frac{1}{2}M\left[\dot{w}\left(\frac{L}{2}, t\right)\right]^2 + \frac{\rho A}{2}\int_0^L [\dot{w}(x,t)]^2 \mathrm{d}x \\
&= \frac{1}{2}M\dot{q}_1^2 + \frac{\rho A}{2}\dot{q}_1^2\int_0^L \varphi_1^2 \mathrm{d}x \\
&= \left(\frac{1}{2}M + \frac{3\rho AL}{16}\right)\dot{q}_1^2 \\
V &= Mgw\left(\frac{L}{2}, t\right) + \rho Ag\int_0^L w(x,t)\mathrm{d}x + \frac{EI}{2}\int_0^L \left(\frac{\partial^2 w}{\partial x^2}\right)^2 \mathrm{d}x \\
&= Mgq_1 + \rho Agq_1\int_0^L \varphi \mathrm{d}x + \frac{EI}{2}q_1^2\int_0^L \left(\frac{\partial^2 \varphi}{\partial x^2}\right)^2 \mathrm{d}x \\
&= Mgq_1 + \frac{\rho Agq_1}{2}\int_0^L \left[1 - \cos\left(\frac{2\pi x}{L}\right)\right]\mathrm{d}x + \frac{EI}{8}q_1^2\left(\frac{2\pi}{L}\right)^4\int_0^L \cos^2\left(\frac{2\pi x}{L}\right)\mathrm{d}x \\
&= \left(M + \frac{\rho AL}{2}\right)gq_1 + \frac{EI\pi^4}{L^3}q_1^2
\end{aligned}$$

Finally, we obtain the equation of motion using Lagrange's approach,

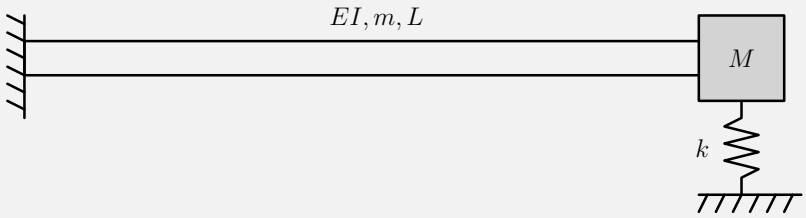
$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_1}\right) + \frac{\partial V}{\partial q_1} = 0$$

$$\Rightarrow \quad \left(M + \frac{3\rho AL}{8}\right)\ddot{q}_1 + \left(M + \frac{\rho AL}{2}\right)g + \frac{2EI\pi^4}{L^3}q_1 = 0$$

A quick comparison with the equation of motion obtained in Example 2.10 shows a lower effective mass and an higher effective stiffness of the beam, due to the clamping. ■

We now complicate the system a bit more by keeping one of the boundary conditions as fixed and adding an elastic foundation to the other end of the beam. The following example illustrates the effect of having an elastic boundary condition at one end of the structure.

Exercise 2.12 Derive an equation of motion for the vibration of a clamped beam with a tip lumped mass M and spring support of stiffness k. The beam has length L, mass m and bending stiffness EI.



Solution The first step is to find a mode shape function that verifies the specified boundary conditions. As the beam is clamped at the left end, which for simplicity is taken at $x = 0$, the boundary conditions are $w(0,t) = w'(0,t) = 0$. A cubic shape function that fulfils the boundary condition and ensure a linear bending moment is chosen such that

$$\varphi_1(x) = \left(\frac{x}{L}\right)^3$$

The L in the denominator is not strictly necessary. However, this makes the shape function non-dimensional (normalised) and has the advantage of making sure that its value at the endpoint where the mass sits is equal to 1. L could have been left out but then the algebra would have resulted more complicated. The deflection of the mass is then $w(L,t) = q_1(t)\varphi_1(L) = q_1(t)$. The spring is compressed by the same amount. The total kinetic energy of the system is

$$\begin{aligned} T &= \frac{1}{2}M\left[\dot{w}(L,t)\right]^2 + \frac{\rho A}{2}\int_0^L \left[\dot{w}(x,t)\right]^2 \mathrm{d}x \\ &= \frac{1}{2}M\dot{q}_1^2 + \frac{\rho A}{2}\dot{q}_1^2\int_0^L \varphi_1^2 \mathrm{d}x \\ &= \frac{1}{2}\left(M + \frac{\rho AL}{7}\right)\dot{q}_1^2 \end{aligned}$$

and the total potential energy is

$$\begin{aligned} V &= Mgw(L,t) + \rho Ag\int_0^L w(x,t)\mathrm{d}x + \frac{EI}{2}\int_0^L \left(\frac{\partial^2 w}{\partial x^2}\right)^2 \mathrm{d}x + \frac{1}{2}k[w(L,t)]^2 \\ &= Mgq_1 + \rho Agq_1\int_0^L \varphi \mathrm{d}x + \frac{EI}{2}q_1^2\int_0^L \left(\frac{\partial^2 \varphi}{\partial x^2}\right)^2 \mathrm{d}x + \frac{1}{2}kq_1^2 \\ &= Mgq_1 + \frac{\rho AL}{4}gq_1 + \frac{EI}{2}q_1^2\int_0^L \frac{36x^2}{L^6}\mathrm{d}x + \frac{1}{2}kq_1^2 \\ &= \left(M + \frac{\rho AL}{4}\right)gq_1 + \left(\frac{6EI}{L^3} + \frac{1}{2}k\right)q_1^2 \end{aligned}$$

In conclusion, the equation of motion becomes

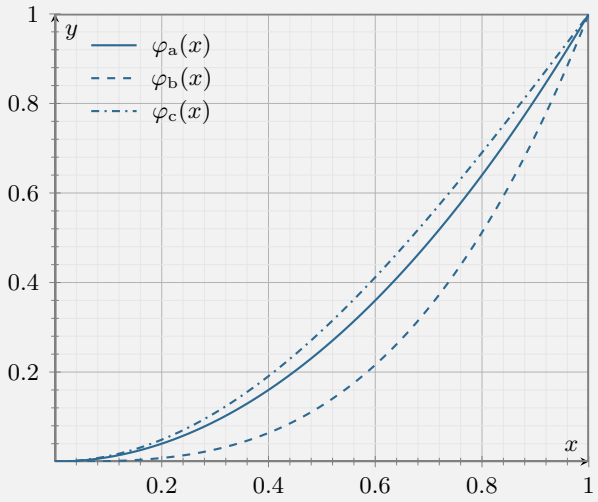
$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_1}\right) + \frac{\partial V}{\partial q_1} = 0 \;\Rightarrow\; \left(M + \frac{\rho AL}{7}\right)\ddot{q}_1 + \left(M + \frac{\rho AL}{4}\right)g + \left(\frac{12EI}{L^3} + k\right)q_1 = 0$$

■

Exercise 2.13 Consider a homogeneous cantilever beam of length L, mass m and bending stiffness EI. Propose three different alternative shape functions for the first mode shape of the beam. Find the first natural frequency of the beam using all three functions and compare/discuss results.

Solution The following three mathematical function can be used as alternative shape functions for the first mode shape of the beam:

$$\varphi_\mathrm{a}(x) = \left(\frac{x}{L}\right)^2, \qquad \varphi_\mathrm{b}(x) = \left(\frac{x}{L}\right)^3 \qquad \text{and} \qquad \varphi_\mathrm{c}(x) = 1 - \cos\left(\frac{\pi x}{2L}\right)$$



As we can see from the plot above, all three functions are normalised and comply with the deflection boundary condition at the left end where $x = 0$ (i.e. $w(0,t) = 0$). Additionally, all also check the zero slope boundary condition $w'(0,t) = 0$ required for a clamped beam.

We can now use the assumed modes method to find the first mode shape natural frequencies as $\omega = \sqrt{K_i/M_i}$ with $K_i = 2V/q_1^2$ and $M_i = 2T/\dot{q}_1^2$.

The effective stiffness K_i and effective inertia M_i (with $i = \mathrm{a}, \mathrm{b}, \mathrm{c}$) for each function are

then, respectively:

$$K_i = EI \int_0^L \left(\varphi_i''\right)^2 \mathrm{d}x \qquad \text{and} \qquad M_i = \rho A \int_0^L \varphi_i^2 \mathrm{d}x$$

the effective inertia for each shape function is:

$$M_\mathrm{a} = \rho A \int_0^L \left(\frac{x^2}{L^2}\right)^2 \mathrm{d}x = \rho A \left(\frac{x^5}{5L^4}\right)\Big|_0^L = \frac{\rho A L}{5} = \frac{m}{5}$$

$$M_\mathrm{b} = \rho A \int_0^L \left(\frac{x^3}{L^3}\right)^2 \mathrm{d}x = \rho A \left(\frac{x^7}{7L^6}\right)\Big|_0^L = \frac{\rho A L}{7} = \frac{m}{7}$$

$$\begin{aligned} M_\mathrm{c} &= \rho A \int_0^L \left[1 - \cos\left(\frac{\pi x}{2L}\right)\right]^2 \mathrm{d}x \\ &= \rho A \int_0^L \left[1 - 2\cos\left(\frac{\pi x}{2L}\right) + \cos^2\left(\frac{\pi x}{2L}\right)\right]^2 \mathrm{d}x \\ &= \rho A \left[x - \frac{4L}{\pi}\sin\left(\frac{\pi x}{2L}\right) + \frac{x}{2} + \frac{L}{2\pi}\sin\left(\frac{\pi x}{L}\right)\right] \\ &= \rho A \left(\frac{3L}{2} - \frac{4L}{\pi}\right) = 0.227 \rho A L = 0.227 m \end{aligned}$$

The effective stiffness for each shape function is:

$$K_\mathrm{a} = EI \int_0^L \left(\frac{4}{L^4}\right) \mathrm{d}x = \frac{4EI}{L^4} x\Big|_0^L = \frac{4EI}{L^3}$$

$$K_\mathrm{b} = EI \int_0^L \left(\frac{36x^2}{L^6}\right) \mathrm{d}x = \frac{36EI}{3L^6} x^3\Big|_0^L = \frac{12EI}{L^3}$$

$$\begin{aligned} K_\mathrm{c} &= EI \left(\frac{\pi}{2L}\right)^2 \int_0^L \sin^2\left(\frac{\pi x}{2L}\right) \mathrm{d}x \\ &= \frac{\pi^2 EI}{4L^2} \left[\frac{x}{2} - \frac{2L}{4\pi}\sin\left(\frac{\pi x}{L}\right)\right]\Big|_0^L \\ &= \frac{\pi^E I}{4L^2} \left(\frac{L}{2} - 0\right) = \frac{\pi^2 EI}{8L} \end{aligned}$$

The first mode shape frequencies can now be calculated as $\omega_i = \sqrt{\frac{K_i}{M_i}}$ $(i = \mathrm{a}, \mathrm{b}, \mathrm{c})$:

$$\omega_\mathrm{a} = \sqrt{\left(\frac{4EI}{L^3}\right)\left(\frac{5}{m}\right)} = 4.47\sqrt{\frac{EI}{mL^3}}$$

$$\omega_\mathrm{b} = \sqrt{\left(\frac{12EI}{L^3}\right)\left(\frac{7}{m}\right)} = 9.17\sqrt{\frac{EI}{mL^3}}$$

$$\omega_\mathrm{c} = \sqrt{\left(\frac{\pi^2 EI}{8L}\right)\left(\frac{1}{0.227m}\right)} = 2.33\sqrt{\frac{EI}{mL}}$$

■

2.5.3 Multiple degree-of-freedom model approximation

To obtain a more flexible and generic description of the dynamic response of a structure we may assume a series of mode shape functions $\varphi_k(x)$ with $k = 1, \ldots, n$, which fulfil the boundary conditions at the beam supports. The approximation for the total beam deflection can then be written as the sum of all these functions weighted by the respective amplitudes,

$$\begin{aligned} w(x,t) &= q_1(t)\varphi_1(x) + q_2(t)\varphi_2(x) + \ldots + q_n(t)\varphi_n(x) \\ &= \sum_{k=1}^{n} q_k(t)\varphi_k(x) \end{aligned} \tag{2.28}$$

The mode amplitudes $q_k(t)$ act as the generalised coordinates of an n-degree-of-freedom model of the beam dynamics.

As an example, if we consider a simply supported beam such as the one shown in Figure 2.11, the approximation for the total beam deflection can be written using the first two mode shapes as

$$\begin{aligned} w(x,t) &= \sum_{k=1}^{2} q_k(t)\varphi_k(x) \\ &= q_1(t)\varphi_1(x) + q_2(t)\varphi_2(x) \end{aligned} \tag{2.29}$$

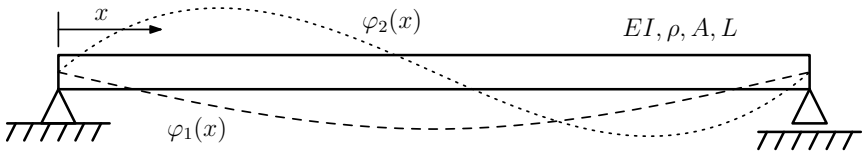


Figure 2.11: Simply supported beam of bending stiffness EI, length L, density ρ and cross-section area A, under a transverse displacement $w(x,t)$. The dashed lines represent the first two mode shapes of the beam, $\varphi_1(x)$ and $\varphi_2(x)$.

The boundary conditions are beam deflection equal to zero at both beam ends and bending moment also equal to zero at both beam ends, that is, $\varphi_1(x=0) = \varphi_1(x=L) = \varphi_2(x=0) = \varphi_2(x=L) = 0$ and

$$\left.\frac{\mathrm{d}^2\varphi_1}{\mathrm{d}x^2}\right|_{x=0} = \left.\frac{\mathrm{d}^2\varphi_1}{\mathrm{d}x^2}\right|_{x=L} = \left.\frac{\mathrm{d}^2\varphi_2}{\mathrm{d}x^2}\right|_{x=0} = \left.\frac{\mathrm{d}^2\varphi_2}{\mathrm{d}x^2}\right|_{x=L} = 0$$

A possible set of mode shape functions that fulfil all the above boundary conditions is

$$\varphi_1(x) = \sin\left(\frac{\pi x}{L}\right)$$

$$\varphi_2(x) = \sin\left(\frac{2\pi x}{L}\right)$$

These mode shape functions are represented by the dashed lines in Figure 2.11. The total kinetic energy of the beam is

$$\begin{aligned} T &= \frac{1}{2}\rho A \int_0^L [\dot{w}(x,t)]^2 \,\mathrm{d}x \\ &= \frac{1}{2}\rho A \int_0^L [\dot{q}_1(t)\varphi_1(x) + \dot{q}_2(t)\varphi_2(x)]^2 \,\mathrm{d}x \\ &= \frac{1}{2}\rho A \left[\dot{q}_1^2 \int_0^L \varphi_1^2 \mathrm{d}x + \dot{q}_2^2 \int_0^L \varphi_2^2 \mathrm{d}x + 2\dot{q}_1\dot{q}_2 \int_0^L \varphi_1\varphi_2 \mathrm{d}x\right] \\ &= \frac{1}{4}\rho A L \left(\dot{q}_1^2 + \dot{q}_2^2\right) \end{aligned} \tag{2.30}$$

and the total potential energy of the beam is

$$\begin{aligned} V &= \rho A g \int_0^L w(x,t)\mathrm{d}x + \frac{EI}{2}\int_0^L \left(\frac{\partial^2 w}{\partial x^2}\right)^2 \mathrm{d}x \\ &= \rho A g \int_0^L (q_1\varphi_1 + q_2\varphi_2)\,\mathrm{d}x + \frac{EI}{2}\int_0^L \left[q_1\frac{\partial^2\varphi_1}{\partial x^2} + q_2\frac{\partial^2\varphi_2}{\partial x^2}\right]^2 \mathrm{d}x \\ &= \frac{\rho L A g}{\pi}(2q_1+q_2) + \frac{\pi^5 EI}{2L^4}\left(\frac{1}{2}q_1^2 + 16q_2^2\right) \end{aligned}$$

Finally, deriving the equations of motion using Lagrange's method leads to

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_1}\right) + \frac{\partial V}{\partial q_1} = 0 \quad &\Rightarrow \quad \frac{1}{2}\rho A L\ddot{q}_1 + \frac{2\rho L A g}{\pi} + \frac{\pi^5 EI}{2L^4}q_1 = 0 \\ \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_2}\right) + \frac{\partial V}{\partial q_2} = 0 \quad &\Rightarrow \quad \frac{1}{2}\rho A L\ddot{q}_2 + \frac{\rho L A g}{\pi} + \frac{16\pi^5 EI}{L^4}q_2 = 0 \end{aligned}$$

2.5.4 Generalised forces in continuous systems

In the following paragraphs we generalise the concept of loads acting on structures such as homogeneous beams. These can either be point loads, distributed loads or a combination of both. We first start by analysing point loads.

Forces due to point loads

To illustrate the treatment of point loads, we consider the simply supported beam in Figure 2.12 with a generic point load $F(t)$, acting at a distance b from the left end, that is, at $x = b$.

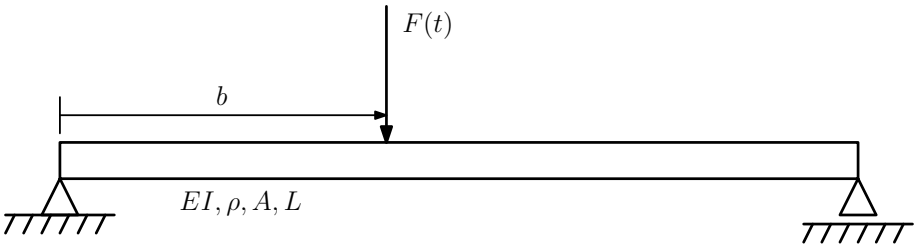


Figure 2.12: Simply supported beam with an applied point force $F(t)$ at coordinate $x = b$ from the left end.

A single degree-of-freedom model approximation can again be used to describe the dynamics of the beam, with mode shape function $\varphi_1(x)$ and generalised coordinate $q_1(t)$. To evaluate the generalised force on this coordinate, we consider a virtual displacement $q_1 \to q_1 + \delta q_1$. Then (i) the virtual displacement at point $x = b$ becomes $\delta q_1\varphi_1(b)$, (ii) the virtual work done by the point force is $\delta q_1\varphi_1(b)F(t)$, and (iii) the corresponding generalised force is $\varphi_1(b)F(t)$. The following important observations can be made:

- The generalised force depends on the point of application of $F(t)$ on the beam.
- The generalised force depends on the value of the mode shape function at this same location.

If multiple mode shapes are considered, the generalised forces are calculated accordingly. Thus, an actual force $F(t)$ acting at $x = b$ creates the following generalised force on the mode with shape function $\varphi_k(x)$:

$$Q_k = \varphi_k(b)F(t) \tag{2.31}$$

Forces due to distributed loads

We now consider a beam with a distributed load characterised by a generic force density $f(x,t)$ (with dimensions of force per unit length of beam), as shown in Figure 2.13, and evaluate the generalised force caused by this type of loading.

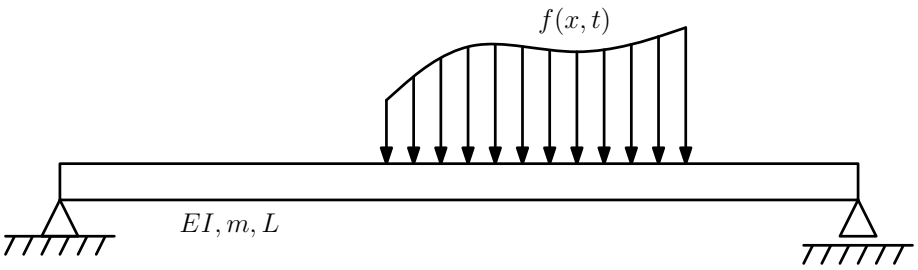


Figure 2.13: Simply supported beam with a generic distributed load $f(x,t)$. Note that the dimensions of this distributed load are force per unit length (e.g. units [N/m] in SI).

We again consider a mode characterised by the mode shape function $\varphi_1(x)$ and a generalised coordinate $q_1(t)$. The virtual displacement of the generalised coordinate is $q_1 \to q_1 + \delta q_1$ and the virtual displacement at coordinate x is $\delta q \varphi(x)$. The virtual work done by the distributed load is

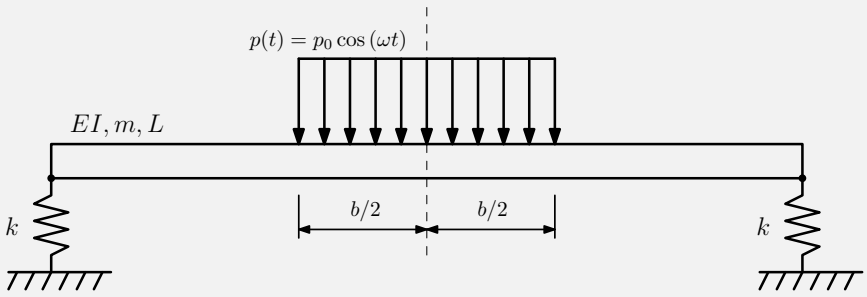
$$\int_0^L \delta q_1 \varphi_1(x) f(x,t) \mathrm{d}x \tag{2.32}$$

and the corresponding generalised force is

$$Q_1(t) = \int_0^L \varphi_1(x) f(x,t) \mathrm{d}x \tag{2.33}$$

Generalised forces for multiple modes are evaluated in a similar way by using the respective mode shape functions in equation 2.33. Appendix A.4 has more information on integration rules.

Exercise 2.14 Choose a set of appropriate mode shape functions and derive the equations of motion for the centre symmetric vibration of the flexibly supported beam with a oscillating pressure load $p(t) = p_0 \cos(\omega t)$, as shown in the figure below.



Solution In this example, no mode shape function is given. As such, the first step should be to carefully analyse the structure and its supports. In doing so it is possible to conclude that:

i) The problem is symmetrical with respect to the central axis, so the mode shape functions should be symmetrical as well;

ii) The beam can oscillate as a rigid body on the supporting springs, corresponding to a mode shape function that is constant along the beam. Assuming that the stiffness of

the supporting springs is significantly lower than the bending stiffness of the beam this will be the first mode shape of the structure (i.e. rigid body motion); and

iii) The beam can bend under the applied load. This mode shape function is zero at the beam ends and has a maximum at mid-span.

Assuming the origin of the coordinate system to be at the centre of the beam, the following two normalised mode shapes verify all the above conditions:

$$\varphi_1(x) = 1$$
$$\varphi_2(x) = \cos\left(\frac{\pi x}{L}\right)$$

The first mode shape function is constant, the second has a maximum in the middle and is zero at the beam ends. The total displacement is then

$$w(x,t) = q_1(t)\varphi_1(x) + q_2(t)\varphi_2(x)$$

The displacement of the endpoints is simply $q_1(t)$ and the kinetic energy function is

$$\begin{aligned} T &= \frac{\rho A}{2}\int_{-\frac{L}{2}}^{\frac{L}{2}} [\dot{w}(x,t)]^2\,\mathrm{d}x = \frac{\rho A}{2}\int_{-\frac{L}{2}}^{\frac{L}{2}} \left(\dot{q}_1^2\varphi_1^2 + 2\dot{q}_1\dot{q}_2\varphi_1\varphi_2 + \dot{q}_2^2\varphi_2^2\right)\mathrm{d}x \\ &= \frac{\rho A}{2}\int_{-\frac{L}{2}}^{\frac{L}{2}} \left[\dot{q}_1^2 + 2\dot{q}_1\dot{q}_2\cos\left(\frac{\pi x}{L}\right) + \dot{q}_2^2\cos^2\left(\frac{\pi x}{L}\right)\right]\mathrm{d}x \\ &= \frac{\rho AL}{2}\left[\dot{q}_1^2 + \frac{4}{\pi}\dot{q}_1\dot{q}_2 + \frac{1}{2}\dot{q}_2^2\right] \end{aligned}$$

The potential energy function has the contributions of gravity, elastic springs and bending,

$$\begin{aligned} V &= \rho Ag\int_{-\frac{L}{2}}^{\frac{L}{2}} w(x,t)\mathrm{d}x + \frac{1}{2}k\left[w\left(\frac{L}{2},t\right)\right]^2 + \frac{1}{2}k\left[w\left(-\frac{L}{2},t\right)\right]^2 + \frac{EI}{2}\int_{-\frac{L}{2}}^{\frac{L}{2}}\left(\frac{\partial^2 w}{\partial x^2}\right)^2\mathrm{d}x \\ &= \rho Ag\int_{-\frac{L}{2}}^{\frac{L}{2}}\left[q_1 + q_2\cos\left(\frac{\pi x}{L}\right)\right]\mathrm{d}x + kq_1^2 + \frac{EI}{2}q_2^2\left(\frac{\pi}{L}\right)^4\int_{-\frac{L}{2}}^{\frac{L}{2}}\cos^2\left(\frac{\pi x}{L}\right)\mathrm{d}x \\ &= \rho ALg\left(q_1 + \frac{2}{\pi}q_2\right) + kq_1^2 + \frac{EI\pi^4}{4L^3}q_2^2 \end{aligned}$$

The next step is to evaluate the generalised forces, which are given by

$$Q_1 = \int_{-\frac{b}{2}}^{\frac{b}{2}} p(x,t)\varphi_1(x)\mathrm{d}x = bp_0\cos(\omega t)$$
$$Q_2 = \int_{-\frac{b}{2}}^{\frac{b}{2}} p(x,t)\varphi_2(x)\mathrm{d}x = \left(\frac{2L}{\pi}\right)\sin\left(\frac{\pi b}{2L}\right)p_0\cos(\omega t)$$

The equations of motion of the beam can be found by combining everything and applying

Lagrange's method,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_1}\right)+\frac{\partial V}{\partial q_1}=Q_1$$
$$\Rightarrow \quad \rho AL\left(\ddot{q}_1+\frac{2}{\pi}\ddot{q}_2\right)+\rho ALg+2kq_1=bp_0\cos(\omega t)$$

and

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_2}\right)+\frac{\partial V}{\partial q_2}=Q_2$$
$$\Rightarrow \quad \rho AL\left(\frac{2}{\pi}\ddot{q}_1+\frac{1}{2}\ddot{q}_2\right)+\frac{2\rho ALg}{\pi}+\frac{EI\pi^4}{2L^3}q_2=\left(\frac{2L}{\pi}\right)\sin\left(\frac{\pi b}{2L}\right)p_0\cos(\omega t)$$

These equations can also be written in matrix form as

$$\rho AL\begin{bmatrix}1 & 2/\pi\\ 2/\pi & 1/2\end{bmatrix}\begin{Bmatrix}\ddot{q}_1\\ \ddot{q}_2\end{Bmatrix}+\begin{bmatrix}2k & 0\\ 0 & EI\pi^4/(2L^3)\end{bmatrix}\begin{Bmatrix}q_1\\ q_2\end{Bmatrix}=$$
$$=p_0\cos(\omega t)\begin{Bmatrix}b\\ (2L/\pi)\sin[\pi b/(2L)]\end{Bmatrix}-\rho ALg\begin{Bmatrix}1\\ 2/\pi\end{Bmatrix}$$

Analysing the equations of motion as written above, it is interesting to note that the mass matrix is now symmetrical while the stiffness matrix is diagonal. ■

2.6 Applications

In this section we present and analyse real-life dynamic systems, using what we learned in this chapter. We start by looking at the dynamic behaviour of a double pendulum and the conditions that may lead to chaotic motion. We later analyse the effects of system properties on its stability, by looking at an example of a tall building.

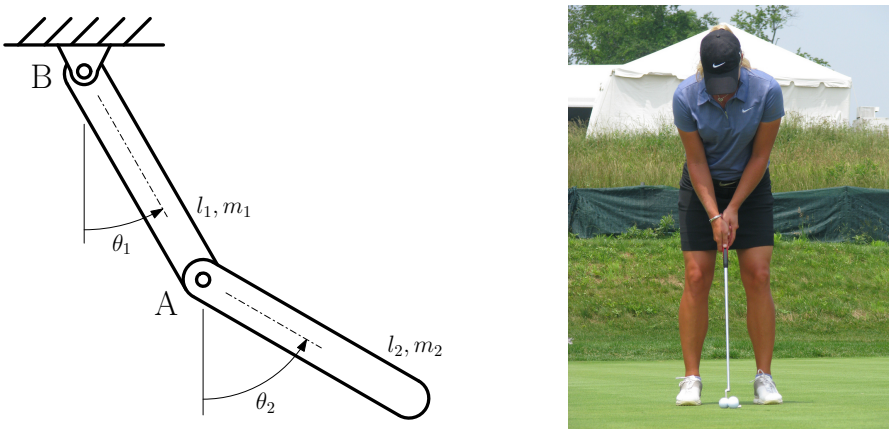
2.6.1 The double pendulum and chaotic motion

The double pendulum is a simple dynamical system that demonstrates high sensitivity to initial conditions, as we will see later. The simplest model of the system consists of two point masses m_1 and m_2, connected in series by rigid mass-less rods of lengths l_1 and l_2, respectively, and acting under gravity. Figures 2.2 and 2.14 show an example of a distributed mass double pendulum, where the masses are not lumped but distributed over the length of each arm.

The generalised coordinates required to describe the dynamic state of the system are the angular positions of the masses θ_1 and θ_2, shown in Figure 2.14(a). The corresponding angular velocities $\dot{\theta}_1$ and $\dot{\theta}_2$ are also relevant when defining the initial conditions. The slightest alteration of these conditions, relative to some reference state, results in motion that comparatively becomes increasingly different and unpredictable as time progresses. The system's increasing unpredictability is perhaps the simplest exhibition of the theory of chaos.

Despite the prevalence of chaos theory within theoretical mathematics and the natural world, not all systems, physical and theoretical, are chaotic. The conditions for chaos in a dynamic system governed by first order differential equations are: (i) the equations must be composed of three or more independent dynamic variables or, in other words, time-dependent variables; (ii) there must exist one or more non-linear terms in the equations of motion that couples several variables.

Therefore, the chaotic dynamics of the double pendulum system may be verified by deriving



(a) (b)

Figure 2.14: The double pendulum: (a) schematic representation of a distributed mass double pendulum with relevant variables represented and (b) Suzann Pettersen putting in Maryland in 2009, a possible application of the double pendulum movement (photograph by *dnkbdotcom*, reproduced under Creative Commons license BY-NC-SA-2.0).

the equations of motion and considering whether or not these conditions for chaos have been met. The double pendulum system is Hamiltonian — meaning there is no energy loss —, and therefore Lagrange's theorem can be used to derive the equations of motion. As was shown before, this method offers the most efficient derivation although it is worth noting that it is possible to derive the equations of motion using Newtonian particle mechanics and free-body-diagrams. However, the derivation is challenging and unnecessary. Returning to the planar double pendulum introduced in Figure 2.14, we write the Cartesian position coordinates (x_1, y_1) and (x_2, y_2) for each mass $(m_1,\ m_2)$ in terms of the system's generalised coordinates θ_1 and θ_2 are:

$$
\begin{aligned}
x_1 &= l_1 \sin\theta_1 \\
y_1 &= -l_1 \cos\theta_1 \\
x_2 &= l_1 \sin\theta_1 + l_2 \sin\theta_2 \\
y_2 &= -l_1 \cos\theta_1 - l_2 \cos\theta_2
\end{aligned}
$$

The position coordinates are differentiated with respect to time such that they may be substituted into the expression for the total kinetic energy T of the system. In addition, the reference for the gravitational potential energy of the system V can be defined such that the expression gives zero potential energy when the system is at equilibrium position, i.e. both masses at rest. After some mathematical manipulation, this leads to

$$
\begin{aligned}
T &= \frac{1}{2}(m_1 + m_2) l_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 l_2^2 \dot{\theta}_2^2 + m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \\
V &= m_1 g l_1 (1 - \cos\theta_1) + m_2 g \left[l_1 (1 - \cos\theta 1) + l_2 (1 - \cos\theta_2) \right]
\end{aligned}
$$

Noting that the system has no generalised forces, the equations of motion for the system can now be derived using Lagrange's theorem applied to a conservative system

$$
\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} + \frac{\partial V}{\partial q_i} = 0
$$

Making $q_1 = \theta_1$ and $q_2 = \theta_2$ we obtain the equations of motion for the pendulum, respectively,

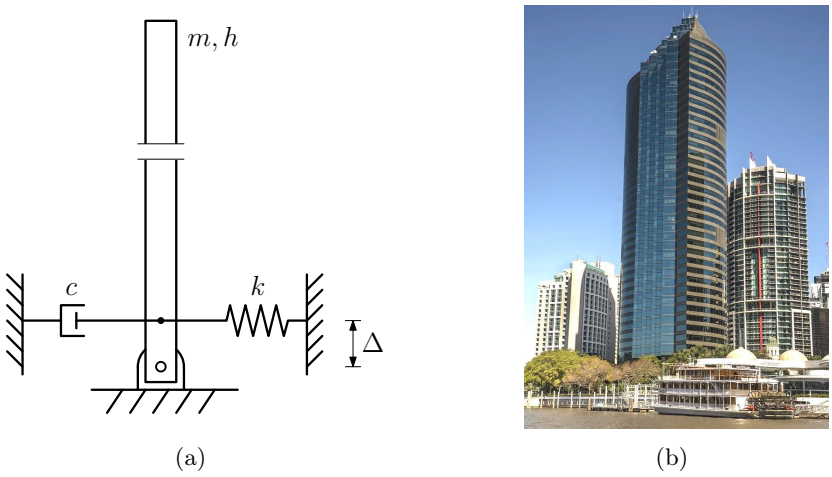
$$
\begin{aligned}
&(m_1 + m_2) l_1 \ddot{\theta}_1 + m_2 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) + m_2 l_2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) + (m_1 + m_2) g \sin\theta_1 = 0 \\
&m_2 l_2 \ddot{\theta}_2 + m_2 l_1 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) - m_2 l_1 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) + m_2 g \sin\theta_2 = 0
\end{aligned}
$$

The above system of first order differential equations of motion contains four independent dynamic variables (θ_1, θ_2, $\dot{\theta}_1$, $\dot{\theta}_2$) and both coupled and non-linear terms, highlighting the system's potential for chaotic motion. An example of chaotic motion of a distributed mass double pendulum can be seen in Figure 2.2(c).

2.6.2 Oscillation of tall buildings and structures

In this example application we look at the dynamic behaviour of a tall building. Let us start by assuming that the simplified model in Figure 2.15(a) can be used to describe the oscillating behaviour of a tall building, such as the one in Figure 2.15(b). The foundation of the building is represented by a spring with stiffness k and a viscous damper with damping constant c. A possible design problem/question could be to investigate the stability of the building and determine foundation properties that ensure stability.

As a starting point we need to assume that the conceptual model in Figure 2.15(a) is acceptable and define the model parameters. We will assume the building is 50 storeys high, each with an approximated mass of 40 tonnes. The total height of the building is $h = 125$ m and $\Delta = h/125$.



(a) (b)

Figure 2.15: Tall building example: (a) simplified conceptual model of the building and (b) high rise building in Brisbane (photograph by *Sheba_Also 43,000 photos*, reproduced under Creative Commons license BY-SA-2.0).

The single generalised coordinate (degree-of-freedom) of the system described by the conceptual model is the rotation angle of the bar (model building), which we will designate by $\theta(t)$. The total kinetic energy of the system can then be described by the relation

$$T = \frac{1}{6} m l^2 \dot{\theta}^2$$

The potential energy of the system has a gravitational and an elastic component, that is,

$$V = V_g + V_e = \frac{1}{2} mgh \cos\theta + \frac{1}{2} k(\Delta \sin\theta)^2$$

The generalised force due to the viscous damper is

$$Q_\theta = -c\Delta^2 \dot{\theta} \cos\theta$$

From the above energy and generalised force relations, and using Lagrange's method, the equation of motion of the system becomes

$$\frac{1}{3} mh^2 \ddot{\theta} - \frac{1}{2} mgh \sin\theta + k\Delta^2 \sin\theta \cos\theta = -c\Delta^2 \dot{\theta} \cos\theta$$

To analyse the stability of the system the equation of motion must be linearised and it is reasonable to assume small angular displacements ($\sin\theta \approx \theta$ and $\cos\theta \approx 1$). The linearised equation of motion is then

$$\frac{1}{3}mh^2\ddot{\theta} - \frac{1}{2}mgh\theta + k\Delta^2\theta = -c\Delta^2\dot{\theta}$$

The building model only has one physically possible stationary solution, which corresponds to $\theta^{\mathrm{s}} = 0$. To study this solution it is necessary to consider a small perturbation in the form $\theta = 0 + \delta\theta$ and, consequently, $\dot{\theta} = \delta\dot{\theta}$ and $\ddot{\theta} = \delta\ddot{\theta}$. Following this, the equation of motion becomes

$$\frac{1}{3}mh^2\delta\ddot{\theta} + \left(k\Delta^2 - \frac{1}{2}mgh\right)\delta\theta + c\Delta^2\delta\dot{\theta} = 0$$

and the corresponding characteristic equation is

$$\frac{1}{3}mh^2\lambda^2 + c\Delta^2\lambda = \left(\frac{1}{2}mgh - k\Delta^2\right)$$

The two roots $\lambda_{1,2}$ of the characteristic equation are then

$$\lambda_{1,2} = \frac{3}{2mh^2}\left[-c\Delta^2 \pm \sqrt{c^2\Delta^4 - \frac{4}{3}mh^2\left(k\Delta^2 - \frac{1}{2}mgh\right)}\right]$$

These roots have a real and positive part if

$$k < \frac{mgh}{2\Delta^2} = 1226.3 \text{ MN/m}$$

Consequently, the state $\theta^{\mathrm{s}} = 0$ is unstable if $k < 1226.3$ MN/m and stable otherwise. This is, as we can see, independent of the damping of the system c. This makes physical sense as the damping forces are dissipative and do not restitute the system to its equilibrium or stationary positions. The stability of the system depends only on its geometrical properties — which in this example are h and Δ —, stiffness k and inertia m.

3 Dynamics of continuous systems

3.1 Introduction

In the final sections of the previous chapter we discussed how to determine equations of motion for continuous systems, such as beams. With Rayleigh's approach (also known as the Assumed Modes Method) we assume mode shapes to calculate the natural frequencies of continuous dynamic systems. These are, however, approximated frequencies and not the exact ones, because the method assumes masses to be lumped and does not consider the distributed nature of the stiffness and inertia of the system.

To describe the real dynamics of a continuous body or structure, displacements must be calculated as a function of time at every point of that same body. In other words, the mechanical response of the system must be described by the displacement field, which is a continuous function of the space and time coordinates. In this chapter we turn towards a more general formulation, which uses partial differential equations as equations of motion for continuous displacement fields.

The dynamic behaviour of a continuous body or structure is characterised by its distributed mass, elasticity and stiffness. Consequently, and due to the continuous nature of the body, the dynamic system has an infinite number of degrees of freedom. This leads to equations of motion that are partial differential equations for displacement fields. As such, the boundary conditions are of key importance for determining the types of motion that may occur.

Only uniform cases — such as bar and beams of uniform cross-section, and constant thickness membranes — can be treated easily. In the scope of this course we focus on the dynamics of uniform slender structures, such as bars and beams, and address issues such as travelling waves and the corresponding dispersion relations, and standing waves with the associated natural frequencies and mode shapes. In particular, we study the wave equations for axial waves in uniform bars and beams, for (longitudinal) sound waves, and for flexural (bending) waves in thin beams. This is called the Euler-Bernoulli equation.

3.2 Longitudinal waves

When dealing with wave propagation in solids, the simplest continuous system is the dynamic response of a long bar, that is, the propagation of a strain wave (due to axial displacements) along a bar with constant cross section and constant material properties. The axial coordinate is x, and the axial displacement of a material point from its equilibrium position is defined by a generic function $u(x,t)$. To derive an equation of motion for the displacement field $u(x,t)$ we first consider the balance of forces on a volume element of the long bar, as shown in Figure 3.1a. The cross-section area of the bar is A, its density and elastic modulus are ρ and E, respectively.

The first step is to consider a small element of the bar with length Δx at initial coordinate x and write the equilibrium equation for this element using Newton's second law, that is, the total balance of forces on element Δx. This leads to the equation of motion for the axial displacement

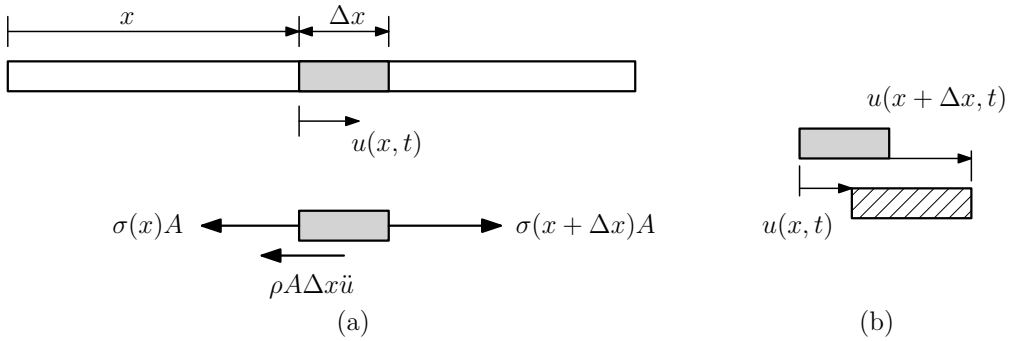


Figure 3.1: Longitudinal waves on a uniform bar, with constant cross-section A, constant density ρ and constant elastic modulus E: (a) balance of forces on a volume element of length Δx of the bar, and (b) stretching of a volume element in a non-uniform displacement field.

$u(x,t)$ from its initial position, which is

$$\rho A \Delta x \ddot{u} + A\sigma(x) - A\sigma(x + \Delta x) = 0 \tag{3.1}$$

As this is a continuous medium, the evolution of the stress between x and $x + \Delta x$ is described by a continuous function. Assuming a linear change in stress along the distance Δx the stress at the right side of the element is

$$\sigma(x + \Delta x) \approx \sigma(x) + \frac{\partial \sigma}{\partial x} \Delta x \tag{3.2}$$

Substituting the above relation into the equilibrium equation 3.1, and noting that the acceleration is $\ddot{u} = \partial^2 u / \partial t^2$, the dynamic force equilibrium condition becomes

$$\rho \frac{\partial^2 u}{\partial t^2} = \frac{\partial \sigma}{\partial x} \tag{3.3}$$

We now write the stress in terms of the displacement. For uniaxial stress and considering that the waves are elastic, we know that according to Hooke's law

$$\sigma = E\varepsilon = E \frac{\Delta l}{l} \tag{3.4}$$

where Δl is the change in length of the segment, which can be expressed in terms of the displacement field $u(x,t)$. The change in length is given by $\Delta l = \Delta u = u(x + \Delta x) - u(x)$, as can be seen in Figure 3.1b. Thus, the strain is

$$\varepsilon = \frac{u(x + \Delta x) - u(x)}{\Delta x} \approx \frac{\partial u}{\partial x} \tag{3.5}$$

Inserting this into the force equilibrium condition gives the final form of the wave equation:

$$\frac{\partial^2 u}{\partial t^2} = \frac{E}{\rho} \frac{\partial^2 u}{\partial x^2} \tag{3.6}$$

Longitudinal waves Longitudinal waves (sometimes also known as *axial waves* or *1-waves*) are waves in which the displacement of the medium is in the same direction as, or the opposite direction to, the direction of travel of the wave. Mechanical longitudinal waves are also called *compression waves* because they produce compression (and tension) when travelling through a medium, and *pressure waves* because they produce increases and decreases in

pressure. These waves can be described by a differential equation in the generic form

$$\frac{\partial^2 \Phi}{\partial t^2} = c^2 \frac{\partial^2 \Phi}{\partial x^2} \tag{3.7}$$

where Φ is the general displacement field and

$$c = \sqrt{\frac{E}{\rho}} \tag{3.8}$$

is called the wave speed or the speed of sound in the solid material. The reason for this will become clear when we discuss travelling wave solutions of this equation.

Other wave equations that can be derived in a similar manner for different types of longitudinal waves are briefly described in the following paragraphs.

Transverse waves on a stretched wire

We now consider a wire (string) with tension T and mass per unit length μ and assume it is infinitesimally thin and completely flexible, as shown in Figure 3.2a. To find the wave equation for this system we start by writing down the force equilibrium equation along the transverse direction, $F_y = ma_y$, for a small section of the string (see Figure 3.2b).

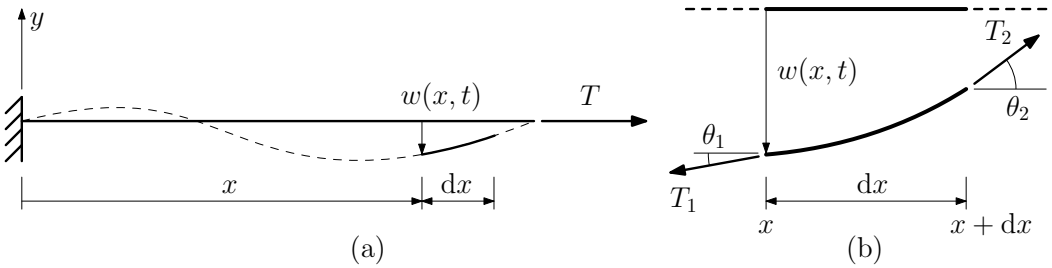


Figure 3.2: Longitudinal transverse wave on a stretched wire or string: (a) transverse displacement $w(x,t)$ of a stretched thin wire of mass per unit length μ under a tension force T and (b) forces acting on a small section of the wire.

Neglecting gravity, T_1 and T_2 are the end tensions in the string section. Noting that the transverse displacement is $w(x,t)$ the slope becomes

$$\frac{\partial w}{\partial x} = w'(x,t)$$

Recalling that this is a stretched wire, the slope is small and thus becomes approximately equal to the angles θ in Figure 3.2b. Consequently, the longitudinal components of the tension are essentially T_1 and T_2. Additionally, considering that the longitudinal motion of the wire can be neglected we know that there is no longitudinal acceleration and the longitudinal forces cancel, implying that $T_1 = T_2 = T$.

The same cannot, however, be said for the transverse components of the tension, which do not cancel. These forces are

$$\begin{aligned} T_{1y} &= -T \sin\theta_1 = -T w'(x,t) \\ T_{2y} &= T \sin\theta_2 = T w'(x + \mathrm{d}x, t) \end{aligned} \tag{3.9}$$

The net force then becomes

$$\begin{aligned} F_y &= Tw'(x+\mathrm{d}x,t) - Tw'(x,t) \\ &= T\mathrm{d}x\frac{w'(x+\mathrm{d}x,t)-w'(x,t)}{\mathrm{d}x} \\ &= T\mathrm{d}x\frac{\partial^2 w(x,t)}{\partial x^2} \end{aligned} \tag{3.10}$$

Since the mass of the section of wire is $\mu \mathrm{d}x$ the equilibrium of transverse forces becomes

$$\begin{aligned} &F_y = ma_y \\ \Rightarrow \quad &T\mathrm{d}x\frac{\partial^2 w}{\partial x^2} = \mu \mathrm{d}x\frac{\partial^2 w}{\partial t^2} \end{aligned} \tag{3.11}$$

Thus, the wave equation for a wire with mass per unit length μ under tensions force T, such as the one shown in Figure 3.2, is

$$\frac{\partial^2 w}{\partial t^2} = \frac{T}{\mu}\frac{\partial^2 w}{\partial x^2} \tag{3.12}$$

The general form of the longitudinal wave equation 3.7 can have different solutions, namely standing and travelling wave solutions. In the former, space and time are uncoupled, while in the latter they are part of the same function. Both these solutions will be discussed in detail in the following sections.

Torsional waves along uniform circular bars

For a uniform circular bar of shear modulus G and density ρ, such as the one shown in Figure 3.3a, the wave equation for angular (torsional) displacements $\theta(x,t)$ is

$$\frac{\partial^2 \theta}{\partial t^2} = \frac{G}{\rho}\frac{\partial^2 \theta}{\partial x^2} \tag{3.13}$$

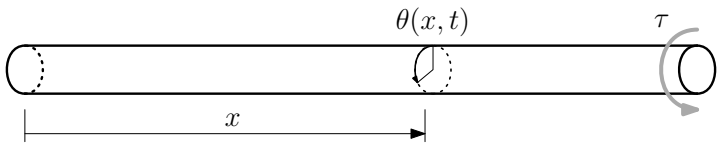


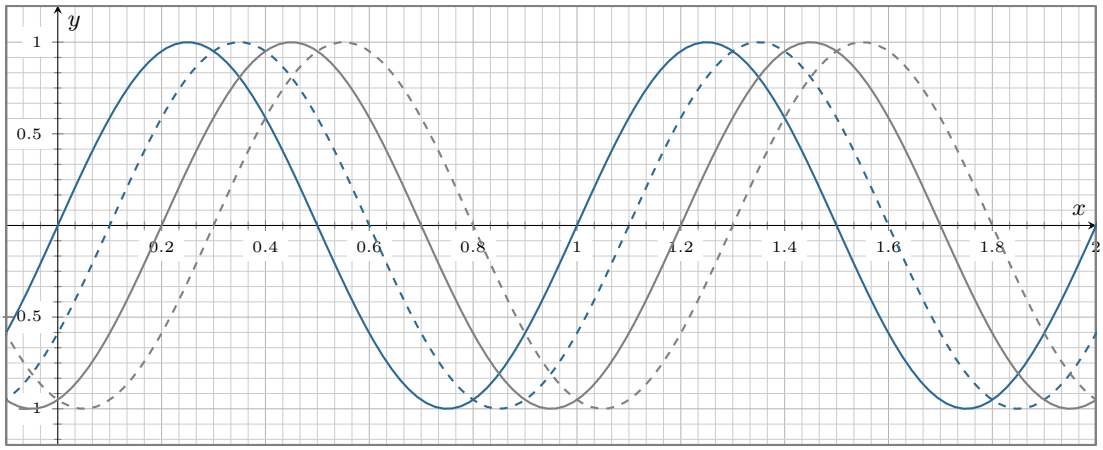
Figure 3.3: Longitudinal torsional wave: angular displacement $\theta(x,t)$ on a circular bar (shaft) under an applied torque τ.

3.2.1 Travelling wave solutions

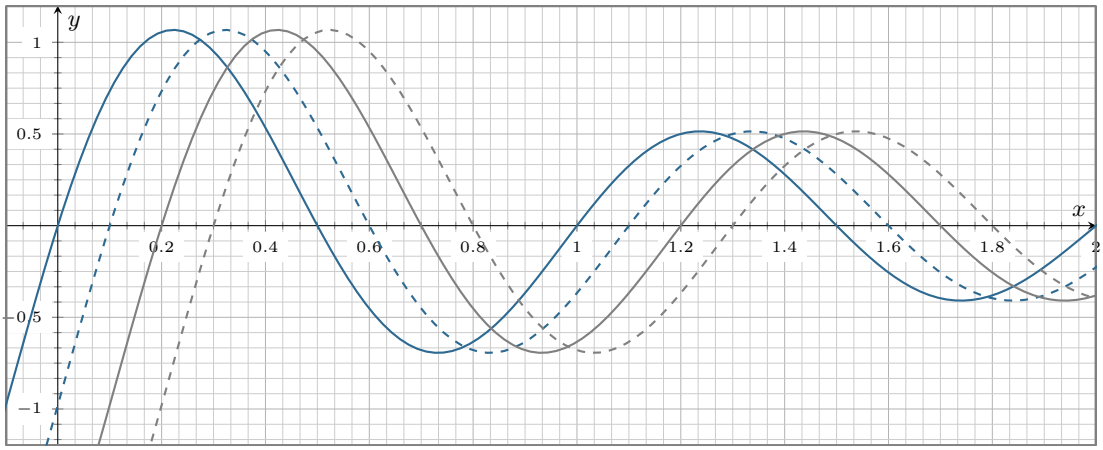
Travelling wave solutions of the partial differential equation 3.7 will be discussed in this section. In general terms, travelling waves are characterised by the fact that time and space are coupled and cannot be separated. A travelling sinusoidal wave can, for example, be described by the mathematical relation

$$u(x,t) = A\sin\left[\frac{2\pi}{\lambda}(x-ct)\right] \tag{3.14}$$

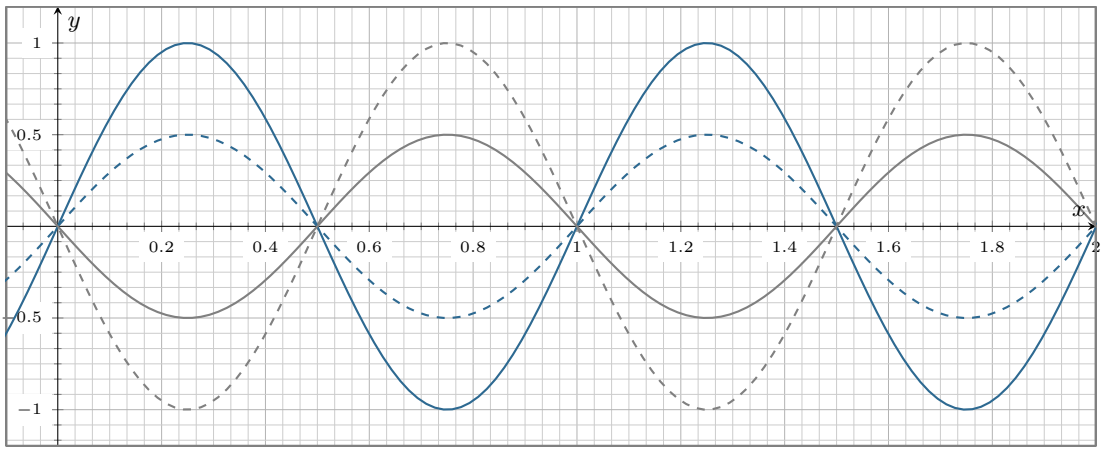
where A is the amplitude of the wave, λ is the wavelength and c the wave speed. To better understand this solution it is useful to look at a series of *snapshots* taken at different time instants. This is illustrated in more detail in Figure 3.4a. For each time t the solution is a sinusoidal function of x with period λ. However, tracing the location of a point where u has a fixed value,



(a) Travelling sine wave.



(b) Generic travelling wave function.



(c) Standing sine wave.

Figure 3.4: Longitudinal wave solutions: (a) propagation of a travelling sine wave, corresponding to equation 3.14 for $A = 1$ and $ct = 0$ (solid blue), 0.1 (dashed blue), 0.2 (solid grey) and 0.3 (dashed grey); (b) generic wave function travelling in the positive x direction. Different plots correspond to snapshots at increasing times; and (c) displacement function of a standing wave oscillation at different times.

for instance where the function has a maximum, we find that this point travels to the right with velocity c. This is why c is called the wave speed or speed of sound in the material.

Conversely, looking at the behaviour of a fixed point in space, we find that the solution oscillates in time with a frequency that is related to the wavelength by

$$\omega = \frac{2\pi c}{\lambda} \qquad \text{and} \qquad f = \frac{c}{\lambda} \tag{3.15}$$

Consequently, short wavelengths correspond to large frequencies and vice-versa.

To determine the wave speed c as a function of the parameters in the wave equation, we simply differentiate the solution 3.14 and substitute the result into the specific wave equation under consideration. For example, by inserting the travelling wave solution (equation 3.14) into the longitudinal wave equation 3.6 for the uniform bar we find

$$\begin{aligned} &\frac{\partial^2 u}{\partial t^2} = \frac{E}{\rho}\frac{\partial^2 u}{\partial x^2} \\ \Rightarrow \quad &-\frac{4\pi^2}{\lambda^2}c^2 A \sin\left[\frac{2\pi}{\lambda}(x-ct)\right] = \frac{E}{\rho}\left(-\frac{4\pi^2}{\lambda^2}\right) A \sin\left[\frac{2\pi}{\lambda}(x-ct)\right] \end{aligned} \tag{3.16}$$

Cancelling the constant terms on both sides of this equation we find the wave speed to be

$$c^2 = \frac{E}{\rho} \quad \Rightarrow \quad c = \sqrt{\frac{E}{\rho}} \tag{3.17}$$

Deriving the wave speeds for the torsional wave (see equation 3.13) and for the transverse wave (see equation 3.12) is a good additional exercise.

From the relation above it can be seen that for this type of equation the wave speed does not depend on the wavelength, only on material properties (density and Young's modulus). The propagation of such waves is called non-dispersive propagation. In the current scope, *dispersion* means that waves of different wavelengths travel at different speeds. This happens in flexural waves in beams, as will be discussed later.

For non-dispersive wave equations, a wave of any shape will travel without shape change at the same speed c. To demonstrate this, consider the general longitudinal waves described by the following functions

$$\begin{aligned} u(x,t) &= f(x-ct) \\ u(x,t) &= g(x+ct) \end{aligned} \tag{3.18}$$

where f and g are arbitrary shape functions. For the first function, the same function value is attained at an x value that increases like ct. We are thus dealing with a wave travelling with speed c in the positive x direction. Conversely, the second function describes a wave travelling with speed c in the negative x direction.

To demonstrate that the functions in equations 3.18 fulfil the wave equation 3.6, it is sufficient to insert them and account for the internal derivative. We then find again that $c^2 = E/\rho$ as previously. An example of a generic wave travelling in the positive x direction is illustrated in Figure 3.4b.

3.2.2 Standing wave solutions

Travelling waves are characterised by a shape that does not change but moves with a given speed — the wave speed c. An alternative form of solution, which describes a different type of wave, is the standing wave. This is a solution which does not move in space but oscillates in time only, as can be seen in Figure 3.4c. Mathematically, this is a reflection of the fact that space and time are uncoupled. For the longitudinal wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

the standing wave solution is the product of a space-dependent function and a separate time-dependent function, that is,

$$u(x,t) = X(x)T(t) \tag{3.19}$$

Inserting the above displacement field function into the wave equation gives

$$\begin{aligned} X\left(\frac{\mathrm{d}^2T}{\mathrm{d}t^2}\right) &= c^2T\left(\frac{\mathrm{d}^2X}{\mathrm{d}x^2}\right) \\ \frac{1}{T}\left(\frac{\mathrm{d}^2T}{\mathrm{d}t^2}\right) &= c^2\frac{1}{X}\left(\frac{\mathrm{d}^2X}{\mathrm{d}x^2}\right) = -\omega^2 \end{aligned} \tag{3.20}$$

Note that in the second equation one side depends only on x and the other only on t. As such, the only way these two terms can be equal is if they are both constant. We shall denote this constant by $-\omega^2$ because, as we will see later, it is the square of the frequency of the standing wave oscillation. Equating both the left and the right-hand side of equation 3.20 to a constant leads to the following two separate differential equations for the space and time functions:

$$\begin{aligned} \frac{\mathrm{d}^2T}{\mathrm{d}t^2} + \omega^2T &= 0 \\ \frac{\mathrm{d}^2X}{\mathrm{d}x^2} + \frac{\omega^2}{c^2}X &= 0 \end{aligned} \tag{3.21}$$

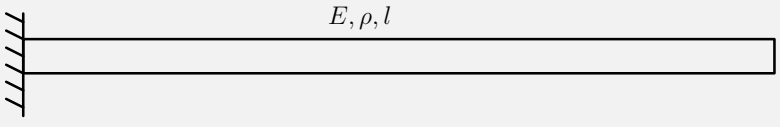
Thus, $X(x)$ and $T(t)$ are spatial and temporal oscillations, respectively, which can be given by

$$\begin{aligned} X(x) &= C_3\cos\left(\frac{\omega}{c}x\right) + C_4\sin\left(\frac{\omega}{c}x\right) \\ T(t) &= C_1\cos(\omega t) + C_2\sin(\omega t) \end{aligned} \tag{3.22}$$

An example with $T(t) = A\cos(\omega t)$, $X(x) = \sin(\omega x/c)$ is shown in Figure 3.4c.

The standing wave is a deformation shape $X(x)$ which oscillates with frequency ω. The constants C_i must be determined from initial and/or boundary conditions. Boundary conditions are of particular importance since, in the presence of boundaries, only certain frequencies and the corresponding deformation shapes are physically possible. These correspond to a discrete set of oscillation modes, each one characterised by its distinct frequency and mode shape function. A few examples will be used to discuss how we can determine the shape functions for given sets of boundary conditions.

Exercise 3.1 Determine the possible longitudinal vibration modes of a uniform bar of length l, density ρ and modulus E which has a clamped and a free end.



Solution In this solution we will assume the longitudinal wave can be described by a standing wave solution (steady vibrations). The displacement field is described by the function $u(x,t)$, the clamped end is at $x = 0$, the free end at $x = l$. These coordinates refer to the non-oscillating stationary bar.

The boundary condition at the clamped end is $u(0,t) = 0$, meaning that there is no displacement at this end. The boundary condition at the free end is $u'(l,t) = 0$, meaning that this is a stress-free end. We seek solutions of the wave equation

$$\frac{\partial^2u}{\partial t^2} = \frac{E}{\rho}\frac{\partial^2u}{\partial x^2} = c^2\frac{\partial^2u}{\partial x^2}$$

in the form of a standing wave $u(x,t) = f(x)g(t)$. Inserting this generic solution into the wave equation gives

$$\frac{\mathrm{d}^2 g}{\mathrm{d}t^2} + \omega^2 g = 0$$
$$\frac{\mathrm{d}^2 f}{\mathrm{d}x^2} + \frac{\omega^2}{c^2} f = 0$$

The general solution of the differential equation for $f(x)$ is

$$f(x) = C_1 \cos\left(\frac{2\pi x}{\lambda}\right) + C_2 \sin\left(\frac{2\pi x}{\lambda}\right) \qquad \text{where} \qquad \lambda = \frac{2\pi c}{\omega}$$

Constants C_1 and C_2 can be determined from the boundary conditions. The boundary conditions for the displacement field u, $u(0,t) = u'(l,t) = 0$, define boundary conditions for the function $f(x)$: $f(0) = 0$ and $f'(l) = 0$. From these boundary conditions it follows that

$$f(0) = C_1 = 0$$
$$f'(l) = \frac{2\pi}{\lambda}\left[-C_1 \sin\left(\frac{2\pi l}{\lambda}\right) + C_2 \cos\left(\frac{2\pi l}{\lambda}\right)\right] = \frac{2\pi}{\lambda} C_2 \cos\left(\frac{2\pi l}{\lambda}\right) = 0$$

Constant C_2 cannot be zero as this would correspond to the static case (no longitudinal wave). Consequently, we find from the second condition that

$$\cos\left(\frac{2\pi l}{\lambda}\right) = 0 \qquad \Rightarrow \qquad \frac{2\pi l}{\lambda} = \frac{\pi}{2} + n\pi \qquad \Leftrightarrow \qquad \lambda_n = \frac{4l}{1+2n} \qquad (n = 0, 1, \ldots)$$

This condition defines a number of possible wavelengths for the longitudinal wave. The corresponding possible oscillation frequencies are shown in Figure 3.5 and can be described by

$$\omega_n = \frac{2\pi c}{\lambda_n} = (1+2n)\frac{\pi c}{2l} = (1+2n)\frac{\pi}{2l}\sqrt{\frac{E}{\rho}}$$

The generic standing wave solution can then be written as

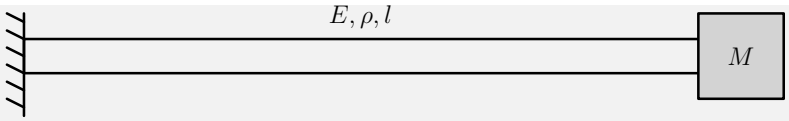
$$u(x,t) = C_2 \sin\left[(1+2n)\frac{\pi}{2l}x\right] \cos\left(\omega_n t - \phi\right)$$

where the constants C_2 and ϕ are the amplitude and phase angle of the longitudinal wave, which can be calculated from initial conditions such as initial velocity and displacement of known cross-sections on the bar.

■

The next example shows how to deal with more complicated boundary conditions. Consider, in this case, the same bar as in the previous example but now with a lumped mass attached to its free end.

Exercise 3.2 A steel bar of 3.5 m length and square cross section (width 50 mm) is clamped at one end and has a mass $M = 50$ kg attached at the other end. Calculate the first two frequencies of longitudinal vibration and determine the associated mode shapes. Compare with the frequencies obtained from a single-degree of freedom lumped parameter model. The material of the bar is steel with elastic modulus $E = 207$ GPa and density $\rho = 7830$ kg/m^3.



Solution The boundary condition at the right end of the bar is different in this example. At the left end it is the same as in the previous exercise, $u(0,t)=0$. At the right end, however, the force balance requires that the stress must balance the inertial force due to acceleration of the mass M. The stress and the inertial force are then, respectively,

$$\sigma(x=l) = E\left.\frac{\partial u}{\partial x}\right|_{x=l}$$
$$F = -M\left.\frac{\partial^2 u}{\partial t^2}\right|_{x=l}$$

Thus, combining these two conditions, the boundary condition at the right end of the bar becomes

$$\left.\frac{\partial u}{\partial x}\right|_{x=l} = -\frac{M}{EA}\left.\frac{\partial^2 u}{\partial t^2}\right|_{x=l}$$

Assuming again the solution can be described by the standard form of a standing wave, $u(x,t)=f(x)g(t)$, the wave equation yields the following two separate differential equations:

$$\frac{\mathrm{d}^2 g}{\mathrm{d}t^2} + \omega^2 g = 0$$
$$\frac{\mathrm{d}^2 f}{\mathrm{d}x^2} + \frac{\omega^2}{c^2} f = 0$$

Inserting these into the boundary conditions gives

$$f(0)g(t) = 0$$
$$f'(x=l)g(t) = -\frac{M}{EA} f(x=l)\ddot{g}(t) = \frac{M\omega^2}{EA} f(x=l)g(t)$$

The boundary condition $f(0)=0$ requires that $C_1=0$. The boundary condition at $x=l$ then becomes

$$\frac{2\pi}{\lambda} C_2 \cos\left(\frac{2\pi l}{\lambda}\right) = \frac{M\omega^2}{EA} C_2 \sin\left(\frac{2\pi l}{\lambda}\right)$$

or

$$\frac{2\pi EA}{M\omega^2\lambda} = \tan\left(\frac{2\pi l}{\lambda}\right)$$

We substitute $\lambda = 2\pi c/\omega$ to obtain the transcendental equation

$$\frac{EA}{Mc\omega} = \tan\left(\frac{\omega l}{c}\right)$$

which can be re-written in terms of the non-dimensional parameter $W = \omega l/c$ as

$$\frac{EAl}{Mc^2 W} = \left(\frac{m}{M}\right)\frac{1}{W} = \tan(W)$$

where $m = \rho Al$ is the mass of the bar. There is no analytical solution to this equation so we look for a graphical solution. For the parameters given, the mass ratio is $m/M = 1.37$ and it is possible to find from the graphical representation of both sides of the above equation (see Figure 3.6a) that the first two intersection points between the curves are located at approximately $W = 0.96$ and $W = 3.51$.

The corresponding frequencies are found with $l = 3.5$ m and $c = 5141$ m/s to be $\omega_1 = 1410$ rad/s, and $\omega_2 = 5171$ rad/s. From $\lambda = 2\pi c/\omega$, the associated mode wavelengths are found to be $\lambda_1 = 22.89$ m and $\lambda_2 = 6.24$ m. The corresponding mode shapes are shown in Figure 3.6b.

It is instructive to compare with the results of a lumped parameter model which neglects the mass of the bar and replaces the bar by a massless spring. In that case the spring constant is $k = EA/l$ and the frequency is given by $\omega = \sqrt{k/M}$. Substituting the parameters gives $\omega = 1719$ rad/s, which is about 20% higher than the true value. If we add half the beam mass to the inertial mass of the system then we get $\omega = \sqrt{k/(M + m/2)} = 1325$ rad/s, which is only 6% smaller. Thus for a rough-and-ready estimate of the bar dynamics the simple lumped parameter model does a reasonable job. As an additional exercise, explain why adding half the mass of the bar improved the result so much. ■

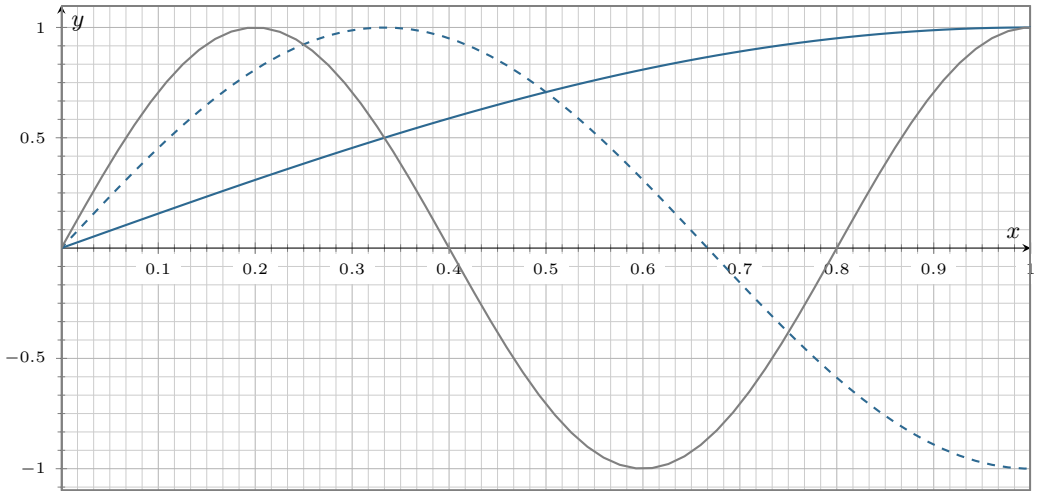


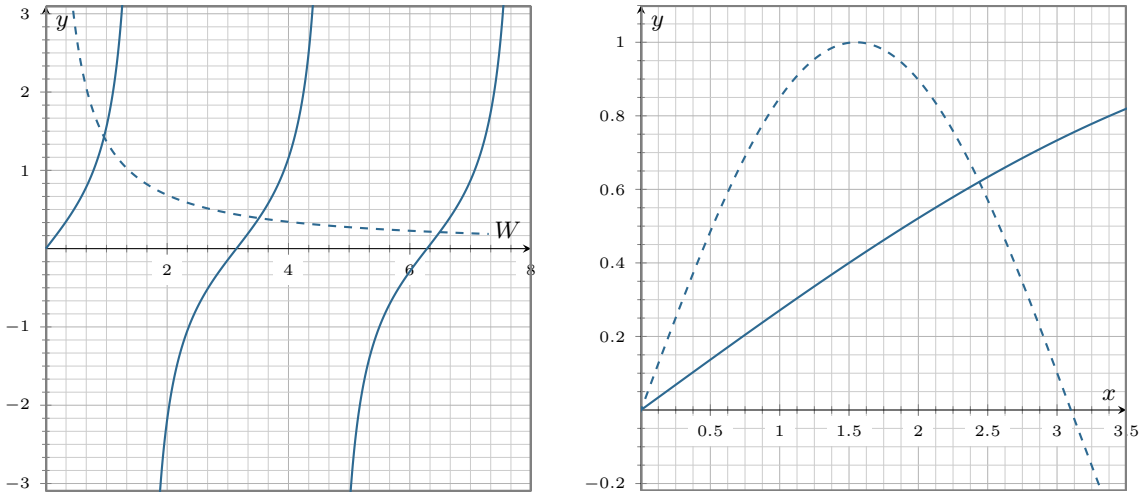
Figure 3.5: First three mode shapes for the longitudinal vibration of the uniform bar in Exercise 3.1. The solid blue line corresponds to $n = 0$, the dashed blue line to $n = 1$ and the solid grey line to $n = 2$.

3.3 Transverse vibration and flexural waves in beams

In the previous sections we have considered longitudinal waves. Now we consider transverse (bending) waves on uniform beams. In transverse waves the displacement is orthogonal to the axis of the beam, as can be seen from the example in Figure 3.7. An equation for the transverse displacement of the beam can be easily obtained by generalising the bending moment equations of beams, which was discussed in second year structural mechanics. This leads to a new type of wave equation that is different from previous examples. To follow on, the following assumptions are made:

- Plane cross-sections of the beam remain plane and at right angles to the neutral axis and
- The slope angle $\partial w/\partial x$ is always small.

These assumptions are often acceptable for small deflections of thin beams, such as when $L/\bar{w} > 10$, where $\bar{w}$ is the maximum deflection on the beam. The wave equation can then be



(a) $\tan(W)$ (solid line) and $m/(MW)$ (dashed line). (b) First (solid line) and second (dashed line) mode shapes.

Figure 3.6: Longitudinal wave on a bar with a lumped mass at one end: (a) graphical solution for the wave equation and (b) mode shapes of the bar for the first (solid line) and second (dashed line) natural frequencies.

derived by combining bending theory with considerations of equilibrium of dynamic forces and moments. Before going through the details of the derivations, it may be useful to revise structural mechanics concepts, in particular regarding sign conventions for forces and bending moments.

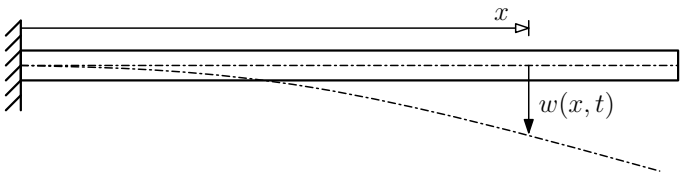


Figure 3.7: Transverse displacement (bending) of a clamped beam.

3.3.1 Euler-Bernoulli wave equation

To derive an equation for the time evolution of transverse displacements of a thin beam, $w(x,t)$, we start with the expression from beam elasto-statics which links the bending moment M to the displacement field, that is,

$$EI\frac{\partial^2 w}{\partial x^2} = -M(x,t) \tag{3.23}$$

Here, EI is the bending stiffness of the beam that has already appeared in the expression for the beam elastic potential energy (see for example equation 2.22). We now consider the equilibrium of forces and moments for a volume element of length Δx as shown in Figure 3.8. Moment balance — the requirement that the volume element does not rotate — gives

$$M(x) + A\tau(x)\frac{\Delta x}{2} + A\left[\tau(x) + \frac{\partial \tau}{\partial x}\Delta x\right]\frac{\Delta x}{2} - \left[M(x) + \frac{\partial M}{\partial x}\Delta x\right] = 0 \tag{3.24}$$

As such, for a small Δx the shear force on a cross section is

$$A\tau(x) = \frac{\partial M}{\partial x} = -EI\frac{\partial^3 w}{\partial x^3} \tag{3.25}$$

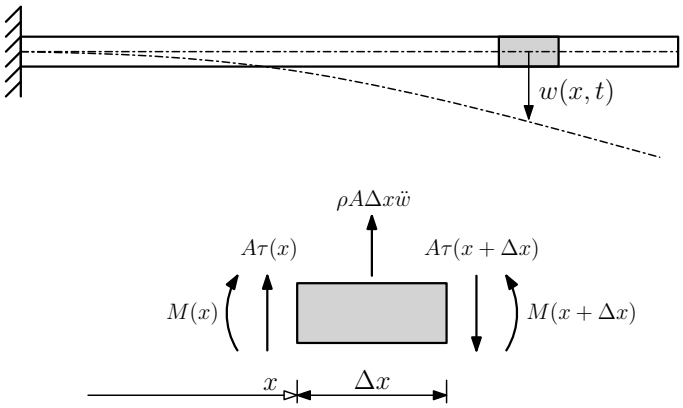


Figure 3.8: Transverse (bending) waves on slender structures: balance of forces and bending moments acting on a volume element of a uniform long beam.

Now we use the requirement of equilibrium of forces, including inertial forces,

$$\rho A \Delta x \frac{\partial^2 w}{\partial t^2} + A[\tau(x) - \tau(x + \Delta x)] = 0 \tag{3.26}$$

By Taylor expanding the stress, the above expression can be rewritten as

$$\rho A \frac{\partial^2 w}{\partial t^2} - A \frac{\partial \tau}{\partial x} = 0 \tag{3.27}$$

and, after substituting equation 3.25, we obtain Euler-Bernoulli's differential wave equation for a uniform beam

$$\rho A \frac{\partial^2 w}{\partial t^2} + EI \frac{\partial^4 w}{\partial x^4} = 0$$

Euler-Bernoulli equation Euler-Bernoulli's wave equation describes the relationship between the beam's deflection and the applied load. In free lateral vibration, when the transverse load is zero, this equation is written as

$$\rho A \frac{\partial^2 w}{\partial t^2} + EI \frac{\partial^4 w}{\partial x^4} = 0 \tag{3.28}$$

where EI is the bending stiffness of the beam, ρ is the material density and A is the area of the cross-section. In the form presented above it is assumed that the beam is uniform and homogeneous. In the more general case, where stiffness and density can change, the equation should be written as

$$\frac{\partial^2 w}{\partial t^2} + \frac{\partial^2}{\partial x^2}\left(\frac{EI}{\rho A}\frac{\partial^2 w}{\partial x^2}\right) = 0 \tag{3.29}$$

3.3.2 Travelling wave solutions of the Euler-Bernoulli equation

We first look for travelling wave solutions to Euler-Bernoulli's wave equation for uniform beams, as shown in equation 3.28, analogous to the ones we obtained for the longitudinal wave equation 3.7. Equation 3.14 is a solution in the standard form of a travelling sine wave with amplitude B, wavelength λ and wave speed c. As such,

$$w(x,t) = B \sin\left[\frac{2\pi}{\lambda}(x - ct)\right] \tag{3.30}$$

can be used as the general form for a travelling transverse wave. Differentiating and inserting into Euler-Bernoulli's equation gives

$$\begin{aligned} &-\rho A\left(\frac{2\pi c}{\lambda}\right)^2 w(x,t) + EI\left(\frac{2\pi}{\lambda}\right)^4 w(x,t) = 0 \\ \Rightarrow \quad &\left(\frac{2\pi}{\lambda}\right)^2 = \frac{\rho A c^2}{EI} \quad \Rightarrow \quad c = \left(\frac{2\pi}{\lambda}\right)\sqrt{\frac{EI}{\rho A}} \end{aligned} \tag{3.31}$$

From this it can be seen that the wave speed depends on the wavelength (they are inversely proportional). This is called dispersive wave propagation. The frequency of the wave is again related to the wavelength by

$$\omega = \frac{2\pi c}{\lambda} \qquad \text{and} \qquad f = \frac{c}{\lambda}$$

and the wave vector is again

$$k = \frac{\omega}{c} = \frac{2\pi}{\lambda}$$

3.3.3 Standing wave solutions of the Euler-Bernoulli equation

As for the longitudinal wave equation, an alternative form of solution is again the standing wave solution, where the space and time functions are uncoupled. This solution is assumed as the following product of a sinusoidal oscillation in time and a function of the space coordinate x:

$$w(x,t) = \phi(x)\cos(\omega t) \tag{3.32}$$

Inserting into the wave equation gives

$$\left[EI\frac{\mathrm{d}^4\phi}{\mathrm{d}x^4}(x) - \rho A\omega^2\phi(x)\right]\cos(\omega t) = 0 \tag{3.33}$$

which must be fulfilled for all t. Consequently,

$$\frac{\mathrm{d}^4\phi}{\mathrm{d}x^4}(x) - \frac{\rho A\omega^2}{EI}\phi(x) = 0 \tag{3.34}$$

To solve this equation, it is convenient to write it in terms of the wave vector k which is related to the frequency by

$$k^4 = \frac{\rho A\omega^2}{EI} \tag{3.35}$$

So the equation for the space dependent function becomes

$$\frac{\mathrm{d}^4\phi}{\mathrm{d}x^4}(x) - k^4\phi(x) = 0 \tag{3.36}$$

This fourth order differential equation on $\phi(x)$ has the general solution

$$\phi(x) = C_1\cos(kx) + C_2\sin(kx) + C_3\cosh(kx) + C_4\sinh(kx) \tag{3.37}$$

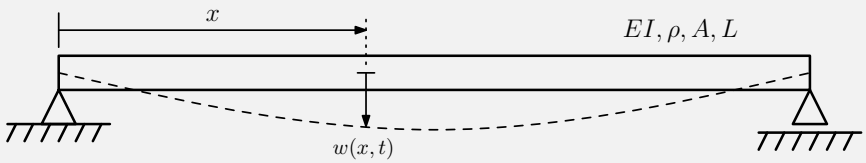
which is the general form for a standing flexural wave on a uniform beam. Different from the longitudinal wave equation, where the standing wave solution is given by equation 3.22, the differential equation for the mode shape function is here of fourth order. Consequently, the solution contains four constants which must be determined from the boundary conditions, for example from support conditions at the beam ends. To determine the transverse vibration behaviour of a beam, additional boundary conditions are necessary. These will be discussed in detail in the following paragraphs and are listed below.

Boundary conditions The standing and travelling wave solutions to Euler-Bernoulli's equation contains four constants which must be determined from the boundary conditions. The simplest case is encountered when the beam ends are either free, simply supported or clamped, as in figures (a) to (c) below. The mathematical conditions describing each one of these are listed in the following table:

Type of boundary condition	Figure	Condition
Clamped	(a)	$w = 0 \;\Rightarrow\; \phi = 0$ $w' = 0 \;\Rightarrow\; \phi' = 0$
Simply supported	(b)	$w = 0 \;\Rightarrow\; \phi = 0$ $M = 0 \;\Rightarrow\; w'' = 0 \;\Rightarrow\; \phi'' = 0$
Free	(c)	$M = 0 \;\Rightarrow\; w'' = 0 \;\Rightarrow\; \phi'' = 0$ $\tau = 0 \;\Rightarrow\; w''' = 0 \;\Rightarrow\; \phi''' = 0$

(a) (b) (c)

Exercise 3.3 Determine the flexural vibration modes of a uniform beam of length L, density ρ and bending stiffness EI, which is simply supported at both ends (see figure below).



Solution In this example we have to evaluate the vibration modes of a beam that is simply supported at both ends. We assume the left end to be at $x = 0$ and the right end at $x = L$. Consequently, the boundary conditions are

$$w(x = 0, t) = w''(x = 0, t) = 0$$
$$w(x = L, t) = w''(x = L, t) = 0$$

We assume the solution in the form $w(x, t) = \varphi(x)\cos(\omega t)$. Inserting in the Euler-Bernoulli equation yields

$$\left[EI\frac{\partial^4 \varphi}{\partial x^4}(x) - A\rho\omega^2\varphi(x)\right]\cos(\omega t) = 0$$

and the equation for the space-dependent mode shape function becomes

$$\frac{\mathrm{d}^4\varphi}{\mathrm{d}x^4}(x) - k^4\varphi(x) = 0$$

with $k^4 = \dfrac{\rho A \omega^2}{EI}$

The general solution of this equation and the derivatives of the general solution are

$$\varphi(x) = C_1 \cos(kx) + C_2 \sin(kx) + C_3 \cosh(kx) + C_4 \sinh(kx)$$
$$\varphi'(x) = k\,[-C_1 \sin(kx) + C_2 \cos(kx) + C_3 \sinh(kx) + C_4 \cosh(kx)]$$
$$\varphi''(x) = k^2\,[-C_1 \cos(kx) - C_2 \sin(kx) + C_3 \cosh(kx) + C_4 \sinh(kx)]$$

The boundary conditions at the simply supported beam ends are

$$\varphi(x=0) = \varphi''(x=0) = 0$$
$$\varphi(x=L) = \varphi''(x=L) = 0$$

Inserting the derivatives of the general solution and its derivatives, the boundary conditions at $x = 0$ yields

$$\varphi(0) = C_1 + C_3 = 0$$
$$\varphi''(0) = k^2(-C_1 + C_3) = 0$$

Solving for the unknown constants we get $C_1 = C_3 = 0$ and the solution to the wave equation becomes

$$\varphi(x) = C_2 \sin(kx) + C_4 \sinh(kx)$$
$$\varphi''(x) = k^2\,[-C_2 \sin(kx) + C_4 \sinh(kx)]$$

The boundary conditions at $x = L$ produce the following system of equations for the two unknown constants C_2 and C_4

$$C_2 \sin(kL) + C_4 \sinh(kL) = 0$$
$$-C_2 \sin(kL) + C_4 \sinh(kL) = 0$$

For this system of equations to have a nontrivial solution the determinant of the matrix of coefficients must be zero, that is,

$$2 \sin(kL) \sinh(kL) = 0$$

Since the sinh function has only one zero at $x = 0$, the zeros of this equation are those of the sine function,

$$\sin(kL) = 0 \qquad \text{with} \qquad kL = n\pi$$

It also follows that $C_4 = 0$, so the solutions are pure sines. The first three mode shapes are shown in Figure 3.9. It can be seen that n simply counts the number of half waves within the length of the bar. The corresponding oscillation frequencies can be obtained from

$$k^4 = \frac{A\rho\omega^2}{EI} \qquad \text{and} \qquad \omega^2 = \frac{EIk^4}{A\rho} = \frac{EI(kL)^4}{A\rho L^4} = \frac{\pi^4 EI}{A\rho L^4} n^4$$
$$\Rightarrow \quad \omega = \left(\frac{\pi n}{L}\right)^2 \sqrt{\frac{EI}{A\rho}}$$

■

The boundary conditions described above are the most common. There are, however, other possibilities and we now turn to more general boundary conditions at the beam ends. In general,

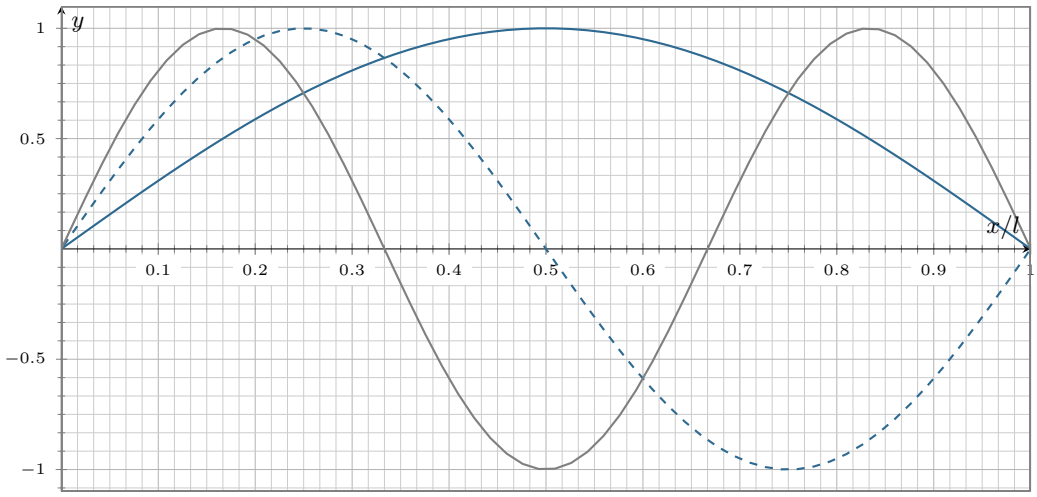


Figure 3.9: Deflection function $y = w(x,t)/C_2$ for a transverse wave on a simply supported long beam for $n = 1$ (solid blue line), $n = 2$ (dashed blue line) and $n = 3$ (solid grey line).

forces may act on the beam ends and/or there may be masses attached as illustrated in Figure 3.10. In these cases, we need to consider the balance of forces acting on the beam end. This force balance requires that the resultant of the force acting on the beam end from the *outside* in the direction perpendicular to the beam and the shear force at the beam end, must add up to zero. Thus, the force on the beam end must be symmetrical to the shear force, that is,

$$-EI \left.\frac{\partial^3 w}{\partial x^3}\right|_{\text{Boundary}} = F_{\text{Boundary}} \tag{3.38}$$

In this equation, the boundary force can be: (i) a spring force (depends on boundary displacement), (ii) an inertial force (depends on boundary acceleration) or (iii) a damping force (depends on boundary velocity). In either case the sign convention is that a force in direction of displacement w is counted as positive. We illustrate the application of equation 3.38 for two cases, namely a beam with one and two spring supports.

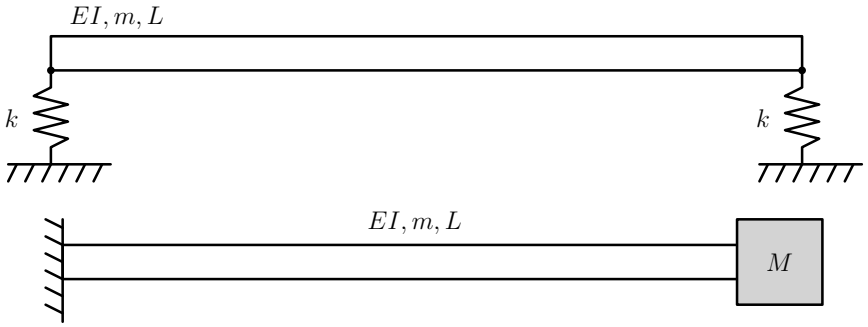
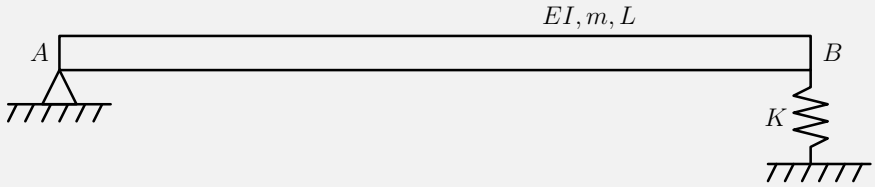


Figure 3.10: Two examples of general boundary conditions at the ends of a uniform long beam. Beam supported on springs (top image), which can represent a flexible foundation, and beam with attached mass (bottom image).

Exercise 3.4 A steel pipe is 10 m long, has a 1.2 m diameter and a wall thickness of 8 mm. End A sits on a rigid base (simply supported) and end B is flexibly supported on a structure with an effective vertical stiffness $K = 1.2 \times 10^6$ N/m. The pipe material has elastic modulus

$E = 20.7 \times 10^{10}$ N/m^2 and density $\rho = 7840$ kg/m^3.

Determine the first two frequencies of flexural vibration of the pipe. What is the frequency obtained by considering the pipe to be rigid?



Solution The first step is to to list the boundary conditions for the pipe. We choose the coordinate system such that the simply supported end is at $x = 0$. The displacement, and therefore also the mode shape function, must fulfil the corresponding boundary conditions, which are: zero vertical displacement (no deflection) and zero bending moment. These can be mathematically stated as

$$w(x = 0, t) = w''(x = 0, t) = 0$$
$$\varphi(x = 0) = \varphi''(x = 0) = 0$$

At the end with the flexible support — connected to the spring — there is no bending moment, thus $w''(x = L) = 0$. However, the spring exerts a force which counteracts the displacement of the beam end. Consequently, the boundary condition at $x = L$ is

$$-EI \left.\frac{\partial^3 w}{\partial x^3}\right|_{x=L} = -Kw(x = L, t)$$
$$\Rightarrow \quad EI\varphi'''(x = L) = K\varphi(x = L)$$

The general solution for the mode shape function is

$$\varphi(x) = C_1 \cos(kx) + C_2 \sin(kx) + C_3 \cosh(kx) + C_4 \sinh(kx)$$

and the boundary conditions at the simply supported end $x = 0$ require that $C_1 = C_3 = 0$, as can be seen from the calculations in example 3.3. As such, the solution and the corresponding first three derivatives are

$$\varphi(x) = C_2 \sin(kx) + C_4 \sinh(kx)$$
$$\varphi'(x) = k\,[C_2 \cos(kx) + C_4 \cosh(kx)]$$
$$\varphi''(x) = k^2\,[-C_2 \sin(kx) + C_4 \sinh(kx)]$$
$$\varphi'''(x) = k^3\,[-C_2 \cos(kx) + C_4 \cosh(kx)]$$

The boundary conditions at the flexible support become

$$\varphi''(x = L) = 0$$
$$\varphi'''(x = L) - \frac{K}{EI}\varphi(x = L) = 0$$

which implies that

$$-C_2 \sin(kL) + C_4 \sinh(kL) = 0$$
$$k^3\,[-C_2 \cos(kL) + C_4 \cosh(kL)] - \frac{K}{EI}\,[C_2 \sin(kL) + C_4 \sinh(kL)] = 0$$

This is a system of two equations for the two unknown constants C_2 and C_4. Rearranging the terms and writing the resulting system of equations as

$$\begin{cases} -C_2 \sin(kL) + C_4 \sinh(kL) = 0 \\ C_2 \left[\frac{KL^3}{EI} \sin(kL) + (kL)^3 \cos(kL) \right] + C_4 \left[\frac{KL^3}{EI} \sinh(kL) - (kL)^3 \cosh(kL) \right] = 0 \end{cases}$$

This system of equations has the solvability condition

$$\det \begin{vmatrix} -\sin(kL) & \sinh(kL) \\ \frac{KL^3}{EI} \sin(kL) + (kL)^3 \cos(kL) & \frac{KL^3}{EI} \sinh(kL) - (kL)^3 \cosh(kL) \end{vmatrix} = 0$$

$$\Rightarrow \quad (kL)^3 \left[\sin(kL)\cosh(kL) - \cos(kL)\sinh(kL) \right] - 2\frac{KL^3}{EI} \sin(kL)\sinh(kL) = 0$$

This is a transcendental equation for which there is no analytical solution. To obtain a numerical solution, we use the notations $(kL) = \beta$ and $2KL^3/(EI) = \lambda$. With the parameters provided and using $I = \pi r^3 d$ for a thin circular tube we find $\lambda = 2.13$. The equation becomes

$$\beta^3 \left[\sin(\beta)\cosh(\beta) - \cos(\beta)\sinh(\beta) \right] - 2.13 \sin(\beta)\sinh(\beta) = 0$$

There is no simple graphical method for determining the roots of this equation. Instead, it is possible to use the MATLAB `fzero` command. The first two roots are

```
z=fzero('x^3*(sin(x)*cosh(x)-cos(x)*sinh(x))-2.13*sin(x)*sinh(x)',2)
z=1.3302
z=fzero('x^3*(sin(x)*cosh(x)-cos(x)*sinh(x))-2.13*sin(x)*sinh(x)',3)
z=3.9443
```

The frequencies are obtained as in the previous example from

$$k^4 = \frac{A\rho\omega^2}{EI}$$
$$\omega^2 = \frac{EIk^4}{A\rho} = \frac{EI(kL)^4}{A\rho L^4}$$

This gives for the first two frequencies with $(k_1 L) = 1.33$ and $(k_2 L) = 3.94$ the values $\omega_1 = 38.6$ rad/s and $\omega_2 = 316.3$ rad/s. For comparison we can create a lumped parameter model where we assume the bar as rigid. For rotation around the bar end we have the equation of motion

$$I\ddot{\theta} + KL^2\theta = 0 \qquad \text{with} \qquad I = \frac{1}{3}mL^2$$
$$\omega^2 = \frac{3K}{M}$$

which, for a mass $M = 2364$ kg, gives a frequency of $\omega = 39.01$ rad/s. Thus the lumped parameter model, which assumes the pipe to be rigid, works perfectly well. The approximate frequency is about 1% higher than the correct one, as should be expected according to Rayleigh's theorem. ■

The next exercise is a variant of the same problem, where both beam ends are now supported by springs.

Exercise 3.5 A steel pipe is 10 m long, has a 1.2 m diameter and a wall thickness of 8 mm. Both ends are flexibly supported on structures with effective vertical stiffness $K = 1.2 \times 10^6$ N/m. The pipe material has elastic modulus $E = 20.7 \times 10^{10}$ N/m^2 and density $\rho = 7840$ kg/m^3.

Determine the first two frequencies of flexural vibration of the pipe. What is the frequency

obtained by considering the pipe to be rigid?

Solution This example is very similar to the previous exercise. However, we now look at symmetrical vibration modes so it is more convenient to take the origin of the coordinate system at the centre of the beam. This ensures that the boundary conditions and the mode shape functions will be symmetric with respect to the origin. The beam boundaries are then at $x = -L/2$ and $x = L/2$ with the corresponding boundary conditions

$$EI\varphi'''(x = -L/2) = K\varphi(x = -L/2) \qquad \text{and} \qquad \varphi''(-L/2) = 0$$
$$EI\varphi'''(x = L/2) = K\varphi(x = L/2) \qquad \text{and} \qquad \varphi''(L/2) = 0$$

The general solution for the mode shape function is

$$\varphi(x) = C_1\cos(kx) + C_2\sin(kx) + C_3\cosh(kx) + C_4\sinh(kx)$$

but the solutions are required to be symmetrical, so $C_2 = C_4 = 0$. The solution is now

$$\varphi(x) = C_1\cos(kx) + C_3\cosh(kx)$$
$$\varphi'(x) = k\left[-C_1\sin(kx) + C_3\sinh(kx)\right]$$
$$\varphi''(x) = k^2\left[-C_1\cos(kx) + C_3\cosh(kx)\right]$$
$$\varphi'''(x) = k^3\left[C_1\sin(kx) + C_3\sinh(kx)\right]$$

Due to symmetry, the boundary conditions at both ends are now the same, that is,

$$\varphi''(x = L/2) = 0$$
$$\varphi'''(x = L/2) - \frac{K}{EI}\varphi(x = L/2) = 0$$

The resulting system of equations can be written as

$$\begin{cases} -C_1\cos(kL/2) + C_3\cosh(kL/2) = 0 \\ C_1\left[-\frac{K}{EI}\cos(kL/2) + k^3\sin(kL/2)\right] + C_3\left[-\frac{K}{EI}\cosh(kL/2) + k^3\sinh(kL/2)\right] = 0 \end{cases}$$

and the solvability condition for this system of equations is

$$\det\begin{vmatrix} -\cos(kL/2) & \cosh(kL/2) \\ \frac{KL^3}{8EI}\cos(kL/2) - (kL/2)^3\sin(kL/2) & \frac{KL^3}{8EI}\cosh(kL/2) - (kL/2)^3\sinh(kL/2) \end{vmatrix} = 0$$

or

$$(kL/2)^3\left[\sin(kL/2)\cosh(kL/2) + \cos(kL/2)\sinh(kL/2)\right] - \frac{KL^3}{4EI}\cos(kL/2)\cosh(kL/2) = 0$$

Using the notations $(kL/2) = \beta$ and $KL^3/(4EI) = \lambda = 0.266$ we can use the MATLAB `fzero` command as previously. The first two roots are

```
z=fzero('x^3*(sin(x)*cosh(x)+cos(x)*sinh(x))-0.266*cos(x)*cosh(x)',0.3)
z=0.6012
z=fzero('x^3*(sin(x)*cosh(x)+cos(x)*sinh(x))-0.266*cos(x)*cosh(x)',2)
z=2.3750
```

The frequencies are obtained as in the previous example from

$$k^4 = \frac{A\rho\omega^2}{EI} \qquad \text{and} \qquad \omega^2 = \frac{EIk^4}{A\rho} = \frac{EI(kL/2)^4}{A\rho(L/2)^4}$$

This gives for the first two frequencies with $(k_1 L/2) = 0.6012$ and $(k_2 L/2) = 2.3750$ the values $\omega_1 = 31.54$ rad/s and $\omega_2 = 490.3$ rad/s. For comparison the lumped parameter model simply gives

$$\omega^2 = \frac{2K}{M}$$

which gives with a mass $M = 2364$ kg a frequency $\omega = 31.86$ rad/s. Thus the lumped parameter model, which assumes the pipe to be rigid, works perfectly well again, and Rayleigh's theorem still holds (note that the simplified model frequency is slightly high). ■

In the above examples it is interesting to see how the complex and lengthy calculations with trigonometric and hyperbolic functions (from beam dynamics and boundary conditions) manage to produce, in the end, the simple lumped parameter result, which should be possible to obtain very quickly. What follows from this observation can be summarised as follows:

a) For simple examples, such as finding the fundamental modes of rigid structures, it is more efficient to use simple models, such as the ones discussed in previous courses.

b) If more information is necessary, e.g. in order to assess higher order vibration modes, one has to make a much more substantial effort. However, the simple model results should always be used to check whether the detailed solutions are correct.

3.4 Applications

In this section we again present and analyse real-life dynamic systems. We apply the concepts we learned in previous sections of this chapter to look at how sound waves propagate in an ideal gas, and at the centrifugal stiffening effect of a beam attached to a rotating hub.

3.4.1 Propagation of sound waves

Sound can propagate through media such as air, water and solids as longitudinal waves. Sound waves are usually generated by a sound source, such as the vibrating diaphragm of a stereo speaker. The sound source creates vibrations in the surrounding medium. As the source continues to vibrate the medium, the vibrations propagate away from the source at the speed of sound, thus forming the sound wave. At a fixed distance from the source, the pressure, velocity, and displacement of the medium vary in time. At an instant in time, the pressure, velocity, and displacement vary in space. Note that the particles of the medium do not travel with the sound wave. This is intuitively obvious for a solid, and the same is true for liquids and gases. The vibrations of particles in the gas or liquid transport the vibrations, while the average position of the particles over time does not change. The behaviour of sound propagation is generally affected by the following:

- The relationship between the density and the pressure of the medium. This relationship, which also depends on the temperature, determines the speed of sound within the medium.
- The motion of the medium itself. If the medium is moving, this movement may increase or decrease the absolute speed of the sound wave depending on the direction of the movement. For example, sound moving through wind will have its speed of propagation increased by

the speed of the wind if the sound and wind are moving in the same direction. If they are moving in opposite directions, the speed of the sound wave will be decreased by the speed of the wind.

- The viscosity of the medium. Medium viscosity determines the rate at which sound is attenuated. For many media, such as air or water, attenuation due to viscosity is negligible.

Longitudinal sound waves are waves of alternating pressure deviations from the equilibrium pressure, causing local regions of compression and rarefaction. As such, we start our analysis of sound waves by going back to the equation for longitudinal waves in a bar, which was derived as

$$\frac{\partial^2 u}{\partial t^2} = \frac{E}{\rho}\frac{\partial^2 u}{\partial x^2}$$

This equation applies to any waves in a material that has finite elasticity and density — irrespective of whether the material is a solid, a liquid, or a gas. In addition, the thickness of the bar does not matter, hence the equation also applies to plane waves in an infinite medium. In the following paragraphs it is assumed that the medium is an ideal gas, and we derive the wave equation for that case — that is, the equation for plane sound waves propagating in air.

Sound waves in an ideal gas

We envisage instead of the bar in Figure 3.1 a gas in a tube. $u(x,t)$ is now the average displacement of the gas molecules. We again consider a small volume (see Figure 3.11) and assume that the density of the gas in this volume does not change much due to the sound wave (i.e. $\rho \approx \rho_0$). The force balance on the small gas volume is

$$\rho A \Delta x \ddot{u} + Ap(x + \Delta x) - Ap(x) = 0 \tag{3.39}$$

where $p(x)$ is the pressure and A is the area of the cross section of the tube. We write the pressure acting on the left on this volume as

$$p(x + \Delta x) \approx p(x) + \frac{\partial p}{\partial x}\Delta x \tag{3.40}$$

Thus, the dynamic force equilibrium condition becomes

$$\rho_0 \frac{\partial^2 u}{\partial t^2} = -\frac{\partial p}{\partial x} \tag{3.41}$$

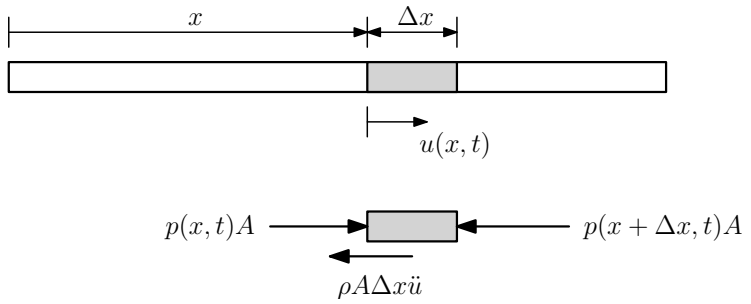


Figure 3.11: Propagation of sound waves: force balance for a volume element in an ideal gas.

So far the argument is exactly the same as for the longitudinal wave in a solid bar. Now however, instead of Hooke's law it is necessary to consider the equation of state of the gas in order to relate the pressure gradient to the displacement field $u(x,t)$. We assume that the expansion and compression of the gas occur in an adiabatic manner. Hence, the dynamic force equilibrium condition becomes

$$\frac{p}{p_0} = \left(\frac{V_0}{V}\right)^{\gamma} \qquad \Rightarrow \qquad \frac{p - p_0}{p_0} \approx -\gamma\left(\frac{V - V_0}{V_0}\right) \tag{3.42}$$

where p_0 and V_0 are reference values of the pressure and volume without the perturbation due to the acoustic wave. It is also assumed that the perturbation is small. The relative volume change is due to the stretching of the considered volume, that is,

$$\frac{V - V_0}{V_0} = \frac{u(x + \Delta x) - u(x)}{\Delta x} \approx \frac{\partial u}{\partial x} \tag{3.43}$$

Combining equations 3.41 to 3.43 we find that

$$\frac{\partial p}{\partial x} = -p_0 \gamma \frac{\partial}{\partial x}\left(\frac{V}{V_0}\right) = -p_0 \gamma \frac{\partial^2 u}{\partial x^2} \tag{3.44}$$

The wave equation then becomes

$$\frac{\partial^2 u}{\partial t^2} = \frac{p_0 \gamma}{\rho_0} \frac{\partial^2 u}{\partial x^2} \tag{3.45}$$

which for the displacement can be converted into an equation for the sound pressure by taking the derivative with respect to x on both sides and substituting the previous relation, leading to

$$\frac{\partial^2 p}{\partial t^2} = c^2 \frac{\partial^2 p}{\partial x^2} \tag{3.46}$$

where the wave speed — the speed of sound in air — is

$$c = \sqrt{\frac{p_0 \gamma}{\rho_0}} \tag{3.47}$$

For an ideal gas

$$pV = nRT \qquad \Rightarrow \qquad \frac{p_0}{\rho_0} = \frac{nRT}{M} = \frac{RT}{m} \tag{3.48}$$

where m is the molar mass. As such, the sound speed in an ideal gas depends only on the molar mass, the temperature and the adiabatic exponent,

$$c = \sqrt{\frac{RT\gamma}{m}} \tag{3.49}$$

As stated earlier, the above derivation of the wave equation applies to planar sound waves moving in the x direction. In three dimensions, the equation generalises to

$$\frac{\partial^2 p}{\partial t^2} = c^2 \left(\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 p}{\partial z^2}\right) \tag{3.50}$$

Sound levels in the workplace are subject to health and safety (H&S) regulations[1]. To calculate the transmission of plane propagating and standing sound waves we can use methods that are very similar to those we have used for longitudinal waves in Section 3.2. This is illustrated below by an example where we evaluate the transmission of sound through a rigid plate.

Sound intensity

Sound pressure or acoustic pressure is the local pressure deviation from the ambient — average, or equilibrium — atmospheric pressure, caused by a sound wave. In air, sound pressure can be measured using a microphone, and in water with a hydrophone. The SI unit of sound pressure is the pascal (Pa). Hearing is sensitive to changes in sound pressure.

[1] See, for example: https://goo.gl/d5mSpt

Sound Sound is usually characterised by the time average of the square pressure deviation, which corresponds to the energy transmitted by the acoustic wave. This can be mathematically described by the relation

$$\left\langle (p - p_0)^2 \right\rangle = \frac{1}{T} \int_t^{t+T} [p(t) - p_0]^2 \, \mathrm{d}t \tag{3.51}$$

The sound (or acoustic) pressure level (SPL) is a logarithmic measure of the effective pressure of a sound wave relative to a reference value, and has units of deciBell (dB). This is commonly used to measure the intensity (or strength) of sound and can be defined as

$$L_p = 10 \log_{10} \left[\frac{\left\langle (p - p_0)^2 \right\rangle}{p_{\mathrm{ref}}^2} \right] \tag{3.52}$$

where the reference pressure p_{ref} is usually defined by the threshold of hearing.

Some examples of sound pressure levels are listed in Table 3.1. Note that for point or line sources the SPL depends on distance. For a line source it decreases with the inverse distance and for a point source it decreases with the inverse square of the distance. Thus, giving an SPL in these cases without specifying distance from the sound source is meaningless.

Table 3.1: Typical sound pressure levels for different sources.

Example	Distance from source [m]	Sound pressure level L_p [dB-SPL]
Jet aircraft	50	140
Threshold of pain	—	130
Threshold of discomfort	—	120
Chainsaw	1	110
Disco (loudspeaker)	1	100
Diesel truck	10	90
Kerbside of busy road	5	80
Vacuum cleaner	1	70
Conversational speech	1	60
Average home	—	50
Quiet library	—	40
Quiet bedroom at night	—	30
Background in TV studio	—	20
Rustling of leafs	—	10
Threshold of hearing	—	0

Exercise 3.6 A plane sound wave of pressure amplitude P impinges on a freely suspended rigid plate of density ρ and thickness d. Calculate the intensity level of the transmitted sound wave as a function of frequency. The boundary conditions at the plate are: (i) continuity of displacements and (ii) force balance.

Solution In this example we have to satisfy the boundary conditions at a freely suspended plate. Freely suspended means that the plate can move in the sound wave without experiencing anything other than pressure and inertial forces. We assume the left face of the plate to be

located at $x = 0$. The plate is rigid, meaning that the displacement of the plate is equal to the displacement of the air on both sides: $u(x = 0, t) = u(x = d, t) = u_{\mathrm{pl}}(t)$. This is the first boundary condition. The second boundary condition results from the balance of forces on the plate and requires that

$$\begin{aligned} p(0,t) - p(d,t) &= \rho d \ddot{u}(0,t) \\ \gamma p_0 \left[u'(0,t) - u'(d,t)\right] &= \rho d \ddot{u}(0,t) \end{aligned}$$

where in the second step we have used equation 3.42. As we are analysing the propagation of sound we should look for travelling wave solutions of the wave equation

$$\frac{\partial^2 u}{\partial t^2} = \frac{E}{\rho}\frac{\partial^2 u}{\partial x^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

which fulfil the boundary conditions. There are two possible approaches to the solution of this example, described in detail below.

First attempt at solution The first approach assumes right-travelling waves on both sides of the plate ('i' stands for incoming, 't' for transmitted and 'r' for reflected). For mathematical simplicity we use a complex exponential notation, noting that the physical wave is described by the real part, which is a cosine. Accordingly, the amplitudes must be understood as complex numbers. The displacement field is then

$$\begin{aligned} u(x,t) &= A_{\mathrm{i}} \exp[i(\omega t - kx)] && x < 0 \\ u(x,t) &= A_{\mathrm{t}} \exp\{i[\omega t - k(x-d)]\} && x > 0 \end{aligned}$$

For the boundary condition

$$u(x = 0, t) = u(x = d, t) = u_{\mathrm{pl}}(t)$$

to be fulfilled at all times, functions

$$u(0,t) = A_{\mathrm{i}} \exp(i\omega t) \qquad \text{and} \qquad u(d,t) = A_{\mathrm{t}} \exp(i\omega t)$$

must be the same at all times. This is possible only if $A_{\mathrm{i}} = A_{\mathrm{t}}$. When the two functions on both sides are the same, however, it is not possible to satisfy the second boundary condition. In conclusion, this approach does not work.

Second attempt at solution We again look for solutions in the form of travelling waves, but this time we allow also a left-travelling (reflected) wave on the left-hand side:

$$\begin{aligned} u(x,t) &= A_{\mathrm{i}} \exp[i(\omega t - kx)] + A_{\mathrm{i}} \exp[i(\omega t + kx)] && \text{for} \quad x < 0 \\ u(x,t) &= A_{\mathrm{t}} \exp\{i[\omega t - k(x-d)]\} && \text{for} \quad x > 0 \end{aligned}$$

The amplitude of the incoming wave is fixed by the prescribed pressure amplitude.

$$\begin{aligned} p(x,t) &= -\gamma p_0 \frac{\partial u}{\partial x} = \gamma p_0 i k A_{\mathrm{i}} \exp[i(\omega t - kx)] \\ &= \gamma p_0 k A_{\mathrm{i}} \left[-\sin(\omega t - kx) + i \cos(\omega t - kx)\right] \end{aligned}$$

The amplitude of the real part must be equal to P, hence

$$P = \gamma p_0 k A_{\mathrm{i}} \qquad \Leftrightarrow \qquad A_{\mathrm{i}} = \frac{P}{\gamma p_0 k}$$

The condition $u(x=0,t) = u(x=d,t) = u_{\mathrm{pl}}(t)$ now becomes

$$A_{\mathrm{i}} \exp(i\omega t) + A_{\mathrm{r}} \exp(i\omega t) = A_{\mathrm{t}} \exp(i\omega t)$$
$$A_{\mathrm{i}} + A_{\mathrm{r}} = A_{\mathrm{t}}$$

Considering the second boundary condition at the plate,

$$-\gamma p_0 \left[u'(0,t) - u'(d,t)\right] = \rho d \ddot{u}(0,t)$$

and inserting the solutions on both sides yields

$$-\gamma p_0 \left[u'(0,t) - u'(d,t)\right] = \rho d \ddot{u}(0,t)$$
$$ik\gamma p_0[(A_{\mathrm{i}} - A_{\mathrm{r}}) \exp(i\omega t) - A_{\mathrm{t}} \exp(i\omega t)] = -\rho d A_{\mathrm{t}} \omega^2 \exp(i\omega t)$$
$$A_{\mathrm{i}} - A_{\mathrm{r}} = \left(-\frac{\rho d \omega^2}{ik\gamma p_0} + 1\right) A_{\mathrm{t}} = \left(i\frac{d\omega\rho}{c\rho_0} + 1\right) A_{\mathrm{t}}$$

Combining the two boundary conditions leads to the following system of linear equations for the amplitudes:

$$-A_{\mathrm{r}} + A_{\mathrm{t}} = A_{\mathrm{i}}$$
$$A_{\mathrm{r}} + \left(i\frac{d\omega\rho}{c\rho_0} + 1\right) A_{\mathrm{t}} = A_{\mathrm{i}}$$

Adding both equations yields

$$A_{\mathrm{t}} = 2A_{\mathrm{i}} \Bigg/ \left(2 + i\frac{d\omega\rho}{c\rho_0}\right) = \left(1 - i\frac{d\omega\rho}{2c\rho_0}\right) \Bigg/ \left[1 + \left(\frac{d\omega\rho}{2c\rho_0}\right)^2\right]$$

and

$$|A_{\mathrm{t}}| = |A_{\mathrm{i}}| \sqrt{1 \Bigg/ \left[1 + \left(\frac{d\omega\rho}{2c\rho_0}\right)^2\right]}$$

The same relationship as for the displacement amplitudes holds for the pressure amplitudes. So the sound pressure levels can be related by

$$\left|P_{\mathrm{t}}^2\right| = \left|P_{\mathrm{i}}^2\right| \Bigg/ \left[1 + \left(\frac{d\omega\rho}{2c\rho_0}\right)^2\right]$$

and

$$L_{\mathrm{t}} = L_{\mathrm{i}} - 10 \log_{10} \left[1 + \left(\frac{d\omega\rho}{2c\rho_0}\right)^2\right]$$

We see that the plate acts as a sound insulation at high frequencies. Low frequencies can only be damped if the plate is heavy. To give a few numbers we assume a steel plate of 0.5 cm thickness, density 7800 kg/m^3, air density 1.2 kg/m^3, sound speed 330 m/s. Then, for a sound frequency of 10 Hz the damping is about 10 dB, for 100 Hz, it is 30 dB, and so on (20 dB more for every factor 10 in frequency). For a wood plate with the same thickness (density 500 kg/m^3): for 10 Hz frequency the damping is only 0.1 dB, for 100 Hz, it is 6.5 dB, for 1000 Hz 25 dB, from then on 20 dB more for every factor 10 in frequency. This is why low frequency repetitive beats are unpopular with neighbours! ■

3.4.2 Centrifugal stiffening of a beam attached to a rotating hub

In this section, an application is discussed where we combine methods from beam vibration and Lagrangian mechanics to analyse the transverse vibration behaviour of beams attached to a rotating hub, such as schematically shown in Figure 3.12. An important application of this model is in the understanding of the vibration behaviour of parts in rotating machinery, such as rotor blades.

The deformation of such parts under dynamic loads is influenced by the inertial forces they experience due to the rotation. This leads to a phenomenon called centrifugal stiffening, where the natural frequencies (e.g. of an individual rotor blade) may significantly increase with the rotor speed. This is an important factor in the design of high-speed compressors and turbines, since the changing natural frequencies can lead to unexpected resonances.

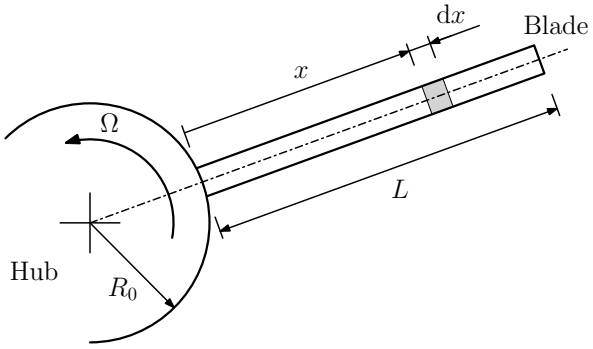


Figure 3.12: Dynamic model of a uniform blade (beam) attached to a rotating hub.

To analyse the effect of rotation on the natural frequencies of bending vibration of the beam, we look at Figure 3.13a which shows the rotational inertial force in a side view. The beam element dx moves on a circle of radius $(R_0 + x)$, its centripetal acceleration is thus $(R_0 + x)\Omega^2$, and the corresponding d'Alembert inertial force is $(\rho A \mathrm{d}x)(R_0 + x)\Omega^2$, acting radially outwards.

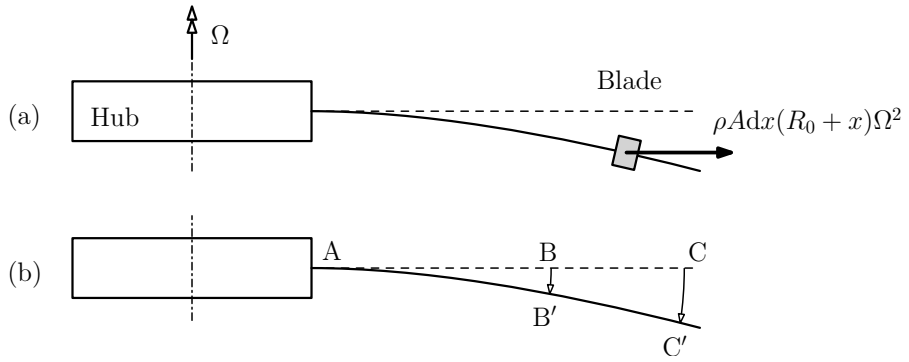


Figure 3.13: Rotor hub and blade model: (a) side view of the rotor hub and blade to show the inertial force on a volume element of the blade and (b) coupling between transversal and axial displacements of a beam with a free end at C and a clamped end at A.

If the rotating beam is simultaneously transversally deflected by a dynamic load, the volume element is displaced towards the axis of rotation. In this case, the inertial force helps to restore the beam to its stationary position. Because of this increase in the restoring force, the effective stiffness of the beam, and consequentially its resonant frequencies, increase with increasing rotation speed.

To include this effect into the model of the beam we use Lagrange's method. The inertial forces are treated as *external* forces and the virtual work is done by these forces, as well as the

corresponding generalised force terms. However, the mode shape functions will not be obtained from simple guesswork but evaluated using the Euler-Bernoulli equation. This ensures that the exact result is obtained in the limit of zero rotation speed.

To understand how the radial inertial forces couple to the transverse vibration, we take a closer look at the geometry of a bent beam. When a beam with a clamped and a free end is deflected transversally, we have a situation as shown in Figure 3.13b. Since the length of the neutral axis does not change during bending, the length of the beam intervals remains unchanged, thus $\overline{\mathrm{AB}} = \overline{\mathrm{AB}}'$ and $\overline{\mathrm{AC}} = \overline{\mathrm{AC}}'$. As a consequence, B′ and C′ are displaced to the left with respect to B and C. Thus, the transversal displacement results in a small axial displacement towards the clamped end of the beam at A.

Using Pythagoras' theorem and the diagram in Figure 3.14 it is possible to evaluate the axial displacement $u(x)$. Using the notation $w'(x) = \mathrm{d}w/\mathrm{d}x$, we find that

$$\begin{aligned} \mathrm{d}u + \mathrm{d}x &= \sqrt{\mathrm{d}x^2 + \mathrm{d}w^2} = \sqrt{\mathrm{d}x^2 + (w'\mathrm{d}x)^2} \\ \mathrm{d}u &= \mathrm{d}x \left[\sqrt{1 - (w')^2} - 1\right] \approx \frac{1}{2}(w')^2 \mathrm{d}x \end{aligned} \tag{3.53}$$

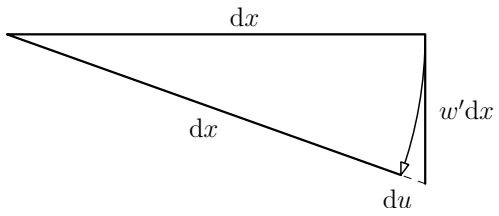


Figure 3.14: Calculation of the axial displacement resulting from a small transverse displacement $\mathrm{d}w = w'\mathrm{d}x$.

The displacement calculated in this manner is not exactly in the axial direction. However, since it is assumed that w and w' are small, the difference from a purely axial displacement can be neglected. From the expression for $\mathrm{d}u$ the total displacement $u(x)$ follows by simple integration from the clamped end at $x = 0$ up to the coordinate x, that is,

$$u(x,t) = \int_0^x \mathrm{d}u = \int_0^x \frac{1}{2}\left(\frac{\partial w}{\partial \xi}\right)^2 \mathrm{d}\xi \tag{3.54}$$

The next step is now to find the appropriate mode shape functions for the blade model.

Mode shape functions

Lagrange's method, as described in Chapter 2, will be used to evaluate the influence of rotational inertia on the dynamic behaviour of the system. This approach requires that the vibration behaviour is described by one or multiple generalised coordinates and, consequently, mode shape functions. In the present section, however, we do not simply guess these functions but obtain them from solving the vibration problem in the absence of rotation.

The blade is modelled as an uniform beam of length L that is clamped to a hub of radius R_0. The free end of the blade (beam) is free. The transverse displacement of the beam is assumed as in equation 3.32, which is

$$w(x,t) = \varphi(x)\cos(\omega t)$$

where the space-dependent function fulfils equation 3.36

$$\frac{\partial^4 \varphi}{\partial x^4} - k^4 \varphi = 0$$

with the solution given by equation 3.37

$$\varphi(x) = C_1 \cos(kx) + C_2 \sin(kx) + C_3 \cosh(kx) + C_4 \sinh(kx)$$

In the present case, the boundary conditions are $\varphi(x = 0, t) = \varphi'(x = 0, t) = 0$ at the clamped end and $\varphi''(x = L, t) = \varphi'''(x = L, t) = 0$ at the free end of the beam. The boundary conditions at the clamped end require $C_3 = -C_1$, $C_4 = -C_2$. Consequently, the solution becomes

$$\varphi(x) = C_1[\cos(kx) - \cosh(kx)] + C_2[\sin(kx) - \sinh(kx)] \tag{3.55}$$

The boundary conditions at the free end then result in the system of equations

$$\begin{aligned} &C_1[\cos(kL) + \cosh(kL)] + C_2[\sin(kL) + \sinh(kL)] = 0 \\ &C_1[-\sin(kL) + \sinh(kL)] + C_2[\cos(kL) + \cosh(kL)] = 0 \end{aligned} \tag{3.56}$$

The solvability equation for this system is that the determinant of the matrix of coefficients must be zero. This leads to the transcendental equation

$$\cos(kL) = -\frac{1}{\cosh(kL)} \tag{3.57}$$

The solutions of this equation are, with good approximation, given by

$$\begin{aligned} &(kL)_n = 1.875 \qquad \text{for } n = 1 \\ &(kL)_n \approx \left(n - \frac{1}{2}\right)\pi \qquad \text{for } n > 1 \end{aligned} \tag{3.58}$$

For the first value, $\cos(kL) = -0.2995$, $\cosh(kL) = 3.3371$, $\sin(kL) = 0.9541$ and $\sinh(kL) = 3.1837$. This yields the mode shape function

$$\varphi_1(x) = [\cos(kx) - \cosh(kx)] - 0.734[\sin(kx) - \sinh(kx)] \tag{3.59}$$

For the values with $n > 1$, $\cos(kL) \approx 1/\cosh(kL) \approx 0$ and $\sinh(kL) \approx \cosh(kL) \gg 1$. Thus, in good approximation $C_1 = -C_2$ and the mode shape functions are given by

$$\varphi_n(x) \approx [\cos(kx) - \cosh(kx)] - [\sin(kx) - \sinh(kx)] \tag{3.60}$$

The first three mode shape functions are graphically shown in Figure 3.15.

A good additional exercise is to work out the detailed derivation of equations 3.55 to 3.60 from the general solution of equation 3.36 and the boundary conditions given above, using a graphical method for solving equation 3.57.

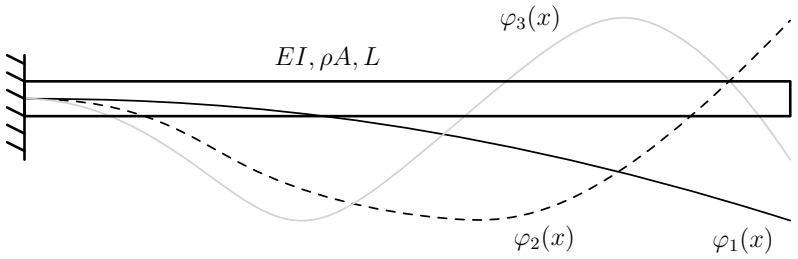


Figure 3.15: Generic representation of the first three mode shape functions for a beam of length L, bending stiffness EI, cross-section area A and density ρ.

Influence of inertial forces

To analyse the influence of centrifugal forces due to rotation of the hub to which the blade is attached, we use the generalised force approach described in Section 2.5.4 by treating the centrifugal force as a distributed external load acting on each volume element of the beam. The different vibration modes are analysed separately. For the $n-$th vibration mode we write

$$w(x,t) = \varphi_n(x) q_n(t) \tag{3.61}$$

where $\varphi_n(x)$ is the $n-$th mode shape function given above.

The kinetic and potential energy functions for oscillation in the $n-$th mode are given by

$$\begin{aligned} T &= \int_0^L \frac{1}{2}\rho A \left(\frac{\partial w}{\partial t}\right)^2 \mathrm{d}x = \frac{1}{2}\rho A \dot{q}_n^2 \int_0^L [\varphi_n(x)]^2 \mathrm{d}x = \frac{1}{2} M_n \dot{q}_n^2 \\ V &= \int_0^L \frac{1}{2} EI \left(\frac{\partial^2 w}{\partial x^2}\right)^2 \mathrm{d}x = \frac{1}{2} EI q_n^2 \int_0^L \left(\frac{\partial^2 \varphi_n}{\partial x^2}\right)^2 \mathrm{d}x = \frac{1}{2} K_n q_n^2 \end{aligned} \tag{3.62}$$

From Lagrange's approach it is then possible to obtain the equations of motion for the mode amplitudes, which is

$$\begin{aligned} &\left[\rho A \int_0^L \varphi_n^2\right] \ddot{q}_j + \left[EI \int_0^L \left(\varphi_n''\right)^2 \mathrm{d}x\right] q_n = Q_n \\ &m_n \ddot{q}_n + k_n q_n = Q_n \end{aligned} \tag{3.63}$$

To evaluate the generalised force, we consider the virtual work done by the centrifugal forces in response to a virtual displacement δq_n. As discussed above, the transverse displacement $w(x)$ causes a displacement $u(x)$ along the direction of the beam axis

$$u(x,t) = -\int_0^x \frac{1}{2}\left(\frac{\partial w}{\partial \xi}\right)^2 \mathrm{d}\xi = -\left[\int_0^x \frac{1}{2}\left(\frac{\partial \varphi_n}{\partial \xi}\right)^2 \mathrm{d}\xi\right] q_n^2 \tag{3.64}$$

Accordingly, the virtual displacement δq_n causes a change δu:

$$\delta u(x,t) = -\left[\int_0^x \left(\frac{\partial \varphi_n}{\partial \xi}\right)^2 \mathrm{d}\xi\right] q_n \delta q_n \tag{3.65}$$

This displacement is opposite to the centrifugal force and, as such, the work done by the centrifugal force is negative. The work done on an element $\mathrm{d}x$ at coordinate x is given by the product of the centrifugal force and the displacement δu,

$$\delta W(x, \mathrm{d}x, t) = -\rho A \mathrm{d}x (R_0 + x)\Omega^2 \left[\int_0^x \left(\frac{\partial \varphi_n}{\partial \xi}\right)^2 \mathrm{d}\xi\right] q_n \delta q_n \tag{3.66}$$

The total virtual work is obtained by integrating this expression over x:

$$\delta W = \int_0^L \delta W(x) \mathrm{d}x = -\rho A \Omega^2 \left[\int_0^L \int_0^x \left(\frac{\partial \varphi_n}{\partial \xi}\right)^2 \mathrm{d}\xi (R_0 + x) \mathrm{d}x\right] q_n \delta q_n \tag{3.67}$$

Dividing this by the virtual displacement gives the virtual force on the mode n

$$Q_n = -\rho A \Omega^2 \left[\int_0^L \int_0^x \left(\frac{\partial \varphi_n}{\partial \xi}\right)^2 \mathrm{d}\xi (R_0 + x) \mathrm{d}x\right] q_n \tag{3.68}$$

or, in short,

$$Q_n = -\rho A \Omega^2 I_n q_n \tag{3.69}$$

The double integral I_n depends on the mode shape function and needs to be evaluated numerically. As a result, we find that the mode amplitudes obey the system of differential equations

$$\begin{aligned} &m_n\ddot{q}_n + k_n q_n = -\rho A\Omega^2 I_n q_n \\ &m_n\ddot{q}_n + \left(k_n + \rho A\Omega^2 I_n\right) q_n = 0 \end{aligned} \tag{3.70}$$

Hence, the centrifugal term adds a contribution to the stiffness proportional to the square of the rotation speed of the blade, increasing the blade natural frequency.

4

Impulsive dynamics

4.1 Introduction

This chapter introduces the mechanics of impact on flexible structures such as beams, the single degree-of-freedom approximation, transverse impact on elastic beams and the local and global response of slender structures upon impact. This is a highly relevant structural engineering topic as it helps understand how complex systems and structures react to localised impacts. Numerous examples of applications can be found in many engineering systems. Some examples are shown in Figure 4.1.

The main concepts that explain the mechanical response of structures on impact will be detailed in the following paragraphs, with the aim of understanding how an impact affects the dynamic response of the structure, namely in terms of vibration — modified vibration amplitudes, frequencies, etc. The main aim is to understand the effect of a single point impact on a slender structure. It is, however, necessary to state the assumptions behind the concepts that will be discussed. In the following sections, we assume that: (i) the mass of the impactor is significantly smaller than the mass of the structure being impacted, (ii) the impacted structure is linear and slender (i.e. it is not curved and its longitudinal dimension is significantly larger than its width and breadth), and (iii) the structure has typical beam behaviour. some of the consequences of breaking these assumptions will also be explored in this chapter.

4.2 Impact on flexible structures

When discussing waves and the dynamic behaviour of continuous structures, such as bars and beams, in previous chapters, we assumed that these slender structures are uniform and homogeneous — i.e. they have constant density ρ, constant bending stiffness EI and constant cross section area A. In a generic structure, however, the geometrical characteristics may not be constant, although the material properties often are. Consequently, a generic beam will have mass per unit length $\rho A(x)$ and bending stiffness $EI(x)$.

In Section 3.3.1 we derived the Euler-Bernoulli equation, which describes the time evolution of the transverse displacements $w(x,t)$ of a slender structure (see equation 3.28). This equation is

$$\rho A \frac{\partial^2 w}{\partial t^2} + EI \frac{\partial^4 w}{\partial x^4} = 0 \tag{4.1}$$

In the generic case where the beam is not uniform, albeit still being homogeneous, the Euler-Bernoulli equation can be rewritten as

$$\frac{\partial^2 w}{\partial t^2} + \frac{E}{\rho} \frac{\partial^2}{\partial x^2} \left[\frac{I(x)}{A(x)} \frac{\partial^2 w}{\partial x^2} \right] = 0 \tag{4.2}$$

The functions defining the change in the moment of inertia $I(x)$ and in the area of the cross section $A(x)$ have to be included in the differential equation above. In this chapter, however, we

Figure 4.1: Examples of engineering systems and structures where impact is relevant. Clockwise from top-left: Marcus Ericsson test drives the Sauber C37 Formula 1 car with the Halo protection system, X-ray image of broken femur, Paul Lawrie at the PGA European Tour Scottish Hydro Challenge, the 9/11 attack on the World Trade Center, springboard synchronised diving at the Rio 2016 Olympics, and a view of the Volvo Side-Impact Protection System (SIPS). All images reproduced under the Creative Commons license CC BY-SA 3.0 (authors, clockwise from top-left: Alberto-g-rovi, N. Dilmen, A. Sneddon, R.J. Fisch, Agência Brasil, F.R. Pozzebom).

again assume the beam to be uniform and homogeneous. Consequently, the wave equations 4.1 and 4.2 can be written as

$$\frac{\partial^2 w}{\partial t^2} + \gamma^2 \frac{\partial^4 w}{\partial x^4} = 0 \tag{4.3}$$

with $\gamma^2 = EI/(\rho A)$. A generic form for a standing wave solution to this wave equations is

$$w(x,t) = \varphi(x)q(t) = \varphi(x)[A_1 \cos(\omega_0 t) + A_2 \sin(\omega_0 t)] \tag{4.4}$$

where $\varphi(x)$ is the shape function and $q(t)$ is the oscillating amplitude function. Substituting this into the wave equation 4.3 we obtain

$$\frac{\mathrm{d}^4\varphi(x)}{\mathrm{d}x^4} - \frac{\omega_0^2}{\gamma^2}\varphi(x) = 0 \tag{4.5}$$

A generic solution — the shape function — to the above fourth order differential equation can be written as

$$\varphi(x) = C_1(\cos kx + \cosh kx) + C_2(\cos kx - \cosh kx) + C_3(\sin kx + \sinh kx) + C_4(\sin kx - \sinh kx) \tag{4.6}$$

where $k = \sqrt{\omega_0/\gamma}$ is the wave number and C_i (with $i = 1, \ldots, 4$) are constants that depend on the particular combination of boundary conditions for the beam. As we have seen before in Section 3.3.3, shape functions for standing waves can be described mathematically in different ways, such as in equation 3.37.

Dynamic response The standing wave transverse vibration of a continuous, uniform and homogeneous slender structure, such as a beam, is defined by the following fourth order differential equation on the mode shape $\varphi(x)$:

$$\frac{\mathrm{d}^4\varphi(x)}{\mathrm{d}x^4} - \frac{\omega_0^2}{\gamma^2}\varphi(x) = 0 \tag{4.7}$$

with a generic solution that can be written in such a way as to explicitly separate its symmetrical and anti-symmetrical parts, that is, $\varphi(x) = \varphi_\mathrm{s}(x)+\varphi_\mathrm{a}(x)$, where the symmetrical part is

$$\varphi_\mathrm{s}(x) = C_1(\cos kx + \cosh kx) + C_2(\cos kx - \cosh kx) \tag{4.8}$$

and the anti-symmetrical part is

$$\varphi_\mathrm{a}(x) = C_3(\sin kx + \sinh kx) + C_4(\sin kx - \sinh kx) \tag{4.9}$$

The global solution to this differential equation is defined in equation 4.6 above.

Figure 4.2 shows three examples of symmetrical and anti-symmetrical shape functions for a fully clamped uniform and homogeneous beam. In normal conditions, the dynamic behaviour of this beam can be described by the Euler-Bernoulli equation 4.3 with the following boundary conditions: (i) zero transverse displacement at the supports, $w(0,t) = w(L,t) = 0$, and (ii) zero rotation at the supports, $w'(0,t) = w'(L,t) = 0$. As we are assuming a standing wave response, these boundary conditions can also be defined as $\varphi(0) = \varphi(L) = 0$ and $\varphi'(0) = \varphi'(L) = 0$, respectively. A more in-depth discussion on this and on different types of boundary conditions can be found in Section 3.3.3.

Enforcing the boundary conditions described above leads to the following relations between the unknown constants C_i $(i = 1, \ldots, 4)$:

$$\begin{aligned} &C_1 = C_3 = 0 \\ &\frac{C_2}{C_4} = \frac{\cos kL - \cosh kL}{\sin kL + \sinh kL} = -\frac{\sin kL - \sinh kL}{\cos kL - \cosh kL} \end{aligned} \tag{4.10}$$

Solving these equations simultaneously and ensuring a non-trivial solution leads to

$$\cos kL \cosh kL = 1$$

from which it is possible to determine the values of the wave number k. Note that this equation does not have an analytical solution so it should be solved either graphically or numerically, similarly to what was shown in examples 3.4 and 3.5.

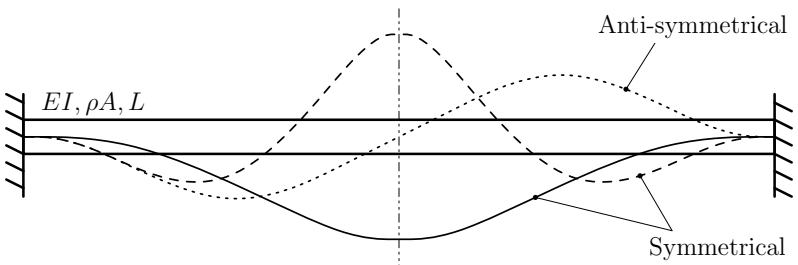


Figure 4.2: Symmetrical (solid and dashed lines) and anti-symmetrical (dotted line) shape functions for a fully clamped uniform and homogeneous beam.

4.2.1 Multiple degree-of-freedom approximation

In the previous section it was assumed that the dynamic response of the beam can be described by a single-mode standing wave (see equation 4.4). If more modes are required, however, the deflection of the impacted structure can be described by superimposing more mode shapes, using the more generic standing wave equation

$$w(x,t) = \sum_{j=0}^{\infty} A_j \varphi_j(x) \sin(\omega_j t - \phi_j) \tag{4.11}$$

where $\varphi_j(x)$ is the j^{th} mode shape function, A_j is the amplitude of mode shape j, and ϕ_j is the corresponding phase angle.

When describing the dynamic behaviour of a slender structure such as a beam using multiple degrees-of-freedom and the deflection function 4.11, the kinetic energy for the j^{th} mode is

$$T_j = \frac{1}{2}\int_0^L \rho A \dot{w}_j^2(x,t)\mathrm{d}x = \frac{1}{2}\left[A_j\omega_j \cos(\omega_j t - \phi_j)\right]^2 \int_0^L \rho A \varphi_j^2(x)\mathrm{d}x \tag{4.12}$$

From this equation it becomes evident that the maximum kinetic energy of mode shape j is reached when the cos function becomes 1, that is,

$$T_j^{\max} = \frac{1}{2}A_j^2\omega_j^2 \int_0^L \rho A \varphi_j^2(x)\mathrm{d}x \tag{4.13}$$

A similar analysis can be done for the elastic potential energy, which is

$$V_j = \frac{1}{2}\int_0^L M_{\mathrm{b}}(x,t)\frac{\partial^2 w_j(x,t)}{\mathrm{d}x^2}\mathrm{d}x \tag{4.14}$$

where $M_{\mathrm{b}}(x,t)$ is the function describing the bending moment on the beam. Substituting again the corresponding term of the wave solution, as defined in equation 4.11, we obtain

$$V_j = \frac{1}{2}\int_0^L EI\left[\frac{\partial^2 w_j(x,t)}{\partial x^2}\right]^2 \mathrm{d}x = \frac{1}{2}\left[A_j \sin(\omega_j t - \phi_j)\right]^2 \int_0^L EI\left[\frac{\mathrm{d}^2\varphi_j(x)}{\mathrm{d}x^2}\right]^2 \mathrm{d}x \tag{4.15}$$

Consequently, the maximum of the elastic potential energy is

$$V_j^{\max} = \frac{1}{2}A_j^2 \int_0^L EI\left[\frac{\mathrm{d}^2\varphi_j(x)}{\mathrm{d}x^2}\right]^2 \mathrm{d}x \tag{4.16}$$

If gravitational effects can be neglected — e.g. when the beam deflections are small — and if it is reasonable to assume that the beam is initially static and that the system is conservative, then the maximum kinetic and maximum potential energies have to be the same, $V_j^{\max} = T_j^{\max}$. Thus, combining relations 4.13 and 4.16 it is possible to obtain the j^{th} mode frequency, which, for a uniform and homogeneous beam is

$$\begin{aligned}\omega_j^2 &= \int_0^L EI\left(\frac{\mathrm{d}^2\varphi_j}{\mathrm{d}x^2}\right)^2 \mathrm{d}x \Big/ \int_0^L \rho A \varphi_j^2 \mathrm{d}x \\ &= \left(\frac{EI}{\rho A}\right)\int_0^L \left(\frac{\mathrm{d}^2\varphi_j}{\mathrm{d}x^2}\right)^2 \mathrm{d}x \Big/ \int_0^L \varphi_j^2 \mathrm{d}x\end{aligned} \tag{4.17}$$

Note that a similar conclusion was reached in Section 2.5.2, when deriving the single degree-of-freedom model approximation for continuous systems.

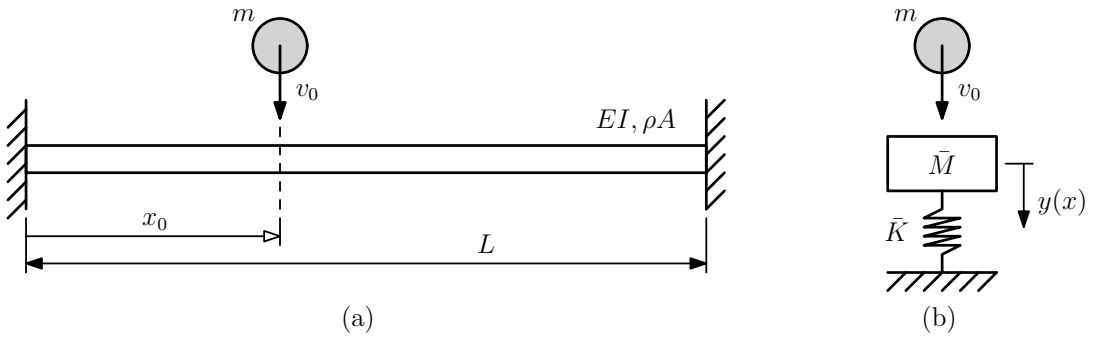


Figure 4.3: (a) Transverse impact of a missile with mass m and velocity v_0 on an elastic fully clamped beam, at a distance x_0 from one end, and (b) corresponding single degree-of-freedom approximation.

4.2.2 Transverse impact on elastic beams

In this section we will analyse the dynamic elastic behaviour of a slender structure, such as the clamped beam in Figure 4.3a, initially at rest, when it is impacted perpendicularly to its neutral axis by a missile of mass m and velocity v_0 at a distance x_0 from one end of the beam.

The single degree-of-freedom approximation in Figure 4.3b is only valid if the mass of the missile, m, is small in comparison to the mass of the structure, ρAL. The mode shape that describes the response of the single degree-of-freedom system is an approximation of the mode shape of the beam, that is, $\varphi(x) \to \bar{\varphi}(x)$. Consequently, it is assumed that

$$w(x,t) = \bar{\varphi}(x)q(t) \tag{4.18}$$

where $q(t)$ is the function describing the oscillatory character of the standing wave. The effective stiffness $\bar{K}$ of the spring on the single degree-of-freedom system is obtained from the strain energy at maximum deflection for a normalised shape function, that is,

$$V^{\max} = \frac{1}{2}\left(q^{\max}\right)^2 \int_0^L EI\left(\frac{\mathrm{d}^2\bar{\varphi}}{\mathrm{d}x^2}\right)^2 \mathrm{d}x = \frac{1}{2}\bar{K}\left(q^{\max}\right)^2 \tag{4.19}$$

from where it is possible to conclude that

$$\bar{K} = \int_0^L EI\left(\frac{\mathrm{d}^2\bar{\varphi}}{\mathrm{d}x^2}\right)^2 \mathrm{d}x \tag{4.20}$$

With a similar analysis, the effective mass (inertia) $\bar{M}$ of the single degree-of-freedom system is obtained from the kinetic energy at maximum deflection, which for a normalised shape function is

$$T^{\max} = \frac{1}{2}\left(\dot{q}^{\max}\right)^2 \int_0^L \rho A\bar{\varphi}^2 \mathrm{d}x = \frac{1}{2}\bar{M}\left(\dot{q}^{\max}\right)^2 \tag{4.21}$$

from where it is also possible to conclude that

$$\bar{M} = \int_0^L \rho A\bar{\varphi}^2 \mathrm{d}x \tag{4.22}$$

A number of structural aspects are relevant when analysing the effect of an impact on an elastic beam. Among these, some of the most significant are (i) the ratio between the mass of the impactor (missile) m and the mass of the beam, $\beta = m/(\rho AL)$, (ii) the exact location of the impact relative to the supports of the structure, and (iii) the deformability of the impacted area compared to the structural compliance. Such an impact will, however, and regardless of these

conditions, always induce forced vibration on the beam. The wave equation of motion for forced vibration of a uniform and homogeneous beam is

$$\rho A \frac{\partial^2 w}{\partial t^2} + EI \frac{\partial^4 w}{\partial x^4} = f(x,t) \tag{4.23}$$

where $f(x,t)$ is a generic distributed external force acting on the structure. Equation 4.23 above can be rewritten including the constant γ as

$$\gamma^2 \frac{\partial^4 w}{\partial x^4} + \frac{\partial^2 w}{\partial t^2} = \frac{f(x,t)}{\rho A} \tag{4.24}$$

In the following paragraphs, it is again assumed that the response of the beam can be described by a generic standing wave solution $w(x,t) = \varphi(x)q(t)$. In most cases where the external force is originated by an impact, it can be approximated by a point load, acting at distance x_0 from one end of the beam, such as shown in figure 4.3. In such a situation, the distributed load becomes

$$f(x,t) = F(t)\delta(x_0) \tag{4.25}$$

where $\delta(x)$ is the Dirac delta function at coordinate x_0, which is defined as

$$\delta(x_0) = \begin{cases} 1 & \text{if} \quad x = x_0 \\ 0 & \text{if} \quad x \neq x_0 \end{cases} \tag{4.26}$$

As we are looking for a single degree-of-freedom approximation for the standing wave dynamic response of the beam — an equivalent lumped-mass model —, we can use Lagrange's method. Consequently, the generic equation of motion of the j^{th} generalised coordinate or mode shape of the beam can be written as

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial T}{\partial q_j} + \frac{\partial V}{\partial q_j} = Q_j \tag{4.27}$$

Neglecting gravitational energy terms, the kinetic and elastic potential energies of the beam can be defined as

$$\begin{aligned} T_j &= \frac{1}{2}\rho A \dot{q}_j^2 \int_0^L \varphi_j^2 \mathrm{d}x \\ V_j &= \frac{1}{2} EI q_j^2 \int_0^L \left(\frac{\mathrm{d}^2\varphi_j}{\mathrm{d}x^2}\right)^2 \mathrm{d}x \end{aligned} \tag{4.28}$$

respectively, and the generalised force acting on mode shape j is

$$Q_j = \int_0^L f(x,t)\varphi_j \mathrm{d}x \tag{4.29}$$

Substituting the kinetic energy T_j and potential energy V_j into Lagrange's equation of motion 4.27, and reorganising terms leads to

$$\ddot{q}_j \int_0^L \varphi_j^2 \mathrm{d}x + \gamma^2 q_j \int_0^L \left(\frac{\mathrm{d}^2\varphi_j}{\mathrm{d}x^2}\right)^2 \mathrm{d}x = \int_0^L \frac{f(x,t)\varphi_j}{\rho A} \mathrm{d}x \tag{4.30}$$

which is a second order differential equation on $q_j(t)$. From Rayleigh's theorem (see Section 2.5.2 on page 46)

$$\begin{aligned} \omega_j^2 = \frac{\bar{K}}{\bar{M}} &= \int_0^L \left(\frac{\mathrm{d}^2\varphi_j}{\mathrm{d}x^2}\right)^2 \mathrm{d}x \left(\frac{EI}{\rho A}\right) \Big/ \int_0^L \varphi_j^2 \mathrm{d}x \\ \Rightarrow \quad \int_0^L \left(\frac{\mathrm{d}^2\varphi_j}{\mathrm{d}x^2}\right)^2 \mathrm{d}x &= k_j^4 \int_0^L \varphi_j^2 \mathrm{d}x \end{aligned} \tag{4.31}$$

Substituting back into the equation of motion and considering the definition of the impact load in equation 4.25 leads to

$$\omega_j^2 q_j + \frac{\mathrm{d}^2 q_j}{\mathrm{d}t^2} = \left[\frac{F(t)}{\rho A}\left|\varphi_j(x_0)\right|\right] \Big/ \int_0^L \varphi_j^2 \mathrm{d}x \tag{4.32}$$

where q_j and $F(t)$ are the time-dependent amplitude and impact load functions, respectively, and $\varphi_j(x)$ is the mathematical function describing the j^{th} mode shape of the beam for the particular combination of boundary conditions. Note that the integral on the right side term yields a constant and, consequently, equation 4.32 is a second order differential equation on q_j. The solution of this differential equation is

$$q_j = \frac{\left|\varphi_j(x_0)\right|}{\rho A \omega_j} \int_0^{t^\star} F(t) \sin[\omega_j(t^\star - t)]\mathrm{d}t \Big/ \int_0^L \varphi_j^2 \mathrm{d}x \tag{4.33}$$

where $t^\star$ is the time during which the force acts on the structure, also know as the time of separation of the impactor from the beam. To completely define the generalised coordinate q_j it is necessary to specify the impact load function $F(t)$. Assuming an orthogonal impact of a perfectly plastic missile and neglecting any local deformation (indentation) on the beam, this impact force can be defined by Newton's second law as

$$\begin{aligned} F(t) &= -ma = -m\frac{\mathrm{d}v}{\mathrm{d}t} = -m\ \ddot{w}|_{x_0} \\ \Rightarrow \quad F(t) &= -m\left|\varphi_j(x_0)\right| \frac{\mathrm{d}^2 q}{\mathrm{d}t^2} \end{aligned} \tag{4.34}$$

Substituting the impact force into the equation of motion (equation 4.32) leads to

$$\omega_j^2 q_j + \frac{\mathrm{d}^2 q_j}{\mathrm{d}t^2} = \frac{-m}{\rho A}\left|\varphi_j(x_0)\right| \frac{\mathrm{d}^2 q}{\mathrm{d}t^2} \Big/ \int_0^L \varphi_j^2 \mathrm{d}x \tag{4.35}$$

The solution to the above second order differential equation on $q(t)$ is

$$q_j = A_j \sin\left(\bar{\omega}_j t + \bar{\phi}_j\right) \tag{4.36}$$

where A_j is the amplitude, $\bar{\phi}_j$ is the modified phase angle and $\bar{\omega}_j$ is modified frequency for the j^{th} mode shape, which accounts for the added mass of the plastic missile. Inserting the solution q_j into the equation of motion 4.35 and reorganising the terms the modified frequency becomes

$$\bar{\omega}_j^2 = \omega_j^2 \Big/ \left(1 + \frac{m\left|\varphi_j(x_0)\right|}{\rho A \int_o^L \varphi_j^2 \mathrm{d}x}\right)$$

Noting that the denominator is a positive constant always greater than 1, the modified frequency is always lower than the corresponding natural frequency of the structure ω_j. This should be expected as the change in frequency is a direct consequence of the added mass to the vibrating system.

If, as assumed above, the post-impact displacement of the beam can be defined by the following standing wave deflection function

$$w(x,t) = \sum_{j=1}^{\infty} A_j \sin\left(\bar{\omega}_j t + \bar{\phi}_j\right) \varphi_j(x) \tag{4.37}$$

the velocity of the beam becomes

$$\dot{w}(x,t) = \frac{\mathrm{d}w(x,t)}{\mathrm{d}t} = \sum_{j=1}^{\infty} A_j \bar{\omega}_j \cos\left(\bar{\omega}_j t + \bar{\phi}_j\right) \varphi_j(x) \tag{4.38}$$

Exact values for the amplitudes A_j and phase angles $\bar{\phi}_j$ are a function of the initial conditions. In the assumption that the beam is at rest before impact, these initial conditions are $w(x,0) = 0$ and $\dot{w}(x,0) = mv_0\delta(x_0)$, where v_0 is the missile velocity just before the impact. Enforcing these initial conditions into the deflection function 4.37, multiplying the above equations by the mode shape $\varphi_j(x)$ and integrating over the length of the beam leads to

$$A_j \sin\bar{\phi}_j = 0 \tag{4.39}$$

and

$$\int_0^L \dot{w}(x,0)\varphi_j(x)\mathrm{d}x = A_j\bar{\omega}_j \cos\bar{\phi}_j \int_0^L \varphi_j^2 \mathrm{d}x \tag{4.40}$$

Solving for the mode amplitude and phase angle we find that the phase angle is $\bar{\phi}_j = 0$ and the j^{th} mode amplitude is

$$A_j = \frac{mv_0\varphi_j(x_0)}{\bar{\omega}_j\rho A\int_0^L \varphi_j^2 \mathrm{d}x}$$

Modified dynamic response When a continuous, uniform and homogeneous slender structure, such as a beam, is impacted by a missile of mass m, it vibrates with a modified frequency $\bar{\omega}_j$ defined by

$$\bar{\omega}_j^2 = \omega_j^2 \Big/ \left(1 + \frac{m\,|\varphi_j(x_0)|}{\rho A\int_o^L \varphi_j^2 \mathrm{d}x}\right) \tag{4.41}$$

where $\varphi_j(x)$ is a mathematical function defining the j^{th} mode shape, ω_j is the corresponding natural frequency, ρ, A and L are the density, area of the cross-section and length of the beam, respectively, and x_0 is the location of the impact on the beam. For an initially static beam and an impact velocity v_0, the j^{th} mode amplitude is

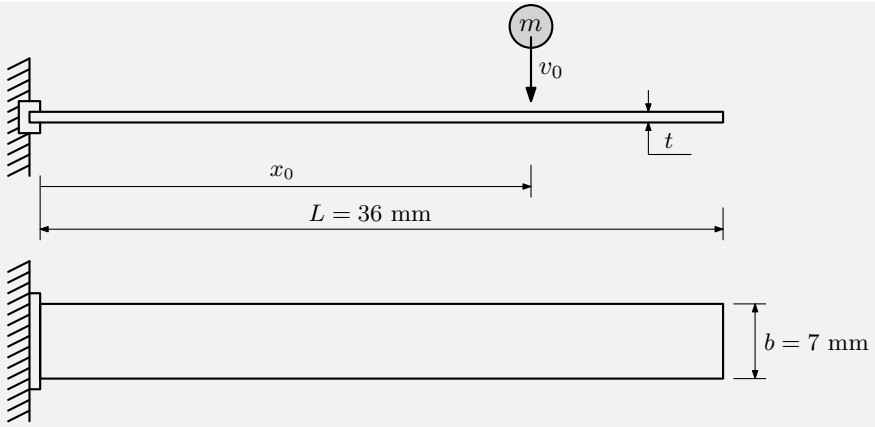
$$A_j = \frac{mv_0\varphi_j(x_0)}{\bar{\omega}_j\rho A\int_0^L \varphi_j^2 \mathrm{d}x} \tag{4.42}$$

and the corresponding phase angle is $\bar{\phi}_j = 0$.

The exercise below illustrates how finding modified frequencies and amplitudes of vibration can be crucial in the design process of engineering systems. One important thing to always note is that in these calculations the shape functions should always be normalised, i.e. their values should be $\varphi(x) \in [-1, 1]$. This ensures that, when calculating amplitudes of vibration, the value of the shape function at the impact location is the correct one.

Exercise 4.1 The thin metallic blade in the figure below is made of brass with a modulus of elasticity $E = 97$ GPa and a density $\rho = 8500$ kg/m^3. The blade is used as a tuning device for musical instruments.

a) Find the thickness of the blade t that will ensure the blade has a natural frequency that's equal to a key of A_4, which is 440 Hz.
b) If the blade is made to vibrate by impacting it with a metal sphere of mass $m = 1$ g with speed $v_0 = 1.4$ m/s at $x_0 = 2L/3$, how much out of tune will the blade resonate? Give your answer in percentage relative to the key of A_4 (440 Hz).
c) Find the corresponding amplitude of vibration of the tuning blade.



Solution To find the thickness of the blade t that ensures the blade has a natural frequency of 440 Hz we need to find the second moment of area of the blade I. For this solution we will assume the following normalised shape function for the beam

$$X(x) = \left(\frac{x}{L}\right)^2$$

which verifies the geometrical boundary conditions at the clamped end: $X(x=0) = 0$ and $X'(x=0) = 0$. It should be noted that this shape function verifies the boundary conditions at $x = 0$ but it does not ensure the correct bending moment on the beam (for this we would need a cubic shape function). The natural frequency of the beam can be found using Rayleigh's method as

$$\omega_1^2 = \int_0^L EI\left(\frac{\mathrm{d}^2X}{\mathrm{d}x^2}\right)^2 \mathrm{d}x \Big/ \int_0^L \rho A X^2 \mathrm{d}x$$

where, using the above shape function,

$$\left(\frac{\mathrm{d}^2X}{\mathrm{d}x^2}\right)^2 = \frac{4}{L^4} \qquad \text{and} \qquad X^2 = \frac{x^4}{L^4}$$

The natural frequency becomes

$$\omega_1^2 = EI\int_0^L \frac{4}{L^4}\mathrm{d}x \Big/ \left(\rho A \int_0^L \frac{x^4}{L^4}\mathrm{d}x\right) = \frac{20EI}{\rho AL}\left(\frac{1}{L^3}\right)$$

As the cross section of the beam is rectangular $I = bt^3/12$ and $A = bt$, leading to

$$\omega_1^2 = \frac{5}{3}\frac{Et^2}{\rho L^4} \quad \Rightarrow \quad t = \omega_1 L^2\sqrt{\frac{3\rho}{5E}}$$

Substituting all known values leads to

$$t = 2\pi \times 440 \times \left(36\times10^{-3}\right)^2\sqrt{\frac{3\times8500}{5\times97\times10^9}} = 0.8\times10^{-3} = 0.8 \text{ mm}$$

We now find the modified frequency after an impact with a metal sphere of mass $m = 1$ g with speed $v_0 = 1.4$ m/s at $x_0 = 2L/3$, which is

$$\bar{\omega}_1^2 = \omega_1^2 \Big/ \left(1 + \frac{mX(x_0)}{\rho A \int_0^L X^2 \mathrm{d}x}\right)$$

Substituting the known values

$$\bar{\omega}_1 = 2\pi \times 440 \Big/ \sqrt{1 + \frac{1 \times 10^{-3} \times X(2L/3)}{8500 \times 7 \times 10^{-3} \times 0.8 \times 10^{-3} \int_0^L X^2 \mathrm{d}x}}$$

and knowing that

$$X\left(\frac{2L}{3}\right) = \frac{4}{9} \qquad \text{and} \qquad \int_0^L X^2 \mathrm{d}x = \frac{L}{5}$$

the modified frequency becomes $\bar{\omega}_1 = 1818.8$ rad/s, which is a 34.2% lower than the natural frequency of the tuning blade. In the impact situation described above, the amplitude of vibration of the tuning blade can be calculated with the following relation

$$A_1 = \frac{mv_0 X(x_0)}{\bar{\omega}_1 \rho A \int_0^L X^2 \mathrm{d}x} = \frac{20mv_0}{9\bar{\omega}_1 \rho A L}$$

Finally, substituting the known values we get

$$A_1 = \frac{20 \times 1 \times 10^{-3} \times 1.4}{9 \times 1818.8 \times 8500 \times 7 \times 10^{-3} \times 0.8 \times 10^{-3} \times 36 \times 10^{-3}} = 0.998 \times 10^{-3} \approx 1.0 \text{ mm}$$

■

A

Useful mathematical derivations

A.1 Total and partial derivatives

A.1.1 Partial derivative

The notation used to indicated a partial derivative is ∂. A partial derivative is the derivative of a function of multiple variables with respect to only one of those variables, while keeping the other variables constant. Partial derivatives of f with respect to x and y can be calculated for $f(x, y)$ as follows:

$$\frac{\partial}{\partial x} f = \frac{\partial f}{\partial x} \mathrm{d}x + 0$$

$$\frac{\partial}{\partial y} f = \frac{\partial f}{\partial y} \mathrm{d}y + 0$$

$$\text{Example 1}: \quad f(x,y) = x + y \quad \text{and} \quad y = x^2 \quad \Rightarrow \quad \frac{\partial f}{\partial x} = \frac{\mathrm{d}x}{\mathrm{d}x} = 1$$

$$\text{Example 2}: \quad g(t) = x(t) + l\sin(\phi(t)) \quad \Rightarrow \quad \frac{\partial g}{\partial x} = \frac{\mathrm{d}x}{\mathrm{d}x} = 1 \quad \text{and} \quad \frac{\partial g}{\partial \phi} = l\cos(\phi)$$

A.1.2 Total derivative

The notation used to indicate a total derivative is d. If using the same examples as before, the total derivative of $f(x, y)$ with respect to x and y can be calculated as:

$$\mathrm{d}f = \frac{\partial f}{\partial x}\mathrm{d}x + \frac{\partial f}{\partial y}\mathrm{d}y$$

$$\text{Example 1}: \quad f(x,y) = x + y \quad \text{and} \quad y = x^2 \quad \Rightarrow \quad \frac{\mathrm{d}f}{\mathrm{d}x} = 1 + 1 \times \frac{\mathrm{d}y}{\mathrm{d}x} = 1 + 2x$$

$$\text{Example 2}: \quad g(t) = x(t) + l\sin(\phi(t)) \quad \Rightarrow \quad \frac{\mathrm{d}g}{\mathrm{d}t} = \dot{x}(t) + l\cos(\phi(t))\dot{\phi}(t)$$

A.2 Basic rules of differentiation

A derivative expresses the rate of change of a function $f(x)$ with respect to a variable x. In case of a linear function this can be represented as the slope $m = \frac{\Delta f(x)}{\Delta x}$. For the general application of this concept, by letting Δx become infinitely small, the slope of any function at a point can be approximated by a tangent at that point, that will be described by the following limit:

$$f'(x) = \lim_{\mathrm{d}x \to 0} \frac{f(x + \mathrm{d}x) - f(x)}{(x + \mathrm{d}x) - x} = \lim_{\mathrm{d}x \to 0} \frac{f(x + \mathrm{d}x) - f(x)}{\mathrm{d}x}$$

Derivatives of space $\frac{\mathrm{d}}{\mathrm{d}x} f(x)$ are noted as $f'(x)$ and derivatives of time $\frac{\mathrm{d}}{\mathrm{d}t} f(t)$ are noted as $\dot{f}(t)$ in these notes.

A.2.1 Basic rules

Power rule

$$f(x) = x^n$$

$$f'(x) = nx^{n-1}$$

$$\text{Example}: \quad f(x) = x^3 \quad \Rightarrow \quad \frac{\mathrm{d}}{\mathrm{d}x} f(x) = 3x^2$$

Chain rule

$$f(t) = g(h(t))$$

$$\frac{\mathrm{d}}{\mathrm{d}t} f(t) = \frac{\mathrm{d}g(h(t))}{\mathrm{d}t} \frac{\mathrm{d}h(t)}{\mathrm{d}t}$$

$$\text{Example}: \quad f(t) = \sin(x(t)) \quad \Rightarrow \quad \frac{\mathrm{d}}{\mathrm{d}t} f(t) = \cos(x(t))\dot{x}(t)$$

Product rule

$$f(t) = g(t)h(t)$$

$$\frac{\mathrm{d}}{\mathrm{d}t} f(t) = \frac{\mathrm{d}g(t)}{\mathrm{d}t} h(t) + \frac{\mathrm{d}h(t)}{\mathrm{d}t} g(t)$$

$$\text{Example}: \quad x(t) = r(t)\cos(\theta(t))$$

$$\Rightarrow \quad \frac{\mathrm{d}}{\mathrm{d}t} x(t) = \dot{x}(t) = \dot{r}(t)\cos(\theta(t)) + r(t)\dot{\theta}(t)[-\sin(\theta(t))]$$

A.2.2 Some useful derivatives

$$\frac{\mathrm{d}}{\mathrm{d}x} \sin(ax) = a\cos(ax)$$

$$\frac{\mathrm{d}}{\mathrm{d}x} \cos(ax) = -a\sin(ax)$$

$$\frac{\mathrm{d}}{\mathrm{d}x} \sin^2(ax) = 2a\sin(ax)\cos(ax) = a\sin(2ax)$$

$$\frac{\mathrm{d}}{\mathrm{d}x} \cos^2(ax) = 2a\cos(ax)[-\sin(ax)] = -a\sin(2ax)$$

A.3 Taylor expansion and linearisation

Taylor Series expansion is used to linearise a non-linear function (*e.g.* $f(x)$) around an equilibrium point $\bar{x}$ as follows:

$$f(x) = f(\bar{x}) + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x=\bar{x}} (x-\bar{x}) + \frac{1}{2}\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_{x=\bar{x}} (x-\bar{x})^2 + \frac{1}{3!}\left.\frac{\mathrm{d}^3 f}{\mathrm{d}x^3}\right|_{x=\bar{x}} (x-\bar{x})^3 + \dots$$

For $\bar{x}$ sufficiently close to x, the higher order terms will approach 0, and we can obtain the approximation:

$$f(x) \approx f(\bar{x}) + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x=\bar{x}} (x-\bar{x})$$

If $\bar{x}$ is chosen to be a point so that $f(\bar{x}) = 0$, this expression can be further simplified to:

$$f(x) = \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x=\bar{x}} (x-\bar{x})$$

We can define a perturbation $\delta x = x - \bar{x}$, which for small perturbations will approach $\delta x = x$ so that:

$$f(\delta x) = \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x=\bar{x}} (\delta x)$$

For example if we want to linearise $\sin(x)$ at $\bar{x} = 0$, we know that $\sin(0) = 0$, and $\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = \cos(x)$, with $\cos(0) = 1$, so that we will be able to approximate the values of sin for values close to $\bar{x} = 0$:

$$\sin(x) \approx \sin(0) + \cos(0)(x - 0) = 0 + 1 \times (x - 0) = x$$

However, for approximations far from the linearisation point, the approximation becomes inexact. For $x = \pi$ we have $\sin(\pi) \approx 0 + 1 \times (\pi - 0) = \pi$. As another example, when linearising $\cos(x + \pi)$ around $\bar{x} = 0$ we get

$$cos(x + \pi) \approx \cos(\bar{x} + \pi) - \sin(\bar{x} + \pi)(x - \bar{x}) = -1$$

A.4 Review on integration

Integration is used to find the area underneath the graph of a function.

A.4.1 Basic rules

Power rule

$$\int x^n \mathrm{d}x = \frac{1}{n+1} x^{n+1} + C$$

Multiplication by a constant

$$\int Kx^n \mathrm{d}x = K \int x^n \mathrm{d}x = K \left(\frac{1}{n+1}\right) x^{n+1} + C$$

Sum and difference rule

$$\int \cos(x) + x^n \mathrm{d}x = \int \cos(x)\mathrm{d}x + \int x^n \mathrm{d}x$$

Definite Integral

$$\int_{x=a}^{x=b} f(x)\mathrm{d}x = F(b) - F(a)$$

Example : $f(x) = x^n$

$$\Rightarrow \quad \int_{x=a}^{x=b} x^n \mathrm{d}x = \left[\frac{1}{n+1} x^{n+1} + C\right]_{x=a}^{x=b} = \frac{1}{n+1} b^{n+1} - \frac{1}{n+1} a^{n+1} + C$$

A.4.2 Some useful integrals

$$\int \sin(ax)\mathrm{d}x = -\frac{1}{a}\cos(ax) + C$$

$$\int \cos(ax)\mathrm{d}x = \frac{1}{a}\sin(ax) + C$$

$$\int \sin^2(ax)\mathrm{d}x = \frac{x}{2} - \frac{1}{4a}\sin(2ax) + C$$

$$\int \cos^2(ax)\mathrm{d}x = \frac{x}{2} + \frac{1}{4a}\sin(2ax) + C$$

A.5 Review on trigonometric functions

A.5.1 Basic trigonometric functions

Figure A.1 shows the graphical representation of the most commonly used trigonometric and hyperbolic functions.

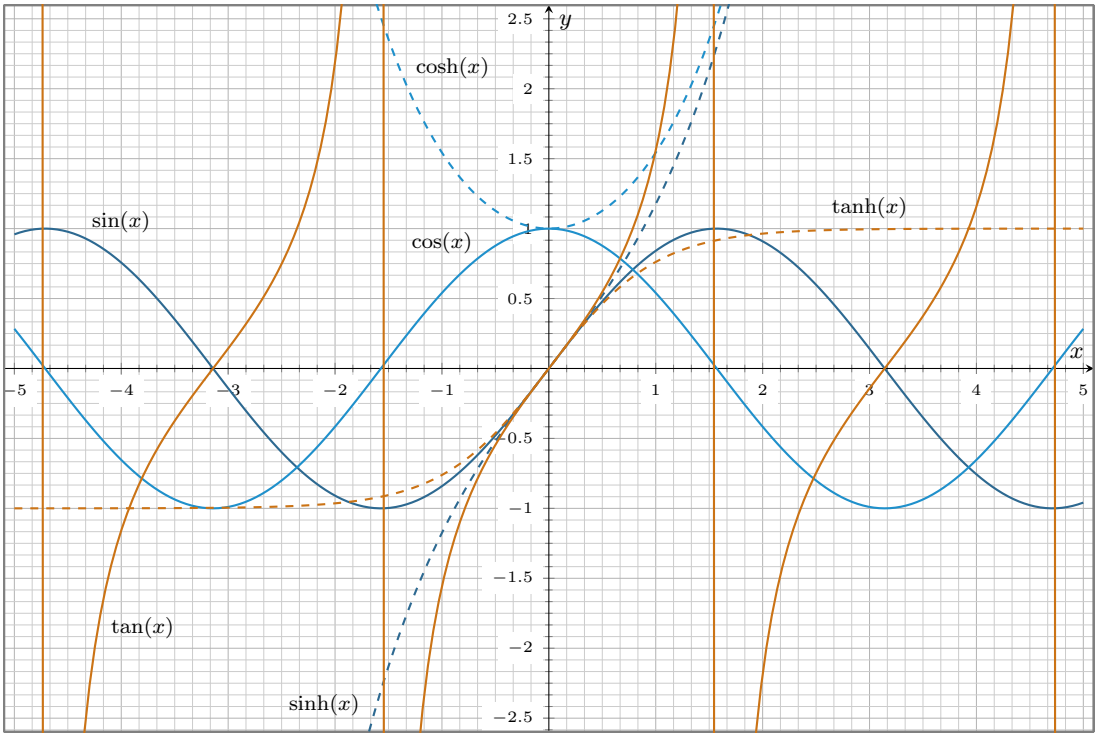


Figure A.1: Plots of some useful trigonometric (solid lines) and hyperbolic (dashed lines) functions.

A.5.2 Some useful identities

$$\sin^2(\alpha) + \cos^2(\alpha) = 1$$

$$\sin(\alpha)\sin(\beta) = \frac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$\cos(\alpha)\cos(\beta) = \frac{1}{2}[\cos(\alpha - \beta) + \cos(\alpha + \beta)]$$

$$\sin(\alpha)\cos(\beta) = \frac{1}{2}[\sin(\alpha + \beta) - \sin(\alpha - \beta)]$$

$$\sin(\alpha) + \sin(\beta) = 2\sin(\frac{\alpha + \beta}{2})\cos(\frac{\alpha - \beta}{2})$$

$$\sin(\alpha) - \sin(\beta) = 2\cos(\frac{\alpha + \beta}{2})\sin(\frac{\alpha - \beta}{2})$$

$$\cos(\alpha) + \cos(\beta) = 2\cos(\frac{\alpha + \beta}{2})\cos(\frac{\alpha - \beta}{2})$$

$$\cos(\alpha) - \cos(\beta) = -2\sin(\frac{\alpha + \beta}{2})\sin(\frac{\alpha - \beta}{2})$$

A.6 Review on vector and matrix operations

Transpose of a matrix

$$\text{If} \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \quad \Rightarrow \quad \mathbf{A}^{\mathrm{T}} = \begin{bmatrix} a_{11} & a_{21} & \dots & a_{n1} \\ a_{12} & a_{22} & \dots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{nn} \end{bmatrix}$$

$$\text{Example}: \quad \mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad \Rightarrow \quad \mathbf{A}^{\mathrm{T}} = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

Adjugate matrix

$$\text{If} \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \quad \Rightarrow \quad \mathrm{adj}\mathbf{A} = \mathbf{C}^{\mathrm{T}}$$

where $C_{ij} = (-1)^{i+j} M_{ij}$ and M_{ij} is the cofactor of element (i, j), that is, the determinant of the $(n-1) \times (n-1)$ matrix that results from deleting row i and column j from matrix $\mathbf{A}$. For example, if

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad \Rightarrow \quad \mathrm{adj}\mathbf{A} = \begin{bmatrix} 45-48 & -(36-42) & 32-35 \\ -(18-24) & 9-21 & -(8-14) \\ 12-15 & -(6-12) & 5-8 \end{bmatrix}^{\mathrm{T}} = \begin{bmatrix} -3 & 6 & -3 \\ 6 & -12 & 6 \\ -3 & 6 & -3 \end{bmatrix}$$

Inverse of a matrix

$$\text{If} \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \quad \Rightarrow \quad \mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}} \mathrm{adj}\mathbf{A}$$

$$\text{Example}: \quad \mathbf{A} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \quad \Rightarrow \quad \mathbf{A}^{-1} = \frac{1}{1} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

Determinant of a 2×2 matrix

$$\text{If} \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \Rightarrow \quad \det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Determinant of a 3×3 matrix

$$\text{If} \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\begin{aligned} \Rightarrow \quad \det \mathbf{A} &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) \\ &\quad + \ a_{13}(a_{21}a_{32} - a_{22}a_{31}) \end{aligned}$$

$$\text{Example}: \quad \mathbf{A} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \quad \Rightarrow \quad \det \mathbf{A} = 2 \times (2 \times 1 - 1) + 1 \times (-1) = 1$$

A.7 Roots of polynomial equations

Second order (quadratic) polynomials

The roots of the second order (quadratic) polynomial $ax^2 + bx + c = 0$ are

$$x_k = \frac{-b + (-1)^k \sqrt{b^2 - 4ac}}{2a} \quad \text{for} \quad k = 1, 2$$

Third order (cubic) polynomials

If the third order polynomial

$$x^3 + px^2 + qx + r = 0$$

solves for the eigenvalues of a dynamic system, the corresponding roots are

$$x_k = 2\sqrt{-\frac{A}{3}} \cos\left(\frac{\phi}{3} + \frac{2}{3}k\pi\right) - \frac{p}{3} \quad \text{for} \quad k = 0, 1, 2$$

where

$$A = \frac{1}{3}\left(3q - p^2\right) \quad \text{and} \quad B = \frac{1}{27}\left(2p^3 - 9qp + 27r\right)$$

and

$$\phi = \begin{cases} \arccos\left(\sqrt{-\frac{27B^2}{4A^3}}\right) & \text{if} \quad B < 0 \\ \arccos\left(-\sqrt{-\frac{27B^2}{4A^3}}\right) & \text{if} \quad B > 0 \end{cases}$$

Glossary

A | C | D | E | F | G | H | M | P | R | S | V

A

Assumed Modes Method gives an approximation to the relevant motion(s) of a system under a set of acting loads, based on an assumed (likely) mode shape of that same system (see also Rayleigh's method). 43

C

centre of gravity an imaginary point in a body of matter where, for convenience in certain calculations, the total weight of the body may be thought to be concentrated. The concept is sometimes useful in designing static structures (e.g., buildings and bridges) or in predicting the behaviour of a moving body when it is acted on by gravity. 20

characteristic equation also known as the eigenvalue equation, is a fundamental concept used to analyse the dynamic behaviour of structures, particularly linear time-invariant systems. It is used to determine the natural frequencies and mode shapes of a structure's dynamic response. The characteristic equation is typically expressed in the form: $\det(K - \omega^2 M) = 0$, where K is the stiffness matrix of the structure, ω is the frequency (rad/s) for which we are solving and M is the mass/inertia matrix of the structure. Solving this equation yields the natural frequencies for which the determinant of the system's characteristic equation equals zero. These natural frequencies represent the frequencies at which the structure will vibrate when subjected to dynamic loads or excitations. Each natural frequency is associated with a mode shape, which describes how the structure vibrates at that frequency. Once the natural frequencies and mode shapes are determined, they can be used to analyse and design structures for various dynamic loads, such as seismic, wind, or machinery-induced vibrations. Additionally, they are crucial for understanding the dynamic response and stability of the structure. The characteristic equation is an essential tool in structural dynamics and helps engineers ensure that structures can withstand dynamic forces while maintaining safety and functionality. 38

conceptual model is a representation of a system, made of the composition of concepts which are used to help people know, understand or simulate a subject the model represents. Some models are physical objects; for example, a toy model which may be assembled, and may be made to work like the object it represents. The term conceptual model may be used to refer to models which are formed after a conceptualisation or generalisation process. Conceptual models are often abstractions of things in the real world. 15

constitutive law is a relation between two physical quantities (especially kinetic quantities as

related to kinematic quantities) that is specific to a material or substance, and approximates the response of that material to external stimuli, usually as applied fields or forces. 42

D

degree-of-freedom of a mechanical system is an independent variable required to describe and define the configuration of the system. The number of degrees-of-freedom of a system is the minimum number of parameters that determine the state of the mechanical system. 17, 19

differential equation is a mathematical equation that relates some function with its derivatives. In applications the functions usually represent physical quantities, the derivatives represent their rates of change, and the equation defines a relationship between the two. Because such relations are extremely common, differential equations play a prominent role in many disciplines including engineering, physics, economics and biology. 15

E

equation of motion is a mathematical equation that describes the behaviour of a physical system in terms of its motion as a function of time. More specifically, the equations of motion describe the behaviour of a physical system as a set of mathematical functions in terms of dynamic variables: normally spatial coordinates and time. 22

F

free-body diagram is a graphical illustration used to visualise applied forces, moments, and resulting reactions on a body in a given condition. Free-body diagrams depict a body or set of connected bodies with all of the applied forces and moments, as well as reactions. In many cases, free-body diagrams are used both to determine the loading of individual structural components as well as calculating internal forces within the structure in many engineering disciplines. 15

G

generalised coordinate is a parameter that describe the configuration of a mechanical system relative to some reference configuration. The set of generalised coordinates must uniquely define the configuration of the system relative to the reference configuration. 17

generalised force is a friction or damping force or an external forces which does work and changes the energy of a system. 30

generalised velocity is the time derivative of the generalised coordinate of the mechanical system. 17

H

homogeneous beam is a beam with constant cross-section, constant density and constant stiffness. 53

M

mathematical model A mathematical description of the conceptual model. 15

moment of inertia of a rigid body, otherwise known as the angular mass or rotational inertia, determines the torque needed for a desired angular acceleration about a rotational axis. It depends on the body's mass distribution and the axis chosen, with larger moments requiring more torque to change the body's rotation. 20

P

potential energy is the energy possessed by an object because of its position relative to other objects, stresses within itself, its electric charge, or other factors. 20

R

Rayleigh's method is a technique for finding natural frequencies by equating the maximum kinetic energy of a system with the maximum potential (often elastic or strain) energy. 43, 107

S

simply supported beam is a type of beam that has pinned support at one end and roller support at the other end. Depending on the load applied, it undergoes shearing and bending. It is the one of the simplest structural elements. 43, 44, 46, 52

V

virtual displacement is an assumed infinitesimal change of system coordinates occurring while time is held constant. It is called virtual rather than real since no actual displacement can take place without the passage of time. Also, virtual displacements are spatial displacements exclusively — time is fixed while they occur. 31

virtual work of a force on a particle along a virtual displacement. The work of a force acting on a particle as it moves along a displacement will be different for different displacements. Among all the possible displacements that a particle may follow, called virtual displacements, one will minimise the action. This displacement is therefore the displacement followed by the particle according to the principle of least action. The work is known as the virtual work. 15, 31

Notation

Variables and symbols

A	–	Cross section area
$A,\ A_i$	–	Wave amplitude
c	–	Damping constant
c	–	Wave speed (axial, longitudinal)
$D,\ d$	–	Diameter
E	–	Elastic modulus (Young's modulus)
f	–	Wave frequency
$f(x,t)$	–	Generic distributed force
φ_i	–	Generic rotation degree of freedom (generalised coordinate)
$\varphi(x)$	–	Generic shape function
G	–	Shear modulus
γ	–	Adiabatic gas constant
I	–	Second moment of area
J	–	Polar moment of inertia
$K,\ k$	–	Stiffness (spring)
$K_t,\ k_t$	–	Torsional (spring) stiffness
k_f	–	Flexural (spring) stiffness
$L,\ l$	–	Length
$\mathcal{L}$	–	Lagrangian operator
λ	–	Wavelength
$M,\ m$	–	Mass (lumped or continuous)
$M,\ M(x)$	–	Bending moment
n	–	Number of degrees-of-freedom
n	–	Number of moles
ω	–	Natural frequency
$p,\ p(x,t)$	–	Pressure
Q_i	–	Generalised force (force, torque)
q_i	–	Generalised coordinate (displacement, angle)
$q(t)$	–	Generic amplitude function
R	–	Universal gas constant
ρ	–	Density

σ	–	Normal stress
T	–	Kinetic energy
T	–	Temperature
$T(t)$	–	Generic time function
t	–	Time
τ	–	Applied torque
$\tau(x)$	–	Shear stress
θ_i	–	Generic rotation degree of freedom (generalised coordinate)
$\dot{\theta}_i$	–	Angular velocity
$\ddot{\theta}_i$	–	Angular acceleration
$u(x,t)$	–	Axial (longitudinal) displacement or wave
V	–	Potential energy (gravitational, elastic, *etc.*)
$V,\ V_0$	–	Volume, initial volume
W	–	Work
w	–	Transverse displacement (deflection)
$X(x)$	–	Generic displacement (shape) function
$x_i,\ y_i,\ z_i$	–	Generic linear degrees of freedom (generalised coordinates)
$\dot{x}_i,\ \dot{y}_i,\ \dot{z}_i$	–	Generic linear velocities
$\ddot{x}_i,\ \ddot{y}_i,\ \ddot{z}_i$	–	Generic linear accelerations
$\bar{z}(t)$	–	Prescribed boundary motion

Acronyms and abbreviations

ACW	–	Antilockwise
AMM	–	Assumed Modes Method
CW	–	Clockwise
FBD	–	Free Body Diagram
RM	–	Rayleigh's Method

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Bibliography

Sources of additional reading

Below is a list of references for additional reading, including books, scientific papers and web-page links. This is not meant to be an extensive list of supporting references and each listed reference addresses in more or less detail one or more of the topics covered in these course notes.

- B.N. Das, A. Kassimali, S. Sami. *Engineering mechanics: Dynamics*, Irwin Publications, Burr Ridge, ISBN 0-256-11450-1, 1994
- S.S. Rao. *Mechanical vibrations* (5^{th} edition in SI units), Prentice Hall, Singapore, ISBN 978-981-06-8712-0, 2011
- V.V. Bolotin. *Nonconservative problems of the theory of elastic stability*, Pergamon Press, Oxford, ISBN 978-008-01-0490-4, 1963
- W. Goldsmith. *Impact: The theory and physical behaviour of colliding solids*, Dover Publications Inc., New York, ISBN 0-486-42004-3, 2001
- M.A. Meyers. *Dynamic behaviour of materials*, John Wiley & sons, Inc., New York, ISBN 0-471-58262-X, 1994
- W.J. Stronge. *Impact mechanics*, Cambridge University Press, Cambridge, ISBN 0-521-60289-0, 2004
- H.F. Olson. *Music, physics and engineering*, Dover Publications Inc., New York, ISBN 0-486-21769-8, 1967
- *Wikipedia: The Free Encyclopedia*, Wikimedia Foundation Inc. https://en.wikipedia.org, 2017